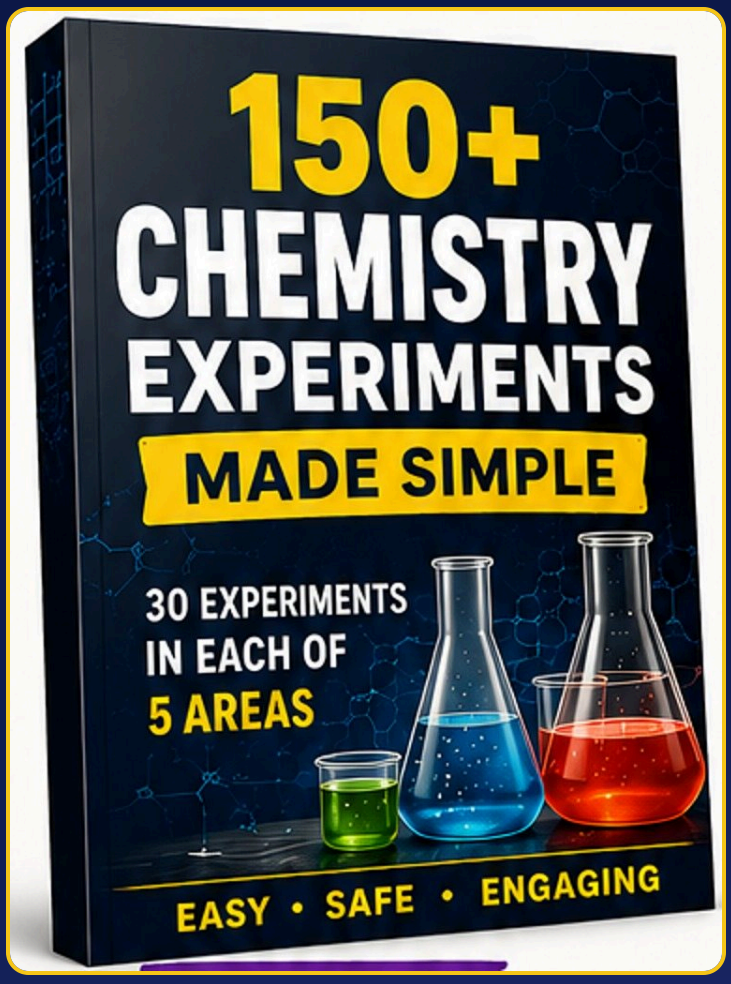
MAIN PRODUCT

150+ Chemistry Experiments Made Simple

30 experiments in each of 5 areas, each one a single printable card



Welcome

> *150 one-page chemistry experiment cards for grades 6-12: 30 experiments in each of 5 areas. No lab required. Open a card, print it and teach it.*

What is on every card

1. **Why does it happen?** The science in plain words your students can follow.
2. **Materials with exact amounts.** Grocery, pharmacy or hardware-store items only.
3. **6 illustrated steps.** A picture for every step, so nobody gets lost.
4. **The equation, simplified.** A balanced equation or the physical process, with a short explanation.
5. **What students will see.** So you know the experiment worked.
6. **Level, time, grade, NGSS and safety.** Plan in minutes.

> **Safety first**
>
> Every student wears splash goggles. An adult handles hot water, candles, cutting and anything marked **Adult handling**. Nothing in the lab is tasted. Read the Classroom Lab Safety Guide (Bonus 2) before your first experiment and follow your school policy.

How to plan

Prep takes about 15 minutes (gather 3-6 simple materials). Most experiments run 20-40 minutes of class time; a few, like crystal growing, run over several days and say so on the card. The NGSS codes are performance expectations the activity supports; use your own standards and pacing to decide how deeply to teach each one.

Illustrations on the cards are simple pictograms for the 6 steps, not photographs of the experiment.

Index of the 150 experiments

General Chemistry (30)

ID	EXPERIMENT	GRADES	LEVEL	NGSS
GEN-01	Elephant Toothpaste	6-12	Medium	MS-PS1-2, HS-PS1-5
GEN-02	Liquid Density Tower	6-12	Easy	MS-PS1-2
GEN-03	Homemade Lava Lamp	6-12	Easy	MS-PS1-2, HS-PS1-2
GEN-04	Oil, Water and Soap	6-12	Easy	MS-PS1-1, MS-PS1-2
GEN-05	Mixtures vs. Solutions	6-8	Easy	MS-PS1-2
GEN-06	Filtering and Evaporation	6-12	Medium	MS-PS1-2, MS-PS1-4
GEN-07	Marker Chromatography	6-12	Easy	MS-PS1-2
GEN-08	Magnetic Separation	6-8	Easy	MS-PS1-2
GEN-09	Dissolving Speed vs. Temperature	6-12	Medium	MS-PS1-4, MS-PS3-4
GEN-10	Saturated Solutions	6-12	Medium	MS-PS1-2, HS-PS1-3
GEN-11	Rock Candy Crystals	6-12	Challenge	MS-PS1-4, HS-PS1-3
GEN-12	Epsom Salt Crystals	6-12	Medium	MS-PS1-2, HS-PS1-3
GEN-13	Physical vs. Chemical Change Stations	6-8	Easy	MS-PS1-2
GEN-14	Yeast Balloon Inflation	6-12	Medium	MS-LS1-7, HS-PS1-2
GEN-15	Conservation of Mass in a Bag	9-12	Challenge	MS-PS1-5, HS-PS1-7
GEN-16	Fizzing Tablet Reaction Rate	9-12	Medium	HS-PS1-5
GEN-17	Cleaning Pennies	6-12	Easy	MS-PS1-2, HS-PS1-2
GEN-18	Magic Milk	6-8	Easy	MS-PS1-2
GEN-19	Diffusion in Hot vs. Cold Water	6-12	Easy	MS-PS1-4, MS-PS3-4
GEN-20	Walking Water	6-8	Easy	MS-PS1-1
GEN-21	Invisible Ink Reveal	6-12	Medium	MS-PS1-2, HS-PS1-2
GEN-22	Floating Egg	6-12	Easy	MS-PS1-2, HS-PS1-3
GEN-23	Sugar Water Density Rainbow	9-12	Challenge	MS-PS1-2, HS-PS1-3
GEN-24	Rusting Steel Wool	9-12	Medium	HS-PS1-2, HS-PS1-4
GEN-25	Cold Fizz: An Endothermic Reaction	9-12	Challenge	HS-PS1-4, MS-PS1-6
GEN-26	Salt and Ice: Lowering the Freezing Point	9-12	Medium	MS-PS3-4, HS-PS3-4
GEN-27	Catalyst Race	9-12	Challenge	HS-PS1-5, MS-LS1-7
GEN-28	Evaporation and Surface Area	6-8	Easy	MS-PS1-4, MS-ESS2-4
GEN-29	Drops on a Penny	6-8	Easy	MS-PS1-1
GEN-30	Candle in a Jar	9-12	Challenge	HS-PS1-2, HS-PS1-7

Organic & Biochemistry (30)

ID	EXPERIMENT	GRADES	LEVEL	NGSS
ORG-01	Strawberry DNA Extraction	6-12	Medium	MS-PS1-2, HS-LS1-6
ORG-02	Pineapple vs Gelatin: Enzymes at Work	6-12	Easy	MS-LS1-7, MS-PS1-2
ORG-03	Heat Switches Off an Enzyme	9-12	Medium	HS-LS1-6, HS-PS1-5
ORG-04	Starch Disappears: Amylase and Iodine	9-12	Challenge	HS-LS1-6, MS-LS1-7
ORG-05	Yeast Balloon Inflator	6-8	Easy	MS-LS1-7, MS-PS1-2
ORG-06	What Does Yeast Like to Eat?	6-12	Medium	MS-LS1-7, HS-LS1-7
ORG-07	Curds and Whey: Cheese Chemistry	6-8	Easy	MS-PS1-2, MS-LS1-7
ORG-08	Plastic from Milk	6-12	Medium	MS-PS1-3, MS-PS1-2
ORG-09	Polymer Slime	6-8	Easy	MS-PS1-3, MS-PS1-2
ORG-10	Oobleck: Liquid or Solid?	6-8	Easy	MS-PS1-1, MS-PS1-2
ORG-11	Melt-and-Pour Soap	6-12	Medium	MS-PS1-4, MS-PS1-2
ORG-12	Emulsifier Showdown	6-12	Medium	MS-PS1-2, HS-PS1-3
ORG-13	Make Mayonnaise: Egg Yolk Emulsion	9-12	Challenge	MS-PS1-2, HS-LS1-6
ORG-14	Butter in a Jar	6-8	Easy	MS-PS1-2, MS-LS1-7
ORG-15	Vitamin C Detective	9-12	Challenge	HS-PS1-2, HS-LS1-6
ORG-16	Starch Detector: Iodine Food Test	6-8	Easy	MS-LS1-7, MS-PS1-2
ORG-17	The Brown Paper Bag Fat Test	6-8	Easy	MS-LS1-7, MS-PS1-2
ORG-18	Sugar Hunter: Glucose Strip Test	9-12	Medium	MS-LS1-7, HS-LS1-6
ORG-19	Egg White Meets Alcohol	9-12	Medium	HS-LS1-6, MS-PS1-2
ORG-20	Cooking Without Heat: Egg White and Acid	6-12	Medium	MS-PS1-2, HS-LS1-6
ORG-21	Fats: Solid or Liquid?	9-12	Medium	HS-LS1-6, MS-PS1-4
ORG-22	Carbon Chains with Gumdrops	6-8	Easy	MS-PS1-1, MS-PS1-5
ORG-23	Isomers: Same Atoms, Different Molecules	9-12	Challenge	HS-PS1-1, MS-PS1-1
ORG-24	Why Do Bananas Turn Brown?	6-12	Easy	MS-PS1-2, HS-PS1-5
ORG-25	Milk Art: Soap and Fat Explosion	6-8	Easy	MS-PS1-2, MS-PS1-1
ORG-26	Chlorophyll Color Detective	9-12	Medium	MS-LS1-6, MS-PS1-2
ORG-27	Biodegradable Plastic from Starch	9-12	Challenge	MS-PS1-3, HS-ESS3-4
ORG-28	Potato Foam: Catalase Enzyme	6-12	Medium	HS-PS1-5, MS-LS1-7
ORG-29	Soap vs Grease	6-12	Easy	MS-PS1-2, MS-PS1-1
ORG-30	Ripening Gas: Ethylene and Fruit	6-8	Easy	MS-LS1-7, MS-PS1-2

Acids, Bases & Metals (30)

ID	EXPERIMENT	GRADES	LEVEL	NGSS
ACI-01	Egg Without a Shell	6-12	Easy	MS-PS1-2, MS-PS1-5
ACI-02	Red Cabbage pH Indicator	6-12	Easy	MS-PS1-2
ACI-03	Household pH Rainbow	6-12	Medium	MS-PS1-2, HS-PS1-2

ID	EXPERIMENT	GRADES	LEVEL	NGSS
ACI-04	Turmeric Test: Soap Turns Red	6-8	Easy	MS-PS1-2
ACI-05	Where Did the Mass Go?	9-12	Medium	HS-PS1-7, HS-PS1-2
ACI-06	Antacid vs. Acid	6-12	Medium	MS-PS1-2, HS-PS1-2
ACI-07	Invisible Ink: Baking Soda and Grape Juice	6-8	Easy	MS-PS1-2
ACI-08	Homemade pH Paper	6-12	Medium	MS-PS1-2, MS-ETS1-3
ACI-09	Buffer Test: Water vs. Baking Soda Water	9-12	Challenge	HS-PS1-6, HS-PS1-2
ACI-10	Clean the Penny	6-8	Easy	MS-PS1-2
ACI-11	Steel Wool Rusting and Warm Jar	9-12	Challenge	HS-PS1-4, MS-PS1-2
ACI-12	Nail Rust Race	6-12	Easy	MS-PS1-2, MS-ETS1-3
ACI-13	Copper Coat on an Iron Nail	9-12	Challenge	HS-PS1-2, HS-PS1-7
ACI-14	Lemon Battery	6-12	Medium	HS-PS1-2, MS-ETS1-2
ACI-15	Penny Pile Battery	9-12	Challenge	HS-PS1-2, MS-ETS1-3
ACI-16	Metals in Vinegar: Activity Series	9-12	Medium	HS-PS1-1, HS-PS1-2
ACI-17	Iron in Your Cereal	6-8	Easy	MS-PS1-1, MS-PS1-2
ACI-18	Aluminum's Invisible Armor	9-12	Medium	HS-PS1-2, HS-PS1-3
ACI-19	Carbonate Detective	6-8	Easy	MS-PS1-2
ACI-20	Chalk Acid Detector	6-8	Easy	MS-PS1-2
ACI-21	Acid Rain Eats the Statue	9-12	Medium	MS-ESS3-3, HS-ESS3-4
ACI-22	Acid Rain and a Leaf	9-12	Medium	MS-ESS3-3, HS-ESS3-4
ACI-23	Hard Water vs. Soap	6-12	Medium	MS-PS1-2, HS-ESS2-5
ACI-24	Why Silver Tarnishes	9-12	Medium	MS-PS1-2, HS-PS1-2
ACI-25	Vitamin C Stops Browning	6-12	Easy	MS-PS1-2, HS-LS1-7
ACI-26	Titration-Lite: Count the Drops	9-12	Challenge	HS-PS1-2, HS-PS1-6
ACI-27	Citric Acid + Baking Soda Dry Fizz	6-12	Easy	MS-PS1-2, HS-PS1-4
ACI-28	Zinc Protects Iron	9-12	Challenge	HS-PS1-2, MS-ETS1-3
ACI-29	Curdling Milk with Acid	6-12	Medium	MS-PS1-2, HS-LS1-6
ACI-30	Rust Remover Showdown	6-12	Medium	HS-PS1-2, MS-ETS1-3

Physical Chemistry (30)

ID	EXPERIMENT	GRADES	LEVEL	NGSS
PHY-01	Ice Cream in a Bag	6-12	Medium	MS-PS1-4, HS-PS3-4
PHY-02	Ice, Water, and Steam	6-12	Easy	MS-PS1-4
PHY-03	Salt, Sugar, or Plain? Melting Ice Race	6-8	Easy	MS-PS1-4, MS-ETS1-3
PHY-04	Instant Ice from Supercooled Water	9-12	Challenge	MS-PS1-4, HS-PS3-4
PHY-05	Cold Pack from Baking Soda and Vinegar	6-12	Easy	MS-PS1-6, HS-PS1-4
PHY-06	Warm Foam from Yeast and Peroxide	9-12	Medium	HS-PS1-4, HS-PS1-5
PHY-07	Steel Wool Hand Warmer	9-12	Medium	MS-PS1-6, HS-PS1-4

ID	EXPERIMENT	GRADES	LEVEL	NGSS
PHY-08	Fizz Speed vs Water Temperature	6-12	Medium	HS-PS1-5, MS-PS1-4
PHY-09	Fizz Speed vs Surface Area	6-12	Easy	HS-PS1-5, MS-PS1-2
PHY-10	Balloon Race: Vinegar Strength	9-12	Medium	HS-PS1-5, HS-PS1-7
PHY-11	Catalyst Showdown: Yeast vs Potato	9-12	Medium	HS-PS1-5, HS-PS1-4
PHY-12	Salt and the Boiling Point	9-12	Challenge	MS-PS1-4, HS-PS1-3
PHY-13	Which Evaporates Faster? Surface Area Test	6-8	Easy	MS-PS1-4, MS-ESS2-4
PHY-14	Rain on a Can: Condensation	6-8	Easy	MS-ESS2-4, MS-PS1-4
PHY-15	Balloon on a Bottle: Charles's Law	9-12	Medium	MS-PS1-4, HS-PS3-4
PHY-16	Marshmallow in a Syringe: Boyle's Law	9-12	Medium	MS-PS1-4, HS-PS3-4
PHY-17	The Collapsing Bottle	6-12	Medium	MS-PS1-4, HS-PS3-4
PHY-18	Does Air Have Mass?	6-8	Easy	MS-PS1-1, MS-PS1-2
PHY-19	Carbon Dioxide Pour	6-12	Medium	MS-PS1-2, HS-PS1-2
PHY-20	Color Spread: Hot vs Cold Diffusion	6-8	Easy	MS-PS1-4, MS-PS3-4
PHY-21	Potato Osmosis	6-12	Medium	MS-PS1-1, HS-PS1-3
PHY-22	Hot or Cold: How Much Sugar Dissolves?	9-12	Medium	HS-PS1-3, MS-PS1-2
PHY-23	Crystal Growth: Fast vs Slow Cooling	6-12	Medium	MS-PS1-2, HS-PS1-3
PHY-24	Liquid Race: Viscosity	6-8	Easy	MS-PS1-1, MS-PS1-4
PHY-25	How Many Drops on a Penny?	6-8	Easy	MS-PS1-1, HS-PS1-3
PHY-26	Heating Race: Water vs Oil	9-12	Challenge	MS-PS3-4, HS-PS3-4
PHY-27	Spoon Heat Race: Thermal Conductivity	6-8	Easy	MS-PS3-4, MS-ETS1-3
PHY-28	The Great Insulation Challenge	6-12	Medium	MS-PS3-4, MS-ETS1-2
PHY-29	Density Tower	6-12	Easy	MS-PS1-2, HS-PS1-3
PHY-30	Boiling Without Heat	9-12	Challenge	MS-PS1-4, HS-ESS2-5

Environmental Chemistry (30)

ID	EXPERIMENT	GRADES	LEVEL	NGSS
ENV-01	Water Filter Challenge	6-12	Medium	MS-ETS1-2, MS-ETS1-3
ENV-02	Acid Rain vs Chalk	6-8	Easy	MS-PS1-2, MS-ESS3-3
ENV-03	Acid Rain and Marble Chips	9-12	Medium	HS-PS1-5, HS-PS1-7
ENV-04	Acid Rain and a Leaf	6-8	Easy	MS-ESS3-3, MS-PS1-2
ENV-05	Greenhouse Effect in Bottles	6-12	Medium	MS-PS3-4, MS-ESS3-3
ENV-06	Breath, Water and Cabbage Indicator	9-12	Medium	HS-PS1-6, HS-ESS2-5
ENV-07	Why Fizzy Water Goes Flat	6-8	Easy	MS-PS1-4, MS-PS1-2
ENV-08	Ocean Acidification and Shells	9-12	Challenge	HS-ESS3-4, HS-PS1-2
ENV-09	Oil Spill Cleanup Challenge	6-12	Challenge	MS-ETS1-2, MS-ETS1-3
ENV-10	Plastic Bags in Soil	6-12	Medium	MS-PS1-3, MS-ESS3-3
ENV-11	Decomposition in Compost Jars	6-8	Easy	MS-LS1-7, MS-ESS3-3

ID	EXPERIMENT	GRADES	LEVEL	NGSS
ENV-12	Soil pH with Cabbage Indicator	6-12	Easy	MS-PS1-2, HS-PS1-2
ENV-13	Soil Filtering Race	6-8	Easy	MS-ESS2-4, MS-ETS1-3
ENV-14	Erosion Slope Challenge	6-8	Medium	MS-ESS3-3, MS-ETS1-2
ENV-15	Water Hardness Soap Shake Test	9-12	Medium	HS-PS1-2, HS-ESS2-5
ENV-16	Homemade Secchi Disk	6-12	Medium	MS-ESS3-3, MS-ETS1-3
ENV-17	Solar Still: Salt Water to Fresh	6-12	Medium	MS-ESS2-4, MS-PS1-4
ENV-18	Water Cycle in a Bag	6-8	Easy	MS-ESS2-4, MS-PS1-4
ENV-19	Algae Bloom Simulation	9-12	Challenge	HS-ESS3-4, HS-LS1-6
ENV-20	Air Particulate Sampling	6-12	Easy	MS-ESS3-3, MS-ETS1-3
ENV-21	Detergent and Seed Germination	6-8	Medium	MS-ESS3-3, MS-LS1-7
ENV-22	Leaf Disc Photosynthesis	9-12	Challenge	MS-LS1-6, HS-LS1-6
ENV-23	Thermos Challenge	6-12	Medium	MS-PS3-4, MS-ETS1-3
ENV-24	Recycle Your Own Paper	6-8	Easy	MS-PS1-3, MS-ESS3-3
ENV-25	Natural Dyes from Onion Skins	6-8	Medium	MS-PS1-3, MS-PS1-2
ENV-26	Road Salt on Ice	6-12	Medium	MS-PS1-4, MS-ESS3-3
ENV-27	Rust Race: Iron and Salt Water	6-12	Medium	MS-PS1-2, HS-PS1-2
ENV-28	Cornstarch Bioplastic Film	9-12	Challenge	MS-PS1-3, HS-ETS1-2
ENV-29	Mulch and Soil Moisture	6-8	Easy	MS-ESS2-4, MS-ETS1-3
ENV-30	Warm Water Loses Its Gas	9-12	Easy	HS-PS3-4, HS-ESS3-4

ELEPHANT TOOTHPASTE

Catalysts and decomposition of hydrogen peroxide

Grades 6-12 | Medium | Prep 15 min | Class 20-30 min | MS-PS1-2 | HS-PS1-5 | Adult handling

WHY DOES IT HAPPEN?

Hydrogen peroxide slowly breaks down into water and oxygen gas. Yeast contains an enzyme that works as a catalyst: it speeds the breakdown up without being used up. The oxygen gets trapped in dish soap, so it comes out as a tall column of foam.

MATERIALS

- 1/2 cup (120 mL) 3% hydrogen peroxide (drugstore bottle)
- 1 packet (2 1/4 tsp, 7 g) dry yeast
- 3 tbsp (45 mL) warm tap water
- 1 tbsp (15 mL) dish soap
- 4-5 drops food coloring
- 1 empty 16 oz (500 mL) plastic bottle
- 1 large baking tray or tub

6 STEPS

1. 
Stand the bottle in the tray and put on splash goggles. An adult pours the hydrogen peroxide into the bottle.

2. 
Add the dish soap and the food coloring to the bottle. Swirl gently to mix.

3. 
In a small cup, stir the yeast into the warm water for 30 seconds.

4. 
Start the timer, then pour the yeast mixture into the bottle all at once.

5. 
Step back and watch the foam rise out of the bottle.

6. 
Touch the bottle with the back of a finger to feel its temperature. Wipe up with paper towels and rinse down the drain.

THE EQUATION, SIMPLIFIED

$$2\,H_2O_2 \rightarrow 2\,H_2O + O_2$$

Hydrogen peroxide splits into water and oxygen; yeast speeds it up as a catalyst.

WHAT STUDENTS WILL SEE

Within 5-10 seconds a thick, colored foam shoots up and out of the bottle. The bottle feels warm because the reaction releases heat. The foam is mostly soap bubbles filled with oxygen.

Safety: Adult handling for the peroxide. Wear splash goggles, never use peroxide stronger than 3%, and keep the foam away from eyes and mouth.

Try next: Repeat with half the yeast and compare the foam height.

LIQUID DENSITY TOWER

Density and layering of liquids

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Density is how much mass is packed into a given volume. Liquids that are denser sink below liquids that are less dense, as long as they do not mix. Stacking them from heaviest to lightest builds a rainbow tower.

MATERIALS

- 1/4 cup (60 mL) honey
- 1/4 cup (60 mL) light corn syrup
- 1/4 cup (60 mL) dish soap
- 1/4 cup (60 mL) water with food coloring
- 1/4 cup (60 mL) vegetable oil
- 1 tall clear glass or jar
- 1 spoon

6 STEPS

1. 
   Put on goggles. Pour the honey into the center of the jar without touching the sides.
2. 
   Slowly pour the corn syrup down the back of a spoon so it settles on top.
3. 
   Repeat with the dish soap, pouring slowly over the spoon.
4. 
   Pour the colored water over the spoon as gently as you can.
5. 
   Finish with the vegetable oil as the top layer.
6. 
   Draw the tower and label each layer from densest to least dense.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Each liquid keeps its identity; they only sort by density.

WHAT STUDENTS WILL SEE

Five clear layers form within a minute and stay separate. The honey sits at the bottom and the oil floats at the top. Shaking the jar briefly mixes some layers, which slowly separate again.

Safety: Wipe spills right away because honey and syrup make floors sticky. Do not taste anything.

Try next: Drop a grape, a small button, and a cork in and see where each stops.

HOMEMADE LAVA LAMP

Gas production and density

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2 | HS-PS1-2

WHY DOES IT HAPPEN?

Oil floats on water because it is less dense. A fizzing tablet reacts with water and makes carbon dioxide gas. Bubbles of gas carry blobs of colored water up, then drop them again when the gas escapes.

MATERIALS

- 1 clear bottle or tall jar
- 3/4 cup (180 mL) vegetable oil
- 1/2 cup (120 mL) water
- 6 drops food coloring
- 2 fizzing antacid tablets (each broken in half)
- 1 flashlight (optional)
- 1 tray

6 STEPS

1. Put on goggles. Fill the bottle about two thirds with oil.
2. Add the water. It sinks below the oil.
3. Add the food coloring and watch the drops sink through the oil.
4. Drop in half a tablet and watch closely.
5. Shine the flashlight from underneath to make the blobs glow.
6. When the bubbling stops, add another half tablet and compare.

THE EQUATION, SIMPLIFIED

$$3\ NaHCO_3 + C_6H_8O_7 \rightarrow Na_3C_6H_5O_7 + 3\ H_2O + 3\ CO_2$$

Sodium bicarbonate and citric acid in the tablet react in water to make carbon dioxide.

WHAT STUDENTS WILL SEE

Colored blobs rise and fall for several minutes after each piece of tablet is added. The motion slows as the tablet is used up.

Safety: Do not swallow any tablet or liquid. Never put a cap on the bottle while it is fizzing.

Try next: Try warm water and see whether the blobs move faster.

OIL, WATER AND SOAP

Polar and nonpolar molecules, emulsions

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-1 | MS-PS1-2

WHY DOES IT HAPPEN?

Oil and water do not mix because water molecules attract each other more than they attract oil. Dish soap molecules have one end that likes water and one end that likes oil. They surround tiny oil droplets and let them spread through the water.

MATERIALS

- 1 clear jar with lid
- 1/2 cup (120 mL) water
- 1/4 cup (60 mL) vegetable oil
- 3 drops food coloring
- 1 tsp (5 mL) dish soap
- 1 spoon
- 1 timer

6 STEPS

1. 
Put on goggles. Pour the water and the food coloring into the jar.

2. 
Add the oil and wait 30 seconds. Note how the layers sit.

3. 
Close the lid and shake hard for 10 seconds.

4. 
Set the jar down and time how fast the oil returns to the top.

5. 
Add the dish soap, close the lid and shake again.

6. 
Time the separation again and compare it with the first try.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Soap lets oil droplets stay spread out in water; nothing is chemically changed.

WHAT STUDENTS WILL SEE

Without soap, the oil separates in under a minute. With soap, the jar stays cloudy for many minutes because the oil droplets are held in the water.

Safety: Make sure the lid is tight, wipe spills right away, and do not drink the mixture.

Try next: Test different amounts of soap and graph the separation time.

MIXTURES VS. SOLUTIONS

Classifying mixtures

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-PS1-2

WHY DOES IT HAPPEN?

A solution is a mixture where the particles are so small they spread out evenly and cannot be seen. In a suspension, bigger bits stay visible and eventually settle. Testing different materials in water shows which behave as solutions.

MATERIALS

- 4 clear cups
- 1 cup (240 mL) water per cup
- 1 tsp (5 g) table salt
- 1 tsp (4 g) sugar
- 1 tsp (5 g) sand
- 1 tsp (3 g) flour
- 1 flashlight

6 STEPS

1. 
Put on goggles and label the cups: salt, sugar, sand, flour.

2. 
Pour 1 cup of water into each cup.

3. 
Add one teaspoon of each material to its cup.

4. 
Stir each cup for 20 seconds.

5. 
Shine the flashlight through each cup and record which look clear and which look cloudy.

6. 
Wait 10 minutes. Record which settled and which stayed mixed.

THE EQUATION, SIMPLIFIED

$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

Salt breaks into ions that spread evenly in water, forming a solution.

WHAT STUDENTS WILL SEE

Salt and sugar vanish into clear liquid (solutions). Sand sinks quickly. Flour makes the water cloudy and settles slowly (suspension).

Safety: Do not taste any of the mixtures. Keep sand off the floor to avoid slips.

Try next: Try baking soda and cornstarch and add them to the chart.

FILTERING AND EVAPORATION

Separating a salt and sand mixture

Grades 6-12 | Medium | Prep 15 min | Class 25-40 min | MS-PS1-2 | MS-PS1-4 | Adult handling

WHY DOES IT HAPPEN?

Sand does not dissolve, but salt does. Filtering catches the sand, and letting the water evaporate leaves the salt behind. Both are physical separations because no new substances form.

MATERIALS

- 2 tbsp (30 g) sand
- 1 tbsp (18 g) table salt
- 1 cup (240 mL) warm water
- 1 coffee filter
- 1 funnel or tall cup
- 2 clear cups
- 1 shallow saucer

6 STEPS

1. 
Put on goggles. Stir the sand and salt together in a cup.

2. 
Add the warm water and stir for one minute.

3. 
Set the coffee filter in the funnel over the second cup.

4. 
Pour the mixture slowly through the filter.

5. 
Pour the clear liquid onto the saucer and set it in a sunny window.

6. 
Check the saucer after a day and look at the crystals with a hand lens.

THE EQUATION, SIMPLIFIED

NaCl(aq) → NaCl(s)

As water evaporates, salt ions come back together and form solid crystals.

WHAT STUDENTS WILL SEE

The filter keeps the sand and the liquid passes through clear. After a day or more the saucer holds small white cubic salt crystals.

Safety: Adult handling of warm water. Do not taste the salt crystals.

Try next: Weigh the salt you recover and compare it with the 18 g you started with.

MARKER CHROMATOGRAPHY

Separating pigments

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Ink is often a mixture of several colored pigments. When water climbs a paper strip, each pigment travels at its own speed. This separates them into a band of colors.

MATERIALS

- 3 coffee filters cut into strips
- 3 washable markers (black, brown, green)
- 3 tall clear cups
- 1/4 cup (60 mL) water
- 3 pencils
- tape
- 1 ruler

6 STEPS

1. 
   Put on goggles. Draw a thick dot 1 inch (2.5 cm) from the bottom of each strip.
2. 
   Tape each strip to a pencil.
3. 
   Pour about 1/2 inch (1 cm) of water into each cup.
4. 
   Hang the strip so the end touches the water but the dot stays above it.
5. 
   Watch for 10 minutes while the water climbs.
6. 
   Dry the strips and measure how far each color traveled.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

The pigments only separate; none of them turns into something else.

WHAT STUDENTS WILL SEE

Black ink usually splits into blue, purple and yellow bands. Brown and green split into several colors as well. The strips show different patterns for each marker.

Safety: Use washable markers only. Do not put markers or strips in the mouth.

Try next: Test the same marker with salt water and compare how far it climbs.

MAGNETIC SEPARATION

Separating mixtures by properties

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Some materials are attracted to a magnet and others are not. Iron-fortified cereal contains tiny pieces of iron. Using a magnet is a physical way to pull a magnetic ingredient out of a mixture.

MATERIALS

- 1 cup (30 g) iron-fortified cereal
- 1/4 cup (50 g) sand
- 2 tbsp (35 g) table salt
- 1 strong magnet in a zipper bag
- 1 bowl
- 1 clear cup of warm water
- 1 spoon

6 STEPS

1. Put on goggles. Crush the cereal in a zipper bag until it is a powder.
2. Mix the cereal powder, sand and salt in the bowl.
3. Move the bagged magnet slowly over the mix and watch what clings.
4. Scrape the clinging bits into a small cup.
5. Stir the rest of the mix in warm water and let the sand settle.
6. Describe how each ingredient was removed in your notebook.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Each material keeps its identity; the magnet only pulls out the one that is attracted.

WHAT STUDENTS WILL SEE

Dark specks of iron stick to the bagged magnet. Salt dissolves in the water and sand sinks to the bottom, so all three can be separated.

Safety: Keep magnets away from phones and cards. Do not eat the cereal used in the lab.

Try next: Design a flowchart to separate a mix of iron filings, sand, salt and plastic beads.

DISSOLVING SPEED VS. TEMPERATURE

Rate of dissolving

Grades 6-12 | Medium | Prep 15 min | Class 25-40 min | MS-PS1-4 | MS-PS3-4 | Adult handling

WHY DOES IT HAPPEN?

Warm water molecules move faster than cold ones. They bump into the sugar more often and with more energy, so the sugar leaves the cube faster. Temperature is a measure of how fast particles move.

MATERIALS

- 3 clear cups
- 3 sugar cubes
- 1 cup (240 mL) cold water with ice
- 1 cup (240 mL) room-temperature water
- 1 cup (240 mL) hot tap water (about 120 F or 50 C)
- 1 thermometer
- 1 timer

6 STEPS

1. 
   Put on goggles. Pour each cup of water and measure its temperature.
2. 
   Write the temperatures in a table.
3. 
   Drop one sugar cube in the cold cup and start the timer.
4. 
   Stop the timer when no solid is left. Record the time.
5. 
   Repeat with the room-temperature cup and the hot cup.
6. 
   Graph time versus temperature.

THE EQUATION, SIMPLIFIED

$$C_6H_{12}O_6(s) \rightarrow C_6H_{12}O_6(aq)$$

Sugar molecules separate from the cube and spread through the water.

WHAT STUDENTS WILL SEE

The cube in the hot water vanishes in under a minute. The room-temperature cube takes several minutes, and the cold one takes the longest.

Safety: Adult handling of hot tap water. Do not taste the sugar water.

Try next: Try crushed sugar versus a whole cube at the same temperature.

SATURATED SOLUTIONS

Solubility limits

Grades 6-12 | Medium | Prep 10 min | Class 25-40 min | MS-PS1-2 | HS-PS1-3 | Adult handling

WHY DOES IT HAPPEN?

Only a certain amount of salt can dissolve in a given amount of water. When the water holds all it can, the solution is saturated and extra salt stays on the bottom. Counting spoonfuls shows the solubility limit.

MATERIALS

- 1 cup (240 mL) warm water
- about 1/2 cup (130 g) table salt
- 1 teaspoon measure
- 1 clear glass
- 1 spoon
- 1 pencil and notebook
- 1 sheet of dark paper

6 STEPS

1. 
   Put on goggles. Pour the warm water into the glass and set it on dark paper.
2. 
   Add one level teaspoon of salt and stir until it disappears.
3. 
   Keep adding one teaspoon at a time, counting each spoon.
4. 
   Stop when salt stays on the bottom after one minute of stirring.
5. 
   Record the number of spoons it took.
6. 
   Compare your number with other groups and explain differences.

THE EQUATION, SIMPLIFIED

$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

Salt keeps dissolving until the water cannot hold more ions.

WHAT STUDENTS WILL SEE

The first spoonfuls vanish quickly. Around 10 to 14 teaspoons, some crystals stay at the bottom no matter how much you stir.

Safety: Adult handling of warm water. Do not taste the solution.

Try next: Repeat with cold water and compare the spoon counts.

ROCK CANDY CRYSTALS

Crystallization from a supersaturated solution

Grades 6-12 | Challenge | Prep 20 min | Class 20-40 min | MS-PS1-4 | HS-PS1-3 | Adult handling

WHY DOES IT HAPPEN?

Hot water can hold much more sugar than cold water. As the solution cools, the extra sugar leaves the liquid and builds up on a string as crystals. Slow cooling makes larger crystals.

MATERIALS

- 3 cups (600 g) sugar
- 1 cup (240 mL) water
- 1 tall clean glass jar
- 1 wooden skewer or cotton string
- 1 clothespin
- extra sugar for coating
- 1 pot

6 STEPS

1. Put on goggles. An adult heats the water until hot and stirs in the sugar a little at a time until no more dissolves.
2. Wet the string and roll it in dry sugar. Let it dry.
3. Pour the warm syrup into the jar.
4. Hang the string in the jar with the clothespin so it does not touch the bottom.
5. Place the jar somewhere still and cool. Check it each day.
6. Record crystal size after 3, 5 and 7 days.

THE EQUATION, SIMPLIFIED

$$C_{12}H_{22}O_{11}(aq) \rightarrow C_{12}H_{22}O_{11}(s)$$

Sugar molecules leave the water and stack into crystals as the solution cools and evaporates.

WHAT STUDENTS WILL SEE

After one day, small crystals appear on the string. After a week they are large and clear. Food-safe only if made with clean food-grade tools and away from the lab area; follow school food policy.

Safety: Adult handling of the hot syrup. Do not touch the jar while the syrup is hot.

Try next: Make a second jar with food coloring and compare the crystals.

EPSOM SALT CRYSTALS

Crystal growth

Grades 6-12 | Medium | Prep 15 min | Class 20-30 min | MS-PS1-2 | HS-PS1-3 | Adult handling

WHY DOES IT HAPPEN?

Hot water dissolves more Epsom salt than cold water. As the liquid cools and some water evaporates, magnesium sulfate comes out and forms needle-like crystals. The shape of the crystals comes from how their particles stack.

MATERIALS

- 1/2 cup (120 mL) hot tap water
- 1/2 cup (130 g) Epsom salt
- 1 sheet black construction paper
- 1 shallow dish
- 1 spoon
- 1 hand lens
- 1 measuring cup

6 STEPS

1. 
Put on goggles. An adult pours the hot water into a cup.

2. 
Stir in the Epsom salt until it dissolves.

3. 
Cut the black paper to fit the dish and place it flat inside.

4. 
Pour the solution over the paper.

5. 
Place the dish in the refrigerator or a cool window for several hours.

6. 
Look at the crystals with the hand lens.

THE EQUATION, SIMPLIFIED

$$Mg^{2+}(aq) + SO_4^{2-}(aq) + 7\,H_2O(l) \rightarrow MgSO_4 \cdot 7H_2O(s)$$

Dissolved ions rejoin with water to form solid crystals as the liquid cools.

WHAT STUDENTS WILL SEE

After a few hours, white needle-like crystals appear on the dark paper. They continue to lengthen overnight.

Safety: Adult handling of hot water. Epsom salt is for lab use only; do not eat it.

Try next: Compare crystals grown in the refrigerator and at room temperature.

PHYSICAL VS. CHEMICAL CHANGE STATIONS

Evidence of chemical change

Grades 6-8 | Easy | Prep 15 min | Class 30-40 min | MS-PS1-2

WHY DOES IT HAPPEN?

In a physical change, the substance stays the same and only its form changes. In a chemical change, new substances form, and you may see gas, color change or temperature change. Stations help students sort changes by evidence.

MATERIALS

- 1 ice cube in a cup
- 1 sheet of paper to tear
- 1 tsp (4 g) sugar and a cup of water
- 1 tbsp (14 g) baking soda
- 1/4 cup (60 mL) white vinegar
- 4 trays
- 1 recording chart

6 STEPS

1. 
Put on goggles. At station 1, watch the ice cube sit in a cup.

2. 
At station 2, tear the paper into small bits.

3. 
At station 3, stir the sugar into water.

4. 
At station 4, put the baking soda in a cup and pour in the vinegar.

5. 
Record what you saw at each station.

6. 
Sort stations into physical or chemical and give evidence.

THE EQUATION, SIMPLIFIED

$$NaHCO_3 + CH_3COOH \rightarrow CH_3COONa + H_2O + CO_2$$

Only station 4 forms new substances; stations 1 to 3 are physical changes.

WHAT STUDENTS WILL SEE

Ice melts, paper tears and sugar dissolves with no new substance. The baking soda and vinegar fizz and bubble because carbon dioxide gas forms.

Safety: Wipe spills right away. Do not taste anything at the stations.

Try next: Add a new station and defend your classification.

YEAST BALLOON INFLATION

Fermentation and gas production

Grades 6-12 | Medium | Prep 10 min | Class 25-40 min | MS-LS1-7 | HS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Yeast are living cells that eat sugar. When they break it down without much oxygen, they make carbon dioxide gas. The gas is trapped by a balloon and shows that a chemical change is happening.

MATERIALS

- 1 empty 16 oz (500 mL) plastic bottle
- 1 packet (2 1/4 tsp, 7 g) dry yeast
- 2 tbsp (25 g) sugar
- 1/2 cup (120 mL) warm water
- 1 balloon
- 1 measuring tape
- 1 spoon

6 STEPS

1. 
Put on goggles. Pour the warm water into the bottle.

2. 
Add the sugar and swirl until dissolved.

3. 
Add the yeast and swirl gently.

4. 
Stretch the balloon over the neck of the bottle.

5. 
Set the bottle in a warm spot for 20 minutes.

6. 
Measure the balloon circumference every 5 minutes.

THE EQUATION, SIMPLIFIED

$$C_6H_{12}O_6 \rightarrow 2\,C_2H_5OH + 2\,CO_2$$

Yeast turns sugar into ethanol and carbon dioxide.

WHAT STUDENTS WILL SEE

A foamy layer forms in 5 to 10 minutes. By 20 minutes the balloon is partly inflated.

Safety: Adult handling of warm water. Do not drink the contents of the bottle.

Try next: Repeat without sugar and compare the balloons.

CONSERVATION OF MASS IN A BAG

Mass before and after a reaction

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | MS-PS1-5 | HS-PS1-7

WHY DOES IT HAPPEN?

Atoms are not created or destroyed in a chemical reaction. A sealed bag traps the gas, so the total mass stays the same before and after. If the bag is open, the gas escapes and the mass appears to drop.

MATERIALS

- 1 digital scale
- 1 sturdy zipper bag
- 1 fizzing antacid tablet
- 1/4 cup (60 mL) water
- 1 small cup
- paper towels
- notebook

6 STEPS

1. 
Put on goggles. Put the water in the cup and place the tablet in a corner of the bag.

2. 
Carefully place the cup upright in the bag without spilling.

3. 
Seal the bag and weigh it. Record the mass.

4. 
Tip the cup so the water meets the tablet.

5. 
Wait until the fizzing stops and weigh the sealed bag again.

6. 
Open the bag, weigh again, and explain the difference.

THE EQUATION, SIMPLIFIED

$$3\ NaHCO_3 + C_6H_8O_7 \rightarrow Na_3C_6H_5O_7 + 3\ H_2O + 3\ CO_2$$

The atoms rearrange; the same number of each atom is on both sides.

WHAT STUDENTS WILL SEE

The bag puffs up, but the sealed mass is the same before and after. After opening, the mass drops a little because carbon dioxide escapes.

Safety: Use a sturdy bag and stop if it swells tight. Never shake it or point the opening at faces.

Try next: Test with half a tablet and predict the gas mass.

FIZZING TABLET REACTION RATE

Surface area and temperature

Grades 9-12 | Medium | Prep 10 min | Class 25-40 min | HS-PS1-5 | Adult handling

WHY DOES IT HAPPEN?

Reactions speed up when particles collide more often and with more energy. Warm water and crushed tablets both increase collisions. Timing the fizzing shows how conditions change the rate.

MATERIALS

- 3 fizzing antacid tablets
- 1 cup (240 mL) cold water
- 1 cup (240 mL) room-temperature water
- 1 cup (240 mL) warm water (about 105 F or 40 C)
- 3 clear cups
- 1 timer
- 1 zipper bag

6 STEPS

1. 
Put on goggles. Pour each water into its own cup.

2. 
Drop a whole tablet in the cold water and start the timer.

3. 
Stop timing when it is gone. Record it.

4. 
Repeat in the room-temperature and warm cups.

5. 
Crush a tablet in the bag and drop it in room-temperature water.

6. 
Compare your four times in a bar graph.

THE EQUATION, SIMPLIFIED

$$3\ NaHCO_3 + C_6H_8O_7 \rightarrow Na_3C_6H_5O_7 + 3\ H_2O + 3\ CO_2$$

The same reaction occurs each time; only its speed changes.

WHAT STUDENTS WILL SEE

The tablet in warm water finishes first. The crushed tablet is much faster than the whole one in the same water.

Safety: Adult handling of warm water. Do not drink the solutions.

Try next: Try a tablet in carbonated water and explain the result.

CLEANING PENNIES

Oxides and acids

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2 | HS-PS1-2

WHY DOES IT HAPPEN?

Old pennies turn dull as copper reacts with oxygen to form a copper oxide coating. Vinegar has acetic acid that reacts with that coating and removes it. Salt helps the cleaning go faster.

MATERIALS

- 6 dull copper-colored pennies
- 1/4 cup (60 mL) white vinegar
- 1 tsp (5 g) table salt
- 2 small cups
- paper towels
- 1 spoon
- 1 timer

6 STEPS

1. 
Put on goggles. Pour the vinegar into a cup and stir in the salt.

2. 
Place three pennies in the cup. Leave the other three dry as a control.

3. 
Wait 5 minutes.

4. 
Remove the pennies and compare them with the control.

5. 
Rinse half of the cleaned pennies and leave the rest unrinsed on a towel.

6. 
Check the pennies again after an hour.

THE EQUATION, SIMPLIFIED

$$CuO + 2\,CH_3COOH \rightarrow Cu(CH_3COO)_2 + H_2O$$

Copper oxide on the surface reacts with acetic acid and dissolves.

WHAT STUDENTS WILL SEE

The pennies in vinegar turn shiny in a few minutes. The unrinsed pennies may develop a green-blue film as the copper compound dries.

Safety: Wash hands after handling pennies. Do not eat or drink the liquid.

Try next: Clean pennies with lemon juice and compare.

MAGIC MILK

Surface tension and soap

Grades 6-8 | Easy | Prep 5 min | Class 15-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Milk has fat and protein that hold the surface tight. Dish soap breaks up the surface tension and pushes the fat around. The food coloring moves with it, showing swirling patterns.

MATERIALS

- 1 shallow plate
- 1/2 cup (120 mL) whole milk
- 4 drops food coloring (different colors)
- 1 cotton swab
- 1 tsp (5 mL) dish soap
- 1 small cup
- 1 paper towel

6 STEPS

1. 
   Put on goggles. Pour the milk to cover the plate bottom.
2. 
   Add one drop of each color near the center.
3. 
   Put dish soap on a clean cotton swab.
4. 
   Touch the swab to the center of the milk and hold still.
5. 
   Watch the colors for 30 seconds.
6. 
   Sketch what you saw and try another spot.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Soap lowers the surface tension; nothing new forms.

WHAT STUDENTS WILL SEE

The colors burst and swirl for a minute or more. Eventually, the motion stops when the soap spreads through the milk.

Safety: Do not drink the milk used in the lab. Wash the plate with warm soapy water.

Try next: Repeat with skim milk and cream and compare.

DIFFUSION IN HOT VS. COLD WATER

Particle motion and temperature

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-4 | MS-PS3-4 | Adult handling

WHY DOES IT HAPPEN?

All particles are always moving, and they move faster when warmer. Food coloring spreads from places of high concentration to low concentration. The spread is quicker in hot water because the molecules bump harder.

MATERIALS

- 2 clear glasses
- 1 cup (240 mL) cold water
- 1 cup (240 mL) hot tap water
- 4 drops food coloring
- 1 thermometer
- 1 timer
- 1 sheet of white paper

6 STEPS

1. 
   Put on goggles. Pour cold water in one glass and hot tap water in the other.
2. 
   Measure and write down each temperature.
3. 
   Place both glasses on white paper.
4. 
   Add two drops of food coloring to each glass at the same time.
5. 
   Time how long until the color spreads fully.
6. 
   Explain the difference using particle motion.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Dye particles spread by random motion; no new substance forms.

WHAT STUDENTS WILL SEE

The color spreads through the hot glass in under a minute. In the cold glass it stays in streaks for several minutes.

Safety: Adult handling of hot water. Do not drink the colored water.

Try next: Try a drop of oil or a drop of milk and compare.

WALKING WATER

Capillary action

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-PS1-1

WHY DOES IT HAPPEN?

Water molecules stick to each other and to paper fibers. This lets water climb up a paper towel against gravity. It carries food coloring from one cup to another.

MATERIALS

- 3 clear cups
- 2 cups (480 mL) water
- red and yellow food coloring
- 3 paper towel strips
- 1 tray
- 1 ruler
- 1 timer

6 STEPS

1. 
Put on goggles. Line up the three cups on the tray.

2. 
Fill the outer two cups with water, one red and one yellow.

3. 
Leave the middle cup empty.

4. 
Fold the paper towel strips and place them between cups.

5. 
Watch for 20 minutes.

6. 
Measure how high the water climbed.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Water only moves along the paper; colors mix to make a new shade but no new substance forms.

WHAT STUDENTS WILL SEE

Colored water climbs the towels and gathers in the middle cup. Red and yellow mix to make orange.

Safety: Wipe spills right away. Do not drink any liquid.

Try next: Try different paper types and compare the height.

INVISIBLE INK REVEAL

Indicators and baking soda

Grades 6-12 | Medium | Prep 15 min | Class 25-40 min | MS-PS1-2 | HS-PS1-2

WHY DOES IT HAPPEN?

Baking soda in water is a weak base. Grape juice contains natural pigments that change color with acidity. Where baking soda is painted, the juice turns blue-green, making the hidden message appear.

MATERIALS

- 1 tbsp (14 g) baking soda
- 1 tbsp (15 mL) warm water
- 1 cotton swab
- 1 sheet of white paper
- 1/2 cup (120 mL) purple grape juice
- 1 paintbrush
- 1 small cup

6 STEPS

1. 
Put on goggles. Stir the baking soda into the water until it dissolves.

2. 
Write a message on the paper using the cotton swab.

3. 
Let the paper dry for 10 minutes.

4. 
Pour grape juice into the cup.

5. 
Brush juice lightly over the paper.

6. 
Describe the message and explain the color change.

THE EQUATION, SIMPLIFIED

$HAn + OH^- \rightarrow An^- + H_2O$

The pigment (HAn) loses a hydrogen ion in base and changes color.

WHAT STUDENTS WILL SEE

The message appears blue-green on a purple background as the juice reaches the baking soda.

Safety: Do not taste. Juice stains clothing; wear an apron.

Try next: Test lemon juice as a different message ink.

FLOATING EGG

Density and salt water

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2 | HS-PS1-3

WHY DOES IT HAPPEN?

An egg is denser than plain water, so it sinks. Dissolving salt makes the water denser than the egg. When the water is dense enough, the egg floats.

MATERIALS

- 1 raw egg
- 1 tall clear glass
- 1 cup (240 mL) water
- 6 tbsp (110 g) table salt
- 1 spoon
- 1 tray
- paper towels

6 STEPS

1. 
   Put on goggles. Fill the glass about two thirds with plain water.
2. 
   Gently lower the egg and note that it sinks.
3. 
   Take the egg out and stir in the salt.
4. 
   Wait until the salt is fully dissolved.
5. 
   Lower the egg again.
6. 
   Add plain water slowly on top and watch what happens.

THE EQUATION, SIMPLIFIED

$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

Dissolved salt raises the density of the water.

WHAT STUDENTS WILL SEE

The egg sinks in plain water and floats in the salty water. Adding plain water on top makes the egg hover in the middle.

Safety: Wash hands after handling raw egg. Do not eat the egg.

Try next: Find the least salt that floats the egg.

SUGAR WATER DENSITY RAINBOW

Concentration and density

Grades 9-12 | Challenge | Prep 20 min | Class 30-40 min | MS-PS1-2 | HS-PS1-3

WHY DOES IT HAPPEN?

Adding sugar to water raises its density. Four solutions with different amounts of sugar can be stacked in order. The more sugar in a solution, the lower in the glass it sits.

MATERIALS

- 4 small cups with 1/4 cup (60 mL) water each
- 0, 1, 2 and 3 tbsp (0, 12, 25, 38 g) sugar
- 4 colors of food coloring
- 1 tall clear glass
- 1 spoon
- 1 measuring spoon
- 1 dropper or straw

6 STEPS

1. 
Put on goggles. Add the different sugar amounts to separate cups of water.

2. 
Stir until each dissolves.

3. 
Color each cup differently.

4. 
Pour the most sugary solution into the glass first.

5. 
Slowly pour the others over the back of a spoon in order.

6. 
Draw and label each layer.

THE EQUATION, SIMPLIFIED

$$C_6H_{12}O_6(s) \rightarrow C_6H_{12}O_6(aq)$$

Sugar dissolves and increases the mass per volume of the solution.

WHAT STUDENTS WILL SEE

Four colored layers sit on top of each other for hours. They slowly blur as sugar spreads between layers.

Safety: Do not drink. Wipe spills on the floor right away.

Try next: Predict where a grape will float.

RUSTING STEEL WOOL

Oxidation of iron

Grades 9-12 | Medium | Prep 10 min | Class 20-40 min | HS-PS1-2 | HS-PS1-4

WHY DOES IT HAPPEN?

Iron reacts with oxygen and water to form rust. Vinegar removes the protective coating on steel wool, so the reaction speeds up. The reaction releases a small amount of heat.

MATERIALS

- 1 pad of plain steel wool (no soap)
- 1/2 cup (120 mL) white vinegar
- 1 thermometer
- 1 clear jar with lid
- 1 bowl
- 1 pair of gloves
- 1 timer

6 STEPS

1. 
Put on goggles and gloves. Soak the steel wool in vinegar for 1 minute.

2. 
Squeeze out the extra vinegar.

3. 
Wrap the steel wool around the thermometer bulb.

4. 
Place it in the jar and cover with a lid.

5. 
Record the temperature every 3 minutes.

6. 
Look at the steel wool after 24 hours.

THE EQUATION, SIMPLIFIED

$$4\,Fe + 3\,O_2 \rightarrow 2\,Fe_2O_3$$

Iron combines with oxygen to form iron oxide (rust).

WHAT STUDENTS WILL SEE

The temperature rises 2 to 5 degrees F over 15 minutes. The wool turns orange-brown after a day.

Safety: Steel wool can splinter; wear gloves. Wash hands afterward.

Try next: Compare dry steel wool with salty steel wool.

COLD FIZZ: AN ENDOTHERMIC REACTION

Energy absorbed in a reaction

Grades 9-12 | Challenge | Prep 10 min | Class 20-30 min | HS-PS1-4 | MS-PS1-6

WHY DOES IT HAPPEN?

In some reactions, energy flows in from the surroundings. This makes the mixture cooler. Baking soda and citric acid in water absorb heat as they react.

MATERIALS

- 1/2 cup (120 mL) water
- 2 tsp (9 g) baking soda
- 2 tsp (8 g) citric acid
- 1 thermometer
- 1 foam cup
- 1 spoon
- 1 timer

6 STEPS

1. 
Put on goggles. Pour water into the foam cup.

2. 
Measure the starting temperature.

3. 
Add the baking soda and stir.

4. 
Add the citric acid and stir gently.

5. 
Record the temperature every 30 seconds.

6. 
Graph temperature versus time.

THE EQUATION, SIMPLIFIED

$$3\ NaHCO_3 + C_6H_8O_7 \rightarrow Na_3C_6H_5O_7 + 3\ H_2O + 3\ CO_2$$

The reaction takes in heat, so the solution cools.

WHAT STUDENTS WILL SEE

The mixture fizzes and the temperature drops by 5 to 10 degrees F in about two minutes.

Safety: Do not drink. Citric acid can irritate eyes; keep goggles on.

Try next: Double the amounts and see if the drop doubles.

SALT AND ICE: LOWERING THE FREEZING POINT

Freezing point depression

Grades 9-12 | Medium | Prep 10 min | Class 25-40 min | MS-PS3-4 | HS-PS3-4

WHY DOES IT HAPPEN?

Salt breaks into ions that get between water molecules. This makes it harder for water to freeze, so the ice melts at a lower temperature. The mixture gets colder than ice alone.

MATERIALS

- 2 cups of crushed ice
- 3 tbsp (55 g) table salt
- 2 foam cups
- 1 thermometer
- 1 spoon
- 1 timer
- 1 notebook

6 STEPS

1. Put on goggles. Fill each foam cup with 1 cup of crushed ice.
2. Measure the temperature of the ice in both.
3. Add salt to one cup and stir.
4. Record temperatures every minute.
5. Observe the liquid in each cup.
6. Graph both lines and compare.

THE EQUATION, SIMPLIFIED

$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

Dissolved ions lower the freezing point of water.

WHAT STUDENTS WILL SEE

The salted ice drops below 32 F (0 C), down to about 15 F (-9 C), and melts more. The plain ice stays near 32 F.

Safety: Cold surfaces can irritate skin. Do not touch the salted ice for long.

Try next: Try sugar instead of salt.

CATALYST RACE

Enzyme catalysts

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | HS-PS1-5 | MS-LS1-7 | Adult handling

WHY DOES IT HAPPEN?

Catalase is an enzyme in living cells that breaks hydrogen peroxide into water and oxygen. Different foods have different amounts. The foam shows which is the faster catalyst.

MATERIALS

- 1/2 cup (120 mL) 3% hydrogen peroxide
- 1 tbsp each of potato, carrot and apple pieces
- 4 clear cups
- 1 tsp (5 mL) dish soap
- 1 timer
- 1 ruler
- 1 tray

6 STEPS

1. 
   Put on goggles. An adult pours 2 tbsp of peroxide into each cup.
2. 
   Add a drop of dish soap to each.
3. 
   Drop the potato in the first cup, carrot in second, apple in third. Keep the fourth plain.
4. 
   Start the timer.
5. 
   Measure the foam height every 30 seconds.
6. 
   Rank the catalysts.

THE EQUATION, SIMPLIFIED

$$2\,H_2O_2 \rightarrow 2\,H_2O + O_2$$

Enzymes in food speed up the breakdown of peroxide.

WHAT STUDENTS WILL SEE

The potato makes the most foam within a minute. Carrot makes a little, apple very little, and the plain cup almost none.

Safety: Adult handling of peroxide. Never use stronger than 3%.

Try next: Boil the potato first and test again.

EVAPORATION AND SURFACE AREA

Evaporation rate

Grades 6-8 | Easy | Prep 10 min | Class 25-40 min | MS-PS1-4 | MS-ESS2-4

WHY DOES IT HAPPEN?

Evaporation happens when water molecules at the surface gain enough energy to escape as vapor. A bigger surface lets more molecules escape. A wide dish dries faster than a narrow glass.

MATERIALS

- 1/4 cup (60 mL) water
- 1 wide plate
- 1 narrow glass
- 1 measuring cup
- 1 marker
- 1 ruler
- 1 warm windowsill

6 STEPS

1. 
   Put on goggles. Measure 1/8 cup (30 mL) of water into the plate and the same into the glass.
2. 
   Mark the water levels.
3. 
   Place both in a warm spot.
4. 
   Check every 30 minutes.
5. 
   Measure the water left.
6. 
   Write a conclusion.

THE EQUATION, SIMPLIFIED

$H_2O(l) \rightarrow H_2O(g)$

Liquid water turns into water vapor.

WHAT STUDENTS WILL SEE

The plate loses water much faster than the glass. After a few hours the plate is nearly dry.

Safety: Wipe spills right away to avoid slips.

Try next: Test with a fan.

DROPS ON A PENNY

Cohesion and surface tension

Grades 6-8 | Easy | Prep 5 min | Class 15-30 min | MS-PS1-1

WHY DOES IT HAPPEN?

Water molecules attract each other strongly. This makes a drop pile up above the edge of a penny. Soap weakens that attraction, so fewer drops fit.

MATERIALS

- 1 penny
- 1 dropper or straw
- 1/4 cup (60 mL) water
- 1/4 cup (60 mL) soapy water
- 1 paper towel
- 1 tray
- 1 notebook

6 STEPS

1. 
   Put on goggles. Place a penny on the tray.
2. 
   Fill the dropper with water.
3. 
   Add drops one at a time, counting.
4. 
   Stop when it spills.
5. 
   Repeat with soapy water.
6. 
   Compare counts.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Surface tension depends on forces between molecules; no new substance forms.

WHAT STUDENTS WILL SEE

Plain water can hold 15 to 30 drops, forming a dome. Soapy water holds far fewer.

Safety: Wipe spills right away.

Try next: Compare heads and tails of the penny.

CANDLE IN A JAR

Combustion needs oxygen

Grades 9-12 | Challenge | Prep 15 min | Class 25-40 min | HS-PS1-2 | HS-PS1-7 | Adult handling

WHY DOES IT HAPPEN?

Burning wax needs oxygen. A jar traps a limited amount of oxygen, so the flame goes out when it is used up. A larger jar holds more oxygen and lets the candle burn longer.

MATERIALS

- 1 tea light candle
- 3 jars of different sizes
- 1 lighter (adult only)
- 1 timer
- 1 tray
- 1 pair of tongs
- 1 notebook

6 STEPS

1. Put on goggles. An adult places the candle on the tray and lights it.
2. Place the smallest jar over the candle.
3. Time how long until the flame goes out.
4. Let the jar cool, then repeat with the medium jar.
5. Repeat with the biggest jar.
6. Graph jar size versus time.

THE EQUATION, SIMPLIFIED

$$C_{25}H_{52} + 38\ O_2 \rightarrow 25\ CO_2 + 26\ H_2O$$

Candle wax burns with oxygen to form carbon dioxide and water.

WHAT STUDENTS WILL SEE

The flame shrinks and goes out in 5 to 25 seconds. Larger jars give longer times. Some fog may show on the glass.

Safety: Adult handling of the flame. Keep hair away and use tongs for hot jars.

Try next: Predict the time for a fourth jar.

STRAWBERRY DNA EXTRACTION

DNA released by soap and salt, clumped by cold alcohol

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-2 | HS-LS1-6 | Adult handling

WHY DOES IT HAPPEN?

Every cell holds DNA, a very long molecule of carbon, hydrogen, oxygen, nitrogen and phosphorus. Dish soap breaks open the fatty cell membranes, and salt helps the DNA strands clump together. DNA does not dissolve in cold alcohol, so it gathers as white strands at the boundary between the two liquids. Strawberries work well because they have many copies of each gene.

MATERIALS

- 4 ripe strawberries (about 3.5 oz / 100 g), fresh or thawed
- 1 quart zip-top bag
- 1 tsp (5 mL) dish soap
- 1/2 tsp (3 g) table salt
- 1/2 cup (120 mL) cold water
- 1/2 cup (120 mL) rubbing alcohol (91%+ isopropyl), chilled in a freezer for 1 hour
- 1 coffee filter and 1 clear cup
- 1 toothpick

6 STEPS

1. 
Remove the strawberry tops and place the berries in the zip-top bag. Seal it and press out the extra air.

2. 
Mash the berries with your fingers for 2 minutes until they are a smooth pulp.

3. 
In a cup, stir the dish soap, salt and cold water gently (avoid foam). Pour it into the bag and mash for 1 more minute.

4. 
Place the coffee filter over a clear cup and pour the pulp through it. Let it drip for 3 minutes.

5. 
Your teacher tilts the cup and slowly pours the cold alcohol down the side so it floats in a layer on top (about 1 inch / 2.5 cm).

6. 
Watch the boundary for 2-5 minutes, then twirl the toothpick at the boundary to spool out the white strands.

THE EQUATION, SIMPLIFIED

DNA(aq) in salt water → DNA(s) clumps in alcohol (no new substance forms: physical change)

The DNA molecules do not react; alcohol simply makes them leave the water and clump together.

WHAT STUDENTS WILL SEE

Within 1-2 minutes cloudy white, stringy, mucus-like strands rise into the alcohol layer. They stick to the toothpick like thin cotton candy.

Safety: Wear splash goggles. 91% isopropyl alcohol is flammable and the vapor is irritating: an adult pours it, with ventilation and no flames or heat sources nearby. Do not taste anything from this experiment.

Try next: Repeat with a banana, a kiwi or split peas and compare how much DNA you collect.

PINEAPPLE VS GELATIN: ENZYMES AT WORK

Protease enzyme (bromelain) cutting protein

Grades 6-12 | Easy | Prep 15 min | Class 20-30 min | MS-LS1-7 | MS-PS1-2

WHY DOES IT HAPPEN?

Gelatin is a protein that forms a net of long chains that traps water. Fresh pineapple and kiwi contain protease enzymes that cut protein chains into small pieces, so the net falls apart. Canned pineapple was heated during canning, which unfolds and disables the enzyme, so it does not work.

MATERIALS

- 4 clear cups of set plain gelatin, prepared the day before by an adult (1 box / 3 oz / 85 g mix makes enough)
- 2 tbsp (about 30 g) fresh pineapple pieces
- 2 tbsp (about 30 g) canned pineapple pieces
- 2 tbsp (about 30 g) fresh kiwi pieces
- 4 spoons or toothpicks
- Marker and tape for labels
- Paper towels

6 STEPS

1. Label the four gelatin cups: Fresh Pineapple, Canned Pineapple, Kiwi, Control (nothing added).
2. Place 2 tbsp of each fruit on the matching cup. The control gets nothing.
3. Set a timer for 30 minutes and leave the cups at room temperature.
4. Check the gelatin under each fruit every 10 minutes and write what you see.
5. After 30 minutes tip each cup gently and compare how firmly each gelatin holds together.
6. Record which fruit liquefied the gelatin and which did not, and explain why.

THE EQUATION, SIMPLIFIED

Protein + n H_2O → smaller peptides (protease enzyme speeds this up)

The enzyme is a catalyst: it is not used up, and it cuts protein chains by adding water at each cut.

WHAT STUDENTS WILL SEE

Under fresh pineapple and kiwi, the gelatin turns runny within 20-30 minutes. The canned pineapple and control cups stay firm.

Safety: Wear splash goggles. Do not eat the gelatin or fruit used in class; check for fruit allergies first and wash hands afterward.

Try next: Test papaya or fresh ginger root and rank all fruits from strongest to weakest.

HEAT SWITCHES OFF AN ENZYME

Enzyme denaturation by temperature

Grades 9-12 | Medium | Prep 15 min | Class 30-40 min | HS-LS1-6 | HS-PS1-5 | Adult handling

WHY DOES IT HAPPEN?

Enzymes are proteins that must keep a precise folded shape to work. Heat makes the protein unfold, a change called denaturation, and the shape of the active site is lost. Here, fresh pineapple juice is split into raw and heated samples, and each is tested on gelatin to see which still works.

MATERIALS

- 1/2 cup (120 mL) fresh pineapple juice (squeezed from fresh fruit, strained)
- 2 small heat-safe cups
- Hot water bath at about 195°F (90°C), adult handled
- 3 clear cups of set plain gelatin, prepared the day before
- 3 spoons
- Bowl of ice water
- Marker, tape and timer

6 STEPS

1. 
Divide the fresh juice equally into two cups. Label one Raw and one Heated.

2. 
An adult heats the Heated cup in the hot water bath for 5 minutes, then removes it.

3. 
Cool the heated juice in a bowl of ice water for 5 minutes until it is at room temperature.

4. 
Spoon 2 tbsp (30 mL) of raw juice on one gelatin cup and 2 tbsp of heated juice on another. Add plain water on a third as a control.

5. 
Wait 30 minutes at room temperature.

6. 
Compare how much gelatin has dissolved in each cup and record the data in a table.

THE EQUATION, SIMPLIFIED

Active folded enzyme → unfolded enzyme (no new substance forms: denaturation)

Heat breaks the weak bonds that hold the enzyme in its working shape; the atoms stay the same.

WHAT STUDENTS WILL SEE

The raw juice dissolves the gelatin into a runny layer in 20-30 minutes. The heated juice leaves the gelatin almost as firm as the water control.

Safety: Wear splash goggles. An adult handles all hot water and hot juice. Do not taste the juice or gelatin.

Try next: Test a cooler temperature such as 120°F (50°C) to find where the enzyme starts to fail.

STARCH DISAPPEARS: AMYLASE AND IODINE

Enzyme digestion of starch, iodine test

Grades 9-12 | Challenge | Prep 20 min | Class 30-40 min | HS-LS1-6 | MS-LS1-7 | Adult handling

WHY DOES IT HAPPEN?

Starch is a long chain of glucose units. Iodine turns blue-black when it sticks to starch, so it works as a detector. Malt powder contains amylase, an enzyme that chops starch into small sugars that iodine cannot color. When the blue-black color fades, the starch is gone.

MATERIALS

- 1 tsp (5 g) cornstarch dissolved in 1 cup (240 mL) warm water, cooled (adult heat)
- 1 tsp (3 g) diastatic malt powder (baking aisle) stirred into 1/4 cup (60 mL) warm water
- Iodine tincture 2% from a pharmacy, diluted 1 part to 10 parts water (adult prepares)
- 5 clear cups
- Dropper
- Marker and tape
- Timer

6 STEPS

1. Label cups: Start, 5 min, 10 min, 20 min, Control.
2. Pour 1/4 cup (60 mL) of starch solution into each of the four timed cups; the control gets plain water.
3. Test the Start cup: an adult adds 3 drops of dilute iodine. Record the color.
4. Stir 1 tbsp (15 mL) of malt solution into the other three starch cups at the same moment and start the timer.
5. At 5, 10 and 20 minutes the adult adds 3 drops of dilute iodine to the matching cup.
6. Record the colors and describe how the amount of starch changed over time.

THE EQUATION, SIMPLIFIED

$(C_6H_{10}O_5)n + n/2\ H_2O \rightarrow n/2\ C_{12}H_{22}O_{11}$ (starch → maltose)

Amylase splits long starch chains into two-glucose sugar units by adding water.

WHAT STUDENTS WILL SEE

The Start cup turns deep blue-black. The 5-minute cup is dark purple, the 10-minute cup is lighter, and the 20-minute cup stays pale yellow-brown.

Safety: Wear splash goggles and gloves. Iodine stains skin and clothing and is poisonous if swallowed: an adult handles the iodine and students never touch it. Do not taste anything from this experiment.

Try next: Repeat with a pinch of crushed cracker in water to see if the starch in food also disappears.

YEAST BALLOON INFLATOR

Fermentation: yeast eats sugar and releases CO2

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-LS1-7 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Yeast is a living fungus. When it has sugar and warmth, it breaks the sugar down for energy and releases carbon dioxide gas and alcohol as waste. The gas gets trapped in the balloon and slowly inflates it.

MATERIALS

- 1 empty 16 oz (500 mL) plastic bottle
- 1 packet (2 1/4 tsp / 7 g) dry yeast
- 2 tbsp (25 g) sugar
- 1/2 cup (120 mL) warm water, about 105°F (40°C)
- 1 balloon
- Measuring tape
- Marker and tape

6 STEPS

1. 
Measure 2 tbsp of sugar and 1 packet of yeast and pour both into the bottle.

2. 
An adult checks the warm water is not hot, and then pours it into the bottle.

3. 
Swirl the bottle gently for 10 seconds to mix the yeast and sugar.

4. 
Stretch the balloon over the bottle mouth so it sits snug and upright.

5. 
Place the bottle in a warm spot and wait. Check every 10 minutes for 40 minutes.

6. 
Measure around the balloon at its widest part at each check and record the data.

THE EQUATION, SIMPLIFIED

$C_6H_{12}O_6 \rightarrow 2\,C_2H_5OH + 2\,CO_2$ (glucose → ethanol + carbon dioxide)

One sugar molecule splits into two alcohol molecules and two carbon dioxide molecules.

WHAT STUDENTS WILL SEE

Foam and small bubbles form in about 5-10 minutes. By 30-40 minutes the balloon stands upright or leans over as it fills with gas.

Safety: Wear splash goggles. An adult checks that the water is warm, not hot. Do not drink the liquid, and keep the room ventilated.

Try next: Place a second setup in a cold spot and compare the balloon size after 40 minutes.

WHAT DOES YEAST LIKE TO EAT?

Fermentation with different foods

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-LS1-7 | HS-LS1-7 | Adult handling

WHY DOES IT HAPPEN?

Yeast can only ferment sugars it can break down. Table sugar and flour starch can both feed it (starch is cut into sugar first, more slowly), but a zero-calorie sweetener cannot be used for energy. By comparing balloon sizes, students see which foods the yeast can use.

MATERIALS

- 4 empty 12 oz (350 mL) bottles
- 4 balloons
- 2 tsp (6 g) dry yeast, split among bottles
- 1 tbsp (12 g) sugar
- 1 tbsp (8 g) flour
- 1 tbsp (about 1 g) powdered zero-calorie sweetener
- 1 cup (240 mL) warm water, about 105°F (40°C)
- Marker and tape

6 STEPS

1. 
Label the bottles: Sugar, Flour, Sweetener and Nothing.

2. 
Add 1/2 tsp of yeast to each bottle, then add the matching food to bottles 1-3. Bottle 4 gets none.

3. 
An adult pours 1/4 cup (60 mL) warm water into each bottle.

4. 
Swirl each bottle for 10 seconds, then stretch a balloon over each mouth.

5. 
Set the bottles in a warm place for 40 minutes.

6. 
Compare the balloons every 10 minutes and sketch the results in a bar chart.

THE EQUATION, SIMPLIFIED

$C_6H_{12}O_6 \rightarrow 2\ C_2H_5OH + 2\ CO_2$ (sugars feed this reaction; the sweetener does not)

Only a sugar that yeast can break down supplies the glucose needed for the reaction.

WHAT STUDENTS WILL SEE

The sugar bottle inflates the most in 20-30 minutes. The flour balloon inflates slowly and less. The sweetener and nothing balloons barely change.

Safety: Wear splash goggles. An adult handles warm water. Do not drink or taste anything.

Try next: Test honey or fruit juice and rank the foods by balloon size.

CURDS AND WHEY: CHEESE CHEMISTRY

Protein coagulation by acid

Grades 6-8 | Easy | Prep 15 min | Class 30-40 min | MS-PS1-2 | MS-LS1-7 | Adult handling

WHY DOES IT HAPPEN?

Milk is a mixture of water, fat and proteins called casein that float as tiny balls. Acid from vinegar changes the charges on the casein so the balls stick together into solid curds. The yellowish liquid left over is whey.

MATERIALS

- 2 cups (480 mL) whole milk
- 3 tbsp (45 mL) white vinegar
- Small pot and a stove or hot plate (adult use only)
- Strainer or coffee filter
- Bowl
- Spoon
- Thermometer

6 STEPS

1. Measure 2 cups of milk into the pot.
2. An adult warms the milk to about 150°F (65°C) on low heat, stirring slowly. Do not let it boil.
3. Remove the pot from the heat. An adult adds 3 tbsp of vinegar.
4. Stir slowly for 10 seconds and then wait 5 minutes without stirring.
5. Pour the mixture through the strainer into the bowl to separate the curds from the whey.
6. Press the curds gently, then compare the solid curds with the liquid whey.

THE EQUATION, SIMPLIFIED

Casein(aq) + H+ (acid) → casein(s) curds (protein clumps; physical change in shape)

The acid does not make a new substance; it makes the milk proteins lose their charge and clump together.

WHAT STUDENTS WILL SEE

As soon as the vinegar goes in, white lumps form and the milk separates into soft white curds and a thin yellowish liquid within a minute.

Safety: Wear splash goggles. An adult handles the hot milk and the heat source. Do not taste anything from this experiment.

Try next: Test lemon juice instead of vinegar and compare how many curds form.

PLASTIC FROM MILK

Casein polymer made from milk and acid

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-3 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Casein in milk is a natural polymer, a long chain of repeating units. Acid makes casein clump into curds, and pressing out the water leaves a moldable material. When dried, the curds harden into a plastic-like solid. This is how some of the first plastics were made.

MATERIALS

- 1 cup (240 mL) whole milk
- 4 tsp (20 mL) white vinegar
- Microwave-safe bowl or small pot (adult heat)
- Strainer
- Cookie cutter or small mold
- Paper towels
- Wax paper
- Food coloring (optional)

6 STEPS

1. 
Measure 1 cup of milk and 4 tsp of vinegar.

2. 
An adult heats the milk until steaming but not boiling, about 150°F (65°C).

3. 
An adult adds the vinegar. Stir for 1 minute until clumps form.

4. 
Pour through the strainer and squeeze the curds in paper towels to remove as much liquid as you can.

5. 
Knead the warm curd into a ball (add a drop of coloring if wanted), then press it into the mold.

6. 
Let the shape dry on wax paper for 48 hours, flipping it once a day.

THE EQUATION, SIMPLIFIED

Casein(aq) + H+ → casein(s); water leaves as it dries (no new substance forms)

The acid makes protein chains clump into a solid, and drying hardens them into a plastic-like object.

WHAT STUDENTS WILL SEE

A soft, pale dough forms in minutes. After 2 days the shape becomes hard and a bit brittle, like a chalky plastic.

Safety: Wear splash goggles. An adult handles hot milk. Keep the shaped objects out of the food area, never eat them, and throw them away if they smell sour.

Try next: Test whether skim or whole milk produces a stronger plastic.

POLYMER SLIME

Cross-linked polymer network (PVA and borate)

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-3 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

White school glue is made of long, loose chains of a polymer called PVA, which slide past each other like cooked spaghetti. A little borate from liquid starch connects the chains at many points, making a stretchy network that behaves like both a solid and a liquid.

MATERIALS

- 1/4 cup (60 mL) washable white school glue
- 1/4 cup (60 mL) water
- 1/4 cup (60 mL) liquid laundry starch (adult dispenses)
- 2 drops food coloring
- Mixing bowl and spoon
- Disposable gloves
- Zip-top bag for storage

6 STEPS

1. 
Put on gloves. Pour 1/4 cup of glue into the bowl.

2. 
Stir in 1/4 cup of water until smooth, then add 2 drops of food coloring.

3. 
An adult slowly pours in the liquid starch while you stir; stop when a lump forms.

4. 
Keep stirring until the mixture pulls away from the bowl and the sides.

5. 
Knead the slime in your gloved hands for 2 minutes until it is less sticky.

6. 
Stretch it slowly, then snap it quickly. Compare the two results.

THE EQUATION, SIMPLIFIED

Many PVA chains + borate cross-links → one stretchy network (cross-linking)

Weak links form between chains, so the chains stay connected but can still move.

WHAT STUDENTS WILL SEE

Within about 30 seconds the sticky liquid thickens into a rubbery lump. Slow pulling stretches it like taffy; a quick pull snaps it.

Safety: Wear splash goggles and gloves. Liquid starch contains borate: an adult dispenses it. Do not eat the slime and wash hands after. Skip this lab for students with skin allergies.

Try next: Change the starch amount and rank the slimes from runniest to stiffest.

OOBLECK: LIQUID OR SOLID?

Non-Newtonian fluid (suspension)

Grades 6-8 | Easy | Prep 5 min | Class 20-30 min | MS-PS1-1 | MS-PS1-2

WHY DOES IT HAPPEN?

Oobleck is a suspension: tiny cornstarch grains float in water. When you press slowly, the grains slide past each other and it flows. When you hit it fast, the grains jam together and it acts like a solid. No new substance forms.

MATERIALS

- 1 cup (130 g) cornstarch
- 1/2 cup (120 mL) water
- Mixing bowl
- Spoon
- Food coloring (optional)
- Tray or baking sheet
- Paper towels

6 STEPS

1. 
Set the bowl on a tray to catch spills.

2. 
Pour the cornstarch into the bowl.

3. 
Add the water a little at a time and add a drop of coloring if wanted.

4. 
Mix slowly with a spoon, then with your hand, until it feels like thick honey.

5. 
Squeeze a handful hard, then open your hand and watch what happens.

6. 
Tap the surface fast, then push your finger in slowly, and compare how it feels.

THE EQUATION, SIMPLIFIED

Cornstarch(s) + water(l) → suspension (no new substance forms: physical mixture)

The grains and water stay separate substances; the grains only change how they pack together.

WHAT STUDENTS WILL SEE

The mixture feels solid when squeezed or struck, but melts into a puddle within seconds when you stop pressing.

Safety: Wear splash goggles. Do not pour oobleck down the drain; let it dry and put it in the trash. Do not taste anything from this experiment.

Try next: Place a small speaker under a plate of oobleck and watch it dance.

MELT-AND-POUR SOAP

Soap from glycerin base, melting and solidifying

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-4 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Soap molecules have a water-loving head and an oil-loving tail. A premade glycerin soap base is solid because its molecules are held in fixed positions. Heat gives the molecules energy so they slide past each other, and cooling takes it away again.

MATERIALS

- 8 oz (225 g) glycerin melt-and-pour soap base, cut in cubes
- Microwave-safe glass measuring cup (adult handled)
- 1 drop soap-safe color or food coloring
- Silicone mold or paper cup
- Spoon
- Tray
- Paper towels

6 STEPS

1. 
An adult cuts the soap base into 1 inch (2.5 cm) cubes and puts them in the cup.

2. 
An adult microwaves it in 20-second bursts, stirring between bursts, until fully liquid.

3. 
Add one drop of color and stir slowly to avoid bubbles.

4. 
An adult pours the liquid soap into the mold.

5. 
Let the soap set for 60 minutes without moving.

6. 
Remove the soap from the mold and compare its look and feel to the starting cubes.

THE EQUATION, SIMPLIFIED

Soap(s) → soap(l) → soap(s) (no new substance forms: melting and freezing)

Heat only changes the state of the soap, and cooling brings it back to a solid.

WHAT STUDENTS WILL SEE

Hard cubes turn into a clear liquid in about 1-2 minutes of heating and then become a solid bar again in about an hour.

Safety: Wear splash goggles. Melted soap is hot: an adult handles the microwave and pouring. Use only a skin-safe soap base.

Try next: Make two bars with different amounts of color and test which lathers more.

EMULSIFIER SHOWDOWN

Oil and water with different emulsifiers

Grades 6-12 | Medium | Prep 10 min | Class 30-40 min | MS-PS1-2 | HS-PS1-3

WHY DOES IT HAPPEN?

Oil and water do not mix because water molecules attract each other more than they attract oil. An emulsifier has one end that likes water and one that likes oil, so it keeps tiny oil droplets spread through the water. Mustard and dish soap are both emulsifiers; this experiment compares them to nothing.

MATERIALS

- 3 clear jars with lids (small, 8 oz / 240 mL)
- 1/2 cup (120 mL) vegetable oil
- 1/2 cup (120 mL) water with a drop of food coloring
- 1 tsp (5 g) yellow mustard
- 1/2 tsp (2.5 mL) dish soap
- Timer
- Marker and tape

6 STEPS

1. 
Label the jars: Nothing, Mustard, Soap.

2. 
Pour 3 tbsp (45 mL) of water and 3 tbsp (45 mL) of oil in each jar.

3. 
Add 1 tsp mustard to the Mustard jar and 1/2 tsp dish soap to the Soap jar.

4. 
Close the lids tightly. Shake each jar for 20 seconds.

5. 
Set the jars down and start the timer.

6. 
Record what each jar looks like at 1, 5 and 30 minutes.

THE EQUATION, SIMPLIFIED

Oil(l) + water(l) + emulsifier → emulsion (no new substance forms: physical mixture)

The emulsifier coats oil droplets so they stay spread out in the water.

WHAT STUDENTS WILL SEE

The jar with nothing separates in under a minute. The mustard and soap jars stay cloudy much longer, for 30 minutes or more.

Safety: Wear splash goggles. Close the lids tightly to prevent leaks, and wipe spills so the floor is not slippery. Do not taste anything from this experiment.

Try next: Test egg yolk or honey as another emulsifier.

MAKE MAYONNAISE: EGG YOLK EMULSION

Emulsion stabilized by lecithin in egg yolk

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | MS-PS1-2 | HS-LS1-6

WHY DOES IT HAPPEN?

Egg yolk contains lecithin, a molecule with a water-loving end and an oil-loving end. When oil is added slowly while whisking, the lecithin surrounds tiny oil droplets and holds them in the water of the yolk and vinegar. Adding oil too fast breaks the emulsion.

MATERIALS

- 1 pasteurized egg yolk
- 1 tsp (5 mL) white vinegar
- 1 tsp (5 g) yellow mustard
- 1/2 cup (120 mL) vegetable oil
- Small bowl
- Whisk or fork
- Pinch of salt

6 STEPS

1. Put the yolk, vinegar, mustard and salt in the bowl.
2. Whisk for 30 seconds until the mixture is smooth.
3. Add the oil one drop at a time while whisking hard, for the first 1/4 cup (60 mL).
4. When the mixture thickens, pour the rest of the oil in a thin stream while whisking.
5. Stop whisking when it is thick and pale. Spoon a bit on a plate.
6. Compare it with the unmixed oil and yolk, and explain what changed.

THE EQUATION, SIMPLIFIED

Oil(l) + water(l) + lecithin → emulsion (no new substance forms: physical mixture)

Lecithin keeps oil droplets from joining together.

WHAT STUDENTS WILL SEE

After 3-5 minutes of whisking the runny yellow liquid becomes a thick, pale cream that holds its shape on a spoon.

Safety: Wear splash goggles. Use pasteurized egg only, wash hands after handling, and do not taste the result; follow school food policy.

Try next: Add the oil all at once in another bowl and compare what happens.

BUTTER IN A JAR

Separating fat from cream by agitation

Grades 6-8 | Easy | Prep 5 min | Class 20-30 min | MS-PS1-2 | MS-LS1-7

WHY DOES IT HAPPEN?

Cream is an emulsion: tiny fat droplets are surrounded by a thin coat of protein and held in water. Shaking breaks the coats so the fat droplets stick together into a lump of butter, leaving buttermilk behind.

MATERIALS

- 1 cup (240 mL) heavy whipping cream, room temperature
- Clean jar with a tight lid (16 oz / 500 mL)
- 2 clean marbles (optional)
- Strainer
- Small bowl
- Paper towels
- Timer

6 STEPS

1. 
   Pour the cream into the jar until it is about half full. Add the marbles if you have them.
2. 
   Close the lid tightly and shake with a steady rhythm.
3. 
   Keep shaking for 5 minutes. It will turn into whipped cream first.
4. 
   Keep shaking for another 3-5 minutes until a solid lump forms and liquid sloshes around it.
5. 
   Pour the contents through a strainer into a bowl.
6. 
   Compare the yellow solid butter with the thin, white buttermilk.

THE EQUATION, SIMPLIFIED

Fat droplets in cream → solid butter + buttermilk (no new substance forms: physical separation)

Shaking only makes the fat droplets stick together.

WHAT STUDENTS WILL SEE

After about 3-5 minutes the cream becomes stiff, and after about 8-10 minutes a yellow lump of butter and thin liquid appear.

Safety: Wear splash goggles. Close the lid tightly. Do not taste the product; follow school food policy.

Try next: Rinse the butter in cold water and compare its texture before and after.

VITAMIN C DETECTIVE

Redox titration with iodine and starch indicator

Grades 9-12 | Challenge | Prep 20 min | Class 30-40 min | HS-PS1-2 | HS-LS1-6 | Adult handling

WHY DOES IT HAPPEN?

Vitamin C (ascorbic acid) reacts with iodine and turns it into colorless iodide. A starch indicator turns blue-black when iodine is left over. Counting how many drops of iodine are needed before the blue color stays tells you which juice has more vitamin C.

MATERIALS

- 1 tsp (3 g) cornstarch boiled in 1 cup (240 mL) water, cooled (adult heat)
- Iodine tincture 2%, diluted 1 part to 10 parts water (adult prepares)
- 3 juices: orange, apple and lemon (1/4 cup / 60 mL each)
- Droppers (one per juice)
- 3 clear cups
- Spoon
- Marker and tape

6 STEPS

1. Label three cups with the juice names.
2. Pour 1/4 cup (60 mL) of juice into each cup and add 2 tbsp (30 mL) of starch solution.
3. Stir each cup.
4. An adult adds dilute iodine one drop at a time, swirling after each drop, and counts the drops.
5. Stop when the blue-black color stays for 20 seconds. Write down the number of drops.
6. Rank the juices from most to least vitamin C and explain your evidence.

THE EQUATION, SIMPLIFIED

$C_6H_8O_6 + I_2 \rightarrow C_6H_6O_6 + 2\ HI$ (vitamin C + iodine → dehydroascorbic acid + hydrogen iodide)

Vitamin C gives up two hydrogen atoms to iodine, which uses up the iodine until the vitamin C is gone.

WHAT STUDENTS WILL SEE

The iodine disappears into the juice with no color change at first. Once the vitamin C is used up, a blue-black color suddenly stays. Fresh lemon or orange juice needs more drops than apple juice.

Safety: Wear splash goggles and gloves. Iodine stains and is poisonous if swallowed: an adult handles it and students do not touch it. Do not taste anything from this experiment.

Try next: Compare fresh juice with juice that has been open for a week.

STARCH DETECTOR: IODINE FOOD TEST

Testing foods for starch

Grades 6-8 | Easy | Prep 15 min | Class 20-30 min | MS-LS1-7 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Iodine turns blue-black in the presence of starch, a long chain of glucose that plants use to store energy. Foods made from grains, potatoes and beans contain starch, but foods such as apples and cheese contain little or none.

MATERIALS

- Iodine tincture 2%, diluted 1 part to 10 parts water (adult prepares)
- Small pieces of potato, bread, apple, cheese and a cracker
- Foil or small plate
- Dropper
- Paper towels
- Marker
- Gloves

6 STEPS

1. Draw a table with the five foods and a column for Prediction and Result.
2. Predict which foods contain starch and write it in your table.
3. Place a small piece of each food on the tray.
4. An adult puts one drop of dilute iodine on each food.
5. Wait 1 minute and observe the color.
6. Record which foods turned blue-black and compare with your predictions.

THE EQUATION, SIMPLIFIED

Starch + I_3^- → starch-iodine complex (blue-black)

Iodine slides inside the coil of the starch chain and the combination absorbs light, so it looks dark.

WHAT STUDENTS WILL SEE

Potato, bread and cracker turn blue-black within seconds. Apple and cheese remain yellow-brown.

Safety: Wear splash goggles and gloves. Iodine stains skin and is poisonous if swallowed: an adult handles it. Do not taste anything from this experiment.

Try next: Test a banana at different stages of ripeness and see how the starch changes.

THE BROWN PAPER BAG FAT TEST

Testing foods for fats and oils

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-LS1-7 | MS-PS1-2

WHY DOES IT HAPPEN?

Fats and oils are nonpolar and soak into paper, making it translucent so that light passes through. Water also darkens paper, but it dries away. A permanent see-through spot means the food contains fat.

MATERIALS

- Brown paper bag cut into 5 squares
- Marker
- 1/2 tsp each of vegetable oil, water, butter, and mashed avocado
- 1/2 tsp of milk
- Cotton swabs or spoons
- Paper towels
- Window or lamp

6 STEPS

1. 
Label each paper square with one of the five samples.

2. 
Rub a little of each sample on its square, using a clean swab each time.

3. 
Let the squares dry for 15 minutes.

4. 
Hold each square up to the window or a lamp.

5. 
Check which squares have a see-through spot.

6. 
Record the results and rank the foods by the size and brightness of the spot.

THE EQUATION, SIMPLIFIED

Fat spreads into paper fibers (no new substance forms: physical change)

Fat remains in the paper while water evaporates, so the translucent spot stays.

WHAT STUDENTS WILL SEE

Oil, butter and avocado leave clear, greasy spots that stay after 15 minutes. Water leaves a spot that disappears, and milk leaves a faint spot.

Safety: Wear splash goggles. Wash hands after handling samples and wipe up oils from the floor. Do not taste anything from this experiment.

Try next: Test a potato chip and a plain cracker and compare.

SUGAR HUNTER: GLUCOSE STRIP TEST

Testing liquids for glucose

Grades 9-12 | Medium | Prep 15 min | Class 30-40 min | MS-LS1-7 | HS-LS1-6

WHY DOES IT HAPPEN?

Glucose test strips contain an enzyme that reacts with glucose and changes the color of a pad. Because only glucose is detected, the strips can show which liquids contain it and roughly how much.

MATERIALS

- Glucose test strips from a pharmacy (1 per sample)
- 1/4 cup (60 mL) each of water, apple juice, sports drink and diet soda
- 1 tsp (4 g) table sugar dissolved in 1/4 cup (60 mL) water
- 5 clear cups
- Marker and tape
- Paper towels
- Color chart from the strip bottle and a timer

6 STEPS

1. Label the five cups with the liquids.
2. Pour 1/4 cup of each liquid in its cup.
3. Stir the table sugar into its cup of water until it dissolves.
4. Dip a new strip into each liquid for 2 seconds and tap off the extra.
5. Wait the time written on the strip package.
6. Compare each strip with the color chart and record the glucose level.

THE EQUATION, SIMPLIFIED

$C_6H_{12}O_6 + O_2 \rightarrow C_6H_{10}O_6 + H_2O_2$ (glucose + oxygen → gluconolactone + hydrogen peroxide; the strip reads this)

The enzyme on the strip changes glucose, and the product causes a color change.

WHAT STUDENTS WILL SEE

Apple juice and sports drink show a color change on the pad. Water and diet soda stay unchanged. Table sugar gives little or no reaction because it is sucrose rather than glucose.

Safety: Wear splash goggles. Used strips go in the trash. Do not taste anything from this experiment.

Try next: Stir a spoonful of honey into water and test it.

EGG WHITE MEETS ALCOHOL

Protein denaturation by alcohol

Grades 9-12 | Medium | Prep 10 min | Class 20-30 min | HS-LS1-6 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Egg white is mostly water and a protein called albumin that is folded into a compact shape. Alcohol pulls water away and disturbs the bonds that hold the folded shape, so the proteins unfold and tangle into a solid white mass, the same thing that happens when egg is cooked.

MATERIALS

- 1 egg white (about 2 tbsp / 30 mL)
- 2 tbsp (30 mL) rubbing alcohol (70% isopropyl), adult pours
- 2 tbsp (30 mL) water
- 2 clear cups
- 2 spoons
- Paper towels
- Marker

6 STEPS

1. Label two cups: Water and Alcohol.
2. Put 1 tbsp (15 mL) of egg white in each cup.
3. Add 1 tbsp (15 mL) of water to the first cup.
4. An adult adds 1 tbsp (15 mL) of rubbing alcohol to the second cup.
5. Stir each cup gently with its own clean spoon.
6. Observe both cups for 5 minutes and compare them.

THE EQUATION, SIMPLIFIED

Folded protein (clear) → unfolded, tangled protein (white): denaturation

The atoms are the same, but the protein changes shape and clumps together.

WHAT STUDENTS WILL SEE

The egg white with water stays clear and runny. The alcohol cup turns white and cloudy within seconds, with stringy lumps in 1-2 minutes.

Safety: Wear splash goggles. Rubbing alcohol is flammable and its vapor is irritating: an adult pours it, with ventilation and no flames. Raw egg: wash hands afterward. Do not taste anything from this experiment.

Try next: Add a small amount of salt water to another sample and test whether it turns white.

COOKING WITHOUT HEAT: EGG WHITE AND ACID

Protein denaturation by acid

Grades 6-12 | Medium | Prep 10 min | Class 20-30 min | MS-PS1-2 | HS-LS1-6

WHY DOES IT HAPPEN?

Acids can change the shape of proteins in the same way heat does. Lemon juice adds hydrogen ions that change the charges on the protein, so the folded chains unwind and clump. This is why lemon juice makes raw fish turn firm and white in ceviche.

MATERIALS

- 1 egg white (about 2 tbsp / 30 mL)
- 1 tbsp (15 mL) lemon juice
- 1 tbsp (15 mL) white vinegar
- 1 tbsp (15 mL) water
- 3 clear cups
- 3 spoons
- Marker and tape

6 STEPS

1. 
   Label three cups: Water, Lemon and Vinegar.
2. 
   Put 2 tsp (10 mL) of egg white in each cup.
3. 
   Add 1 tbsp (15 mL) of water to the first cup, lemon juice to the second and vinegar to the third.
4. 
   Stir each cup with its own clean spoon.
5. 
   Wait 10 minutes.
6. 
   Compare how white and thick each sample has become and write a conclusion.

THE EQUATION, SIMPLIFIED

Folded protein + H+ → unfolded protein clumps (denaturation)

Acid changes how the protein chains stick to each other, but no new atoms are added.

WHAT STUDENTS WILL SEE

The water cup stays clear and runny. The lemon and vinegar cups turn cloudy white and thicken in about 5-10 minutes.

Safety: Wear splash goggles. Raw egg can carry germs: wash hands and clean surfaces. Do not taste anything from this experiment.

Try next: Test a baking soda solution (a mild base) to see if it also changes the egg white.

FATS: SOLID OR LIQUID?

Saturated and unsaturated fats and cooling

Grades 9-12 | Medium | Prep 10 min | Class 30-40 min | HS-LS1-6 | MS-PS1-4

WHY DOES IT HAPPEN?

Fat molecules have long carbon chains. Saturated fats have straight chains that pack tightly and stay solid at room temperature. Unsaturated fats have bends, so the chains cannot pack well and they stay liquid. Cooling slows molecules so even oils can start to thicken.

MATERIALS

- 1 tbsp (15 g) coconut oil (solid)
- 1 tbsp (14 g) butter
- 1 tbsp (15 mL) olive oil
- 1 tbsp (15 mL) vegetable oil
- 4 small clear cups
- Bowl of ice water
- Timer and marker

6 STEPS

1. 
Label four cups with the fats.

2. 
Put 1 tbsp of each fat in its cup.

3. 
Observe each sample at room temperature and record its state.

4. 
Set the cups in the ice water bath, making sure no water enters them.

5. 
Wait 20 minutes.

6. 
Record which samples turned cloudy, thick or solid and relate it to their chains.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change. Straight (saturated) chains pack tightly, bent (unsaturated) chains do not.

Cooling only slows the molecules, so the fats change state without changing composition.

WHAT STUDENTS WILL SEE

Coconut oil and butter stay solid. Olive oil turns cloudy and thicker after 20 minutes, while the vegetable oil stays mostly liquid.

Safety: Wear splash goggles. Wipe up any oil spills right away because they cause slips. Do not taste anything from this experiment.

Try next: Place the olive oil in the refrigerator overnight to see how far it solidifies.

CARBON CHAINS WITH GUMDROPS

Molecular models of alkanes and combustion

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-1 | MS-PS1-5

WHY DOES IT HAPPEN?

Carbon atoms can form four bonds, so they can join into long chains. Using different colors for carbon, hydrogen and oxygen lets students see how atoms are rearranged and how none are lost. Count the atoms on both sides to see a balanced equation.

MATERIALS

- 6 black gumdrops or mini marshmallows for carbon
- 14 white gumdrops for hydrogen
- 6 red gumdrops for oxygen
- 40 toothpicks
- Paper plate
- Pencil and paper
- Models are for building only, not eating

6 STEPS

1. Sort the gumdrops by color on the plate.
2. Build methane: one carbon with four hydrogens attached using toothpicks.
3. Build ethane: two carbons joined together, each with three hydrogens.
4. Take the models apart and build two oxygen molecules (O2) as two reds joined by a toothpick.
5. Model the burning of methane: break the old molecules and make 1 CO2 and 2 H2O.
6. Count the atoms on each side and record that they match.

THE EQUATION, SIMPLIFIED

$CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$ (1 C, 4 H, 4 O on each side)

The same atoms are rearranged into new molecules, so the totals do not change.

WHAT STUDENTS WILL SEE

Students find 1 carbon, 4 hydrogen and 4 oxygen atoms on both sides of the model, showing that atoms are conserved.

Safety: Wear splash goggles. Candy models must not be eaten; toothpicks are sharp, so keep them pointing down. No real flame is used.

Try next: Build propane and write the balanced combustion equation with models.

ISOMERS: SAME ATOMS, DIFFERENT MOLECULES

Butane and isobutane models

Grades 9-12 | Challenge | Prep 10 min | Class 20-30 min | HS-PS1-1 | MS-PS1-1

WHY DOES IT HAPPEN?

Isomers have the same chemical formula but different arrangements of atoms. Butane has a straight chain and isobutane has a branch, so they have different boiling points and properties even though both are C4H10.

MATERIALS

- 4 black gumdrops or marshmallows (carbon) per student
- 10 white gumdrops (hydrogen) per student
- Toothpicks (16 per student)
- Paper plate
- Pencil and paper
- Models are for building only, not eating

6 STEPS

1. Sort 4 carbons and 10 hydrogens onto the plate.
2. Build butane: connect four carbons in a line, then add hydrogens until each carbon has four bonds.
3. Count the atoms and write the formula.
4. Take it apart and build isobutane: three carbons around one central carbon, then add hydrogens.
5. Count the atoms and write its formula.
6. Compare the two models and explain why they are different molecules.

THE EQUATION, SIMPLIFIED

C_4H_{10} (butane) → C_4H_{10} (isobutane): 4 C and 10 H on both sides

Both molecules have the same atoms, just connected in different ways.

WHAT STUDENTS WILL SEE

Both models use 4 carbons and 10 hydrogens, yet one is a straight chain and the other has a branch.

Safety: Wear splash goggles. Candy models must not be eaten. Keep toothpicks pointed down.

Try next: Build the 3 isomers of pentane and check that each carbon has four bonds.

WHY DO BANANAS TURN BROWN?

Enzymatic browning and how to slow it

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2 | HS-PS1-5 | Adult handling

WHY DOES IT HAPPEN?

When a banana or apple is cut, an enzyme meets phenols and oxygen from the air and makes brown pigments. Lemon juice has acid that slows the enzyme, and vitamin C reacts with the oxygen first, so browning is delayed. Cold also slows the reaction.

MATERIALS

- 1 banana or 2 apples
- 4 small plates
- 1 tbsp (15 mL) lemon juice
- 1 tbsp (15 mL) water
- Bowl of ice (for the Cold plate)
- Butter knife (adult cuts)
- Marker and timer

6 STEPS

1. Label four plates: Nothing, Water, Lemon and Cold.
2. An adult cuts the fruit into slices of equal thickness.
3. Place two slices on each plate.
4. Brush water on the Water plate slices and lemon juice on the Lemon plate slices. Set the Cold plate on top of the bowl of ice.
5. Wait 30 minutes and check every 10 minutes.
6. Score the browning from 0-5 for each plate and compare.

THE EQUATION, SIMPLIFIED

$2\,C_6H_6O_2 + O_2 \rightarrow 2\,C_6H_4O_2 + 2\,H_2O$ (a phenol such as catechol + oxygen → quinone + water)

The enzyme speeds up the reaction in which phenols are oxidized into brown quinones.

WHAT STUDENTS WILL SEE

The untreated slices start to turn brown in 10-20 minutes. Lemon-treated slices stay lighter, and the cold plate browns more slowly.

Safety: Wear splash goggles. An adult handles the knife. Do not taste anything from this experiment.

Try next: Test pineapple juice, honey water and salt water as other anti-browning dips.

MILK ART: SOAP AND FAT EXPLOSION

Surface tension and soap with fat in milk

Grades 6-8 | Easy | Prep 5 min | Class 20-30 min | MS-PS1-2 | MS-PS1-1

WHY DOES IT HAPPEN?

Milk is mostly water with floating fat and protein. Dish soap lowers the surface tension of the water, and its molecules attach to the fat. This starts a rush of motion that carries the food coloring along.

MATERIALS

- 1/2 cup (120 mL) whole milk
- Shallow plate or pie dish
- Food coloring in 3 colors
- Cotton swab
- 1 tsp (5 mL) dish soap in a cup
- Paper towels
- Timer

6 STEPS

1. Pour the whole milk into the plate so it covers the bottom.
2. Add one drop each of 3 colors, spaced apart.
3. Wait for the milk to settle completely.
4. Dip the cotton swab in dish soap.
5. Touch the swab to the center of the milk and hold it still for 10 seconds.
6. Watch the colors for 1-2 minutes and record your drawings.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change (soap molecules lower surface tension and bind to fat)

The soap moves around, and the colors move with it.

WHAT STUDENTS WILL SEE

The colors swirl and race away from the swab for about 30-60 seconds, then slowly stop moving as the soap spreads out.

Safety: Wear splash goggles. Wipe spills so floors do not become slippery. Do not taste anything from this experiment.

Try next: Repeat with skim milk and compare the swirling.

CHLOROPHYLL COLOR DETECTIVE

Paper chromatography of spinach pigments

Grades 9-12 | Medium | Prep 15 min | Class 30-40 min | MS-LS1-6 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Green leaves contain several pigments: chlorophyll a and b, carotene and xanthophyll. When alcohol creeps up paper it carries each pigment at a different speed, depending on how strongly it sticks to the paper, so the colors separate.

MATERIALS

- 5 fresh spinach leaves
- Coffee filter cut into strips (1 inch / 2.5 cm wide)
- 1/4 cup (60 mL) rubbing alcohol (70% isopropyl), adult handled
- Tall clear jar
- Pencil
- Coin
- Tape

6 STEPS

1. 
Crush a spinach leaf with the coin on a strip of filter paper to make a green line 1 inch (2.5 cm) from one end.

2. 
An adult pours 1/2 inch (1.5 cm) of rubbing alcohol into the jar.

3. 
Tape the strip to the pencil and rest it across the jar so the end with the line just touches the alcohol but the green line stays above it.

4. 
Let the alcohol travel up the paper for 20-30 minutes.

5. 
Remove the strip before the liquid reaches the top and lay it flat.

6. 
Mark each color band and record the order.

THE EQUATION, SIMPLIFIED

No new substance forms: physical separation by chromatography

The pigments do not react; they just travel different distances on the paper.

WHAT STUDENTS WILL SEE

The green line separates into bands of yellow-orange, yellow-green and blue-green over 20-30 minutes.

Safety: Wear splash goggles. Rubbing alcohol is flammable and the vapor irritates: an adult handles it, with ventilation and no flames. Do not taste anything from this experiment.

Try next: Repeat with a red or purple leaf and compare the pigments.

BIODEGRADABLE PLASTIC FROM STARCH

Making a bioplastic from cornstarch and glycerin

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | MS-PS1-3 | HS-ESS3-4 | Adult handling

WHY DOES IT HAPPEN?

Starch is a natural polymer. Heating it in water makes the chains swell and spread out, and glycerin acts as a plasticizer that slips between chains so the film can bend. A little vinegar helps the gel set smoothly. When dried, the film is flexible and can biodegrade.

MATERIALS

- 1 tbsp (8 g) cornstarch
- 4 tbsp (60 mL) water
- 1 tsp (5 mL) glycerin (from a pharmacy)
- 1 tsp (5 mL) white vinegar
- Small nonstick pot and a stove or hot plate (adult use only)
- Aluminum foil on a flat tray
- Spoon

6 STEPS

1. Put the cornstarch, water, glycerin and vinegar in the pot and stir.
2. An adult heats the mixture on low, stirring constantly.
3. When it turns clear and thick (about 3-5 minutes), the adult removes it from the heat.
4. An adult pours the gel onto the foil and spreads it into a thin layer about 1/8 inch (3 mm).
5. Let it dry for 24-48 hours away from the sun.
6. Peel off the dried film and bend and stretch it. Compare it with a plastic bag.

THE EQUATION, SIMPLIFIED

Starch + glycerin + water → flexible film (no major new substance forms: physical change)

Glycerin spaces the starch chains apart, which keeps the film from becoming brittle.

WHAT STUDENTS WILL SEE

The milky mixture becomes a clear gel in 3-5 minutes and dries into a thin, bendable film in 1-2 days. It is weaker than store-bought plastic.

Safety: Wear splash goggles. The mixture is hot: an adult handles the stove and pot. Never eat the plastic.

Try next: Bury a piece in soil and check it weekly to see how fast it breaks down.

POTATO FOAM: CATALASE ENZYME

Enzyme breaking down hydrogen peroxide

Grades 6-12 | Medium | Prep 10 min | Class 20-30 min | HS-PS1-5 | MS-LS1-7 | Adult handling

WHY DOES IT HAPPEN?

Cells make hydrogen peroxide as a harmful by-product, and the enzyme catalase breaks it into water and oxygen. Potato is rich in catalase, and the oxygen bubbles show how fast the reaction runs. Boiling the potato denatures the enzyme.

MATERIALS

- 3 tbsp (45 mL) of 3% hydrogen peroxide (drugstore)
- 1 raw potato
- 1 boiled potato (cooled)
- 3 small clear cups
- Knife (adult cuts)
- Spoon
- Tray

6 STEPS

1. Label the cups: Raw Potato, Boiled Potato and Control.
2. An adult cuts a 1/2 inch (1.3 cm) cube from the raw potato and one from the boiled potato.
3. Pour 1 tbsp (15 mL) of hydrogen peroxide into each cup, standing on the tray.
4. Drop the raw cube into the first cup, the boiled cube into the second, and nothing into the third.
5. Watch for bubbles for 2 minutes.
6. Rank the three cups by bubbling and explain the difference.

THE EQUATION, SIMPLIFIED

$$2\,H_2O_2 \rightarrow 2\,H_2O + O_2 \text{ (catalase enzyme)}$$

The enzyme speeds up the breakdown but is not used up.

WHAT STUDENTS WILL SEE

The raw potato fizzes and foams right away. The boiled potato makes few bubbles, and the control has almost none.

Safety: Wear splash goggles. Use only 3% hydrogen peroxide and keep it away from eyes. An adult handles the knife. Do not taste anything from this experiment.

Try next: Test carrot, apple and mushroom pieces to compare how much catalase each has.

SOAP VS GREASE

How soap lifts oil

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2 | MS-PS1-1 | Adult handling

WHY DOES IT HAPPEN?

Water alone cannot wash away oil because oil molecules are nonpolar. A soap molecule has an oil-loving tail and a water-loving head. Tails grab the grease and heads face the water, so tiny balls of grease are wrapped and carried away.

MATERIALS

- 2 clear cups
- 1 cup (240 mL) warm water for each cup (adult handled)
- 1 tbsp (15 mL) vegetable oil
- 1 tsp (5 mL) dish soap
- Black pepper
- Paper towels
- Spoon

6 STEPS

1. 
Label the cups: Water and Soapy Water.

2. 
An adult pours 1 cup of warm water into each cup.

3. 
Add 1 tsp of dish soap to the second cup and stir gently.

4. 
Add 1/2 tbsp (7 mL) of oil to each cup and sprinkle a pinch of pepper on top of the oil.

5. 
Stir each cup for 10 seconds.

6. 
Observe each cup after 2 minutes.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change (soap molecules surround oil droplets)

Soap helps oil mix with water without any reaction.

WHAT STUDENTS WILL SEE

In the plain water cup the oil stays in large floating drops. In the soapy cup the oil breaks into tiny droplets that turn the water cloudy.

Safety: Wear splash goggles. An adult handles warm water. Wipe up oil spills right away. Do not taste anything from this experiment.

Try next: Test cold water and compare how well soap works.

RIPENING GAS: ETHYLENE AND FRUIT

Plant hormone gas speeds ripening

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-LS1-7 | MS-PS1-2

WHY DOES IT HAPPEN?

Ripe fruit gives off ethylene, a small gas molecule made of carbon and hydrogen. Ethylene acts as a signal that triggers other fruit to soften and sweeten. Trapping the gas in a bag with ripe fruit speeds the ripening of green fruit nearby.

MATERIALS

- 3 green bananas or green tomatoes
- 1 ripe apple
- 2 paper bags
- Marker
- Camera or paper
- Counter space for 5 days
- Tape

6 STEPS

1. Label two bags: A (green fruit alone) and B (green fruit + apple), and label the counter spot C (open air).
2. Place one green banana alone in bag A and fold the top closed.
3. Place a green banana and a ripe apple together in bag B and fold it closed.
4. Leave the third banana out on the counter at spot C as the control.
5. Leave everything for 5 days.
6. Photograph or draw each banana daily and compare color and softness.

THE EQUATION, SIMPLIFIED

No equation for this change; ethylene (C_2H_4) acts as a signal molecule, not a reactant

Ethylene tells the fruit to start making the enzymes that soften and sweeten it.

WHAT STUDENTS WILL SEE

After 2-3 days the banana in bag B turns yellow and soft, while the banana alone or in open air changes more slowly.

Safety: Wear splash goggles. Wash hands after handling fruit and discard it when finished. Do not taste anything from this experiment.

Try next: Test a green banana next to an onion to see if a vegetable also affects ripening.

EGG WITHOUT A SHELL

Acetic acid + calcium carbonate

Grades 6-12 | Easy | Prep 10 min | Class 20-40 (plus 24-48 h wait) min | MS-PS1-2 | MS-PS1-5

WHY DOES IT HAPPEN?

An eggshell is mostly calcium carbonate. Vinegar is a weak acid called acetic acid. When they meet, the shell breaks down into dissolved calcium acetate, water, and carbon dioxide gas. The gas makes the bubbles. The thin membrane under the shell is made of protein, so it stays behind.

MATERIALS

- 1 raw egg
- 1 clear jar or tall cup, 12 oz (355 mL) or larger
- 1 cup (240 mL) white vinegar, enough to cover the egg
- 1 spoon
- Paper towels
- 1 marker and 1 strip of tape

6 STEPS

1. 
Label the jar with your name and the date, then gently place the raw egg in it.

2. 
Pour about 1 cup (240 mL) of white vinegar over the egg until it is fully covered.

3. 
Look closely at the shell after 1 minute and sketch the tiny bubbles.

4. 
Set a timer and check again after 8 hours. Use a spoon to turn the egg gently so the bubbles can escape.

5. 
After 24 hours, pour off the old vinegar and add fresh vinegar to finish the job.

6. 
After 48 hours, lift the egg out with a spoon, rinse it under cool water, and hold it up to a lamp to see through it.

THE EQUATION, SIMPLIFIED

$$CaCO_3 + 2CH_3COOH \rightarrow Ca(CH_3COO)_2 + H_2O + CO_2$$

One calcium carbonate unit plus two acetic acid units makes calcium acetate, water, and carbon dioxide gas.

WHAT STUDENTS WILL SEE

Bubbles start within seconds and cover the shell in 1 to 2 minutes. After 24 to 48 hours the shell is gone and the egg feels like a rubbery, slightly larger ball. A lamp shines through it and the yolk shows as a dark shadow.

Safety: Wear splash goggles. Do not eat the egg. Wash hands after handling raw egg and cover the jar with a plate to reduce vinegar smell. Squeeze gently because the membrane can burst.

Try next: Try soaking a shell-free egg in corn syrup or salty water and measure how its size changes.

RED CABBAGE PH INDICATOR

Acids, bases and natural indicators

Grades 6-12 | Easy | Prep 15 min | Class 20-40 min | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Red cabbage contains pigments called anthocyanins that change color with acidity. In acids they look pink or red, in neutral liquids purple, and in bases blue or green. This makes cabbage juice a free, natural pH indicator you can make in the classroom.

MATERIALS

- 1/4 head red cabbage, chopped (about 2 cups / 300 g)
- 2 cups (480 mL) hot tap water
- 1 bowl and 1 strainer
- 3 clear cups, 6 oz (180 mL) each
- 1 tbsp (15 mL) white vinegar
- 1 tsp (5 g) baking soda dissolved in 1/4 cup (60 mL) water

6 STEPS

1. 
Chop the cabbage and place it in the bowl, then have an adult pour the hot water over it.

2. 
Wait 10 minutes, stirring now and then, until the water turns deep purple.

3. 
Strain the liquid into a clean cup and throw the cabbage away. This is your indicator.

4. 
Pour 1/4 cup (60 mL) of indicator into each of three cups.

5. 
Add 1 tbsp (15 mL) vinegar to cup 1, the baking soda solution to cup 2, and nothing to cup 3 as your control.

6. 
Compare the three colors side by side and record them with a drawing or photo.

THE EQUATION, SIMPLIFIED

HIn <→ H+ + In-

The pigment (HIn) gives off or takes up hydrogen ions (H+), and each form shows a different color.

WHAT STUDENTS WILL SEE

The vinegar cup turns pink or red right away. The baking soda cup turns blue or green within seconds. The control stays purple. Colors vary a little with the cabbage.

Safety: Wear splash goggles. Hot water is handled only by an adult. Do not taste any liquid in the lab. Cabbage juice can stain clothing.

Try next: Freeze some indicator in an ice cube tray and watch the color change as each cube melts into a different liquid.

HOUSEHOLD PH RAINBOW

Testing everyday liquids with cabbage indicator

Grades 6-12 | Medium | Prep 20 min | Class 30-40 min | MS-PS1-2 | HS-PS1-2

WHY DOES IT HAPPEN?

Different liquids contain different amounts of hydrogen ions, which is what pH measures. Cabbage indicator shows a whole rainbow of colors as the pH changes from acidic to basic. Lining up the colors lets students rank everyday liquids on a pH scale.

MATERIALS

- 1 cup (240 mL) red cabbage indicator (see ACI-02)
- 6 clear cups, 4 oz (120 mL) each
- 2 tbsp (30 mL) lemon juice
- 2 tbsp (30 mL) white vinegar
- 2 tbsp (30 mL) clear soda, left to go flat
- 2 tbsp (30 mL) baking soda solution (1 tsp in 1/2 cup water)
- 2 tbsp (30 mL) soapy water (a few drops of dish soap)
- 2 tbsp (30 mL) plain tap water

6 STEPS

1. Label six cups: lemon juice, vinegar, soda, baking soda, soapy water, and water.
2. Pour 2 tbsp (30 mL) of each test liquid into its own cup.
3. Add 2 tbsp (30 mL) of cabbage indicator to every cup.
4. Swirl each cup gently and wait 30 seconds.
5. Line the cups up from most red to most blue and sketch the color order.
6. Write a prediction for a new liquid, then test it and check if you were right.

THE EQUATION, SIMPLIFIED

$H_2O \rightleftharpoons H^+ + OH^-$

Water splits slightly into hydrogen ions and hydroxide ions; acids add H+ and bases add OH-, which changes the indicator color.

WHAT STUDENTS WILL SEE

Lemon juice and vinegar turn pink or red. Soda is pink to purple-pink. Water stays purple. Baking soda turns blue and soapy water turns blue-green. The colors form a clear rainbow within 30 seconds.

Safety: Wear splash goggles. Never use drain cleaners or bleach for this test. Wash hands after handling soapy water, and do not drink any test liquids.

Try next: Test fruit juices or laundry detergent solution to add them to the rainbow.

TURMERIC TEST: SOAP TURNS RED

Turmeric as a base indicator

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Turmeric contains a yellow pigment called curcumin. Curcumin stays yellow in neutral and acidic liquids but turns reddish orange in bases. Soap is a mild base, so it makes a turmeric stripe change color, which gives students a quick test for basic substances.

MATERIALS

- 1 tbsp (7 g) turmeric powder
- 3 tbsp (45 mL) warm tap water
- 1 coffee filter or white paper
- 1 small paintbrush or cotton swab
- 1 tsp (5 mL) liquid hand soap
- 1 tsp (5 mL) white vinegar
- 1 paper plate

6 STEPS

1. 
Stir the turmeric into the water to make a loose yellow paste.

2. 
Paint three yellow stripes on the paper and let them dry for 10 minutes.

3. 
Label the stripes: soap, vinegar, and water.

4. 
Dab a drop of liquid soap on the first stripe.

5. 
Dab vinegar on the second stripe and water on the third.

6. 
Wait 1 minute and compare all three stripes.

THE EQUATION, SIMPLIFIED

$$HCur + OH^- \rightleftharpoons Cur^- + H_2O$$

Yellow curcumin (HCur) loses a hydrogen ion to a base and becomes the red-orange form (Cur-).

WHAT STUDENTS WILL SEE

The soap stripe turns red-orange in seconds. The vinegar and water stripes stay yellow. A drop of vinegar on the red spot can bring the yellow back.

Safety: Wear splash goggles. Turmeric stains skin, clothing and counters, so cover the table. Do not taste anything.

Try next: Make a secret message by writing with soapy water on a turmeric-painted sheet.

WHERE DID THE MASS GO?

Open-cup mass loss in a gas-forming reaction

Grades 9-12 | Medium | Prep 10 min | Class 30-40 min | HS-PS1-7 | HS-PS1-2

WHY DOES IT HAPPEN?

Baking soda and vinegar react to make carbon dioxide gas. In an open cup the gas drifts into the air, so a scale reads less mass afterward. The atoms are not destroyed; they are in the gas that left the cup, which supports conservation of mass.

MATERIALS

- 1 kitchen scale reading to 1 g
- 1 tall plastic cup, 12 oz (355 mL)
- 2 tbsp (28 g) baking soda in a small cup
- 1/2 cup (120 mL) white vinegar
- 1 spoon
- Notebook and pencil

6 STEPS

1. Put the cup of vinegar and the small cup of baking soda together on the scale and record the total mass.
2. Pour the baking soda into the vinegar slowly while both cups stay on the scale.
3. Watch the foam rise and stir gently once.
4. Wait until the fizzing stops, about 2 minutes.
5. Record the new mass with both cups still on the scale.
6. Calculate the difference and explain where the missing mass went.

THE EQUATION, SIMPLIFIED

$$NaHCO_3 + CH_3COOH \rightarrow CH_3COONa + H_2O + CO_2$$

Baking soda plus acetic acid makes sodium acetate, water, and carbon dioxide gas that escapes.

WHAT STUDENTS WILL SEE

Foam fills about half of the cup in 10 to 20 seconds. The scale drops by roughly 1 to 3 g depending on the amounts. The loss is the CO2 gas that left.

Safety: Wear splash goggles. Use an open cup only; never seal the reaction in a closed container. Wipe spills right away so the scale does not get sticky.

Try next: Repeat with double the baking soda and predict the mass loss first.

ANTACID VS. ACID

Neutralization with a crushed antacid tablet

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-2 | HS-PS1-2

WHY DOES IT HAPPEN?

Antacid tablets often contain calcium carbonate, a base that reacts with acid. In this activity vinegar stands in for acid and cabbage indicator shows when the acid has been used up. Students see the red color fade toward purple as the antacid neutralizes the acid.

MATERIALS

- 1/2 cup (120 mL) red cabbage indicator (see ACI-02)
- 1/2 cup (120 mL) water
- 1/4 cup (60 mL) white vinegar
- 2 calcium carbonate antacid tablets, crushed in a zip bag
- 1 clear cup, 12 oz (355 mL)
- 1 spoon for stirring and crushing

6 STEPS

1. Crush the antacid tablets inside a zip bag using the back of a spoon.
2. Pour the water and indicator into the cup and stir.
3. Add the vinegar and watch the color turn pink.
4. Sprinkle in one-quarter of the antacid powder and stir.
5. Keep adding powder, one quarter at a time, until the color turns purple or blue.
6. Record how much antacid was needed.

THE EQUATION, SIMPLIFIED

$$CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$$

The base calcium carbonate uses up hydrogen ions and releases carbon dioxide.

WHAT STUDENTS WILL SEE

The mixture fizzes as soon as the powder meets the acid. The pink liquid fades to purple and then blue-purple after most of the powder is added. The liquid may stay cloudy.

Safety: Wear splash goggles. This is a model only; students must never eat or drink any material from the experiment.

Try next: Compare two kinds of antacid and see which needs more powder.

INVISIBLE INK: BAKING SODA AND GRAPE JUICE

Base reveals an anthocyanin indicator

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Baking soda solution is a mild base and dries nearly invisible on paper. Purple grape juice contains anthocyanin pigments, the same family as in red cabbage. When the juice meets the baking soda, the pigments shift toward blue-green and the hidden message appears.

MATERIALS

- 2 tbsp (28 g) baking soda
- 1/4 cup (60 mL) warm water
- 1 cotton swab
- 1 sheet white paper or a coffee filter
- 1/2 cup (120 mL) purple grape juice
- 1 wide paintbrush
- 1 tray or plate

6 STEPS

1. 
Stir the baking soda into the warm water until it is mostly dissolved.

2. 
Dip the cotton swab in the solution and write a message on the paper.

3. 
Let the paper dry for 10 minutes until the writing is nearly invisible.

4. 
Pour the grape juice onto a plate.

5. 
Brush the juice evenly over the paper.

6. 
Watch the message appear and record the color.

THE EQUATION, SIMPLIFIED

$$HCO_3^- + H_2O \leftrightarrow H_2CO_3 + OH^-$$

Baking soda in water makes a few hydroxide ions, a mild base that changes the juice pigment color.

WHAT STUDENTS WILL SEE

The writing shows up as blue-green letters on a purple background within about 30 seconds. Stronger solutions show darker letters.

Safety: Wear splash goggles. Work over a tray because juice stains. Do not taste the solutions.

Try next: Test other plant juices such as blueberry and red cabbage.

HOMEMADE PH PAPER

Making and using test strips

Grades 6-12 | Medium | Prep 25 min | Class 30-40 min | MS-PS1-2 | MS-ETS1-3

WHY DOES IT HAPPEN?

Commercial pH paper is paper soaked with an indicator. Students can do the same using cabbage juice, then dry the strips and test liquids. This shows how a design (the strip) turns a color change into measurable data.

MATERIALS

- 1 cup (240 mL) red cabbage indicator (see ACI-02)
- 3 coffee filters
- 1 shallow dish
- 1 pair of scissors
- Paper towels
- 3 clear cups
- 2 tbsp (30 mL) each of vinegar, baking soda solution, and tap water

6 STEPS

1. Cut the coffee filters into strips about 1 inch by 4 inches (2.5 cm by 10 cm).
2. Soak the strips in the cabbage indicator for 5 minutes.
3. Lift the strips out and lay them flat on paper towels.
4. Let them dry for at least 30 minutes, or overnight.
5. Dip one strip into each test liquid for 2 seconds.
6. Compare the strip colors with a drawn color chart and rank the liquids.

THE EQUATION, SIMPLIFIED

HIn <→ H+ + In-

The dried indicator on the paper still changes between its acid and base forms when liquid touches it.

WHAT STUDENTS WILL SEE

Dry strips are blue-purple. The vinegar strip turns pink, the water strip stays purple, and the baking soda strip turns blue-green within seconds.

Safety: Wear splash goggles. Do not dip strips into anything unknown. Keep strips out of mouths and wash hands after handling.

Try next: Use your strips to test rainwater or sink water and add it to your chart.

BUFFER TEST: WATER VS. BAKING SODA WATER

Resisting a pH change

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | HS-PS1-6 | HS-PS1-2

WHY DOES IT HAPPEN?

A buffer is a solution that resists changes in pH when small amounts of acid are added. Baking soda dissolved in water acts as a simple buffer. Counting the drops of vinegar needed to change the indicator color shows why buffers matter.

MATERIALS

- 1 cup (240 mL) red cabbage indicator (see ACI-02)
- 2 clear cups, 8 oz (240 mL)
- 1/2 cup (120 mL) tap water
- 1/2 cup (120 mL) water with 1/2 tsp (3 g) baking soda
- 2 tbsp (30 mL) white vinegar in a small cup
- 1 dropper
- 1 stir stick

6 STEPS

1. 
Pour 1/2 cup (120 mL) plain water into cup A and the baking soda water into cup B.

2. 
Add 2 tbsp (30 mL) cabbage indicator to each cup and stir.

3. 
Note the starting colors of both cups.

4. 
Add vinegar one drop at a time to cup A, stirring and counting drops until it turns pink.

5. 
Do the same for cup B and keep counting.

6. 
Compare the drop counts and write a claim with evidence.

THE EQUATION, SIMPLIFIED

$$HCO_3^- + H^+ \rightarrow H_2O + CO_2$$

Bicarbonate ions use up added hydrogen ions, so the pH stays steady for a while.

WHAT STUDENTS WILL SEE

Plain water turns pink after only a few drops. The baking soda water stays blue-green for many more drops and fizzes slightly before it finally turns purple.

Safety: Wear splash goggles. Keep vinegar in a small cup so it cannot spill in large amounts. Do not drink any solutions.

Try next: Try a pinch of table salt in plain water to see if it is a buffer.

CLEAN THE PENNY

Acid dissolves copper oxide

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Old pennies look dull because the copper on the surface has reacted with oxygen to form copper oxide. Vinegar is a weak acid and salt helps it work. The acid reacts with the oxide and washes it away, leaving shiny copper underneath.

MATERIALS

- 6 dull pennies
- 1/4 cup (60 mL) white vinegar
- 1 tsp (6 g) table salt
- 1 small glass or plastic bowl
- 1 spoon
- Paper towels
- 1 cup of plain water for rinsing

6 STEPS

1. 
Choose three dull pennies and set the other three aside as your control.

2. 
Stir the salt into the vinegar until it dissolves.

3. 
Dip half of one penny into the mixture and leave the other half dry.

4. 
Drop two more pennies in completely and wait 5 minutes.

5. 
Remove the pennies, rinse them in plain water, and dry them on a towel.

6. 
Compare the cleaned pennies with the control pennies.

THE EQUATION, SIMPLIFIED

$$CuO + 2CH_3COOH \rightarrow Cu(CH_3COO)_2 + H_2O$$

Copper oxide reacts with acetic acid to make copper acetate and water.

WHAT STUDENTS WILL SEE

The cleaned parts turn bright copper within 1 to 5 minutes. The control pennies stay dull. The liquid may turn faintly blue-green.

Safety: Wear splash goggles. Wash hands after handling coins and liquid. Pour the used vinegar down the sink with plenty of water, and do not taste it.

Try next: Test other acids such as lemon juice or tomato sauce.

STEEL WOOL RUSTING AND WARM JAR

Oxidation of iron releases heat

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | HS-PS1-4 | MS-PS1-2

WHY DOES IT HAPPEN?

Iron slowly combines with oxygen and water to form rust, and this reaction gives off heat. Steel wool has a lot of surface area, and vinegar removes its protective coating, so the reaction speeds up enough to measure a small temperature rise.

MATERIALS

- 1 pad of plain steel wool (no soap)
- 1/2 cup (120 mL) white vinegar
- 1 small bowl
- 1 jar, 8 oz (240 mL), wide enough for a thermometer
- 1 digital thermometer
- 1 paper towel
- Timer

6 STEPS

1. 
Soak the steel wool in vinegar for 2 minutes to remove its coating.

2. 
Squeeze it out and wrap it around the thermometer bulb.

3. 
Place the wrapped thermometer in the jar so it does not touch the glass.

4. 
Record the starting temperature.

5. 
Check and record the temperature every 3 minutes for 15 minutes.

6. 
Take out the steel wool and note the orange rust on it.

THE EQUATION, SIMPLIFIED

$$4Fe + 3O_2 + 6H_2O \rightarrow 4Fe(OH)_3$$

Iron, oxygen, and water make rust (simplified as iron hydroxide) and release a little heat.

WHAT STUDENTS WILL SEE

The temperature rises about 2 to 5 degrees F (1 to 3 degrees C) over 10 to 15 minutes. Orange-brown rust appears on the steel wool.

Safety: Wear splash goggles. Use plain steel wool and wear gloves because the fibers can splinter. Rinse hands after handling, and throw the rusty wool in the trash.

Try next: Compare with a second jar of steel wool soaked only in water.

NAIL RUST RACE

Which liquid rusts iron fastest?

Grades 6-12 | Easy | Prep 15 min | Class 20-40 (plus 5-day check) min | MS-PS1-2 | MS-ETS1-3

WHY DOES IT HAPPEN?

Rust forms when iron meets oxygen and water. Salt makes water conduct better, so it speeds rusting, while oil keeps air away and slows it. Students design a fair test, change one variable, and compare the nails over several days.

MATERIALS

- 5 identical iron nails (not galvanized)
- 5 clear cups
- 1/2 cup (120 mL) each of tap water, salt water (1 tsp in 1/2 cup), vinegar, and cooking oil
- 1 empty cup for the dry nail
- Masking tape and marker
- Notebook

6 STEPS

1. 
Label the cups: dry, water, salt water, vinegar, and oil.

2. 
Pour 1/2 cup (120 mL) of each liquid into its cup.

3. 
Place one nail in every cup, covering the nail fully except for the dry one.

4. 
Take a photo of the nails on day 0.

5. 
Check the nails each day for five days and record rust color and amount.

6. 
Rank the cups from most to least rust and explain why.

THE EQUATION, SIMPLIFIED

$$2Fe + O_2 + 2H_2O \rightarrow 2Fe(OH)_2$$

Iron reacts with oxygen and water to form iron hydroxide, the first step of rust.

WHAT STUDENTS WILL SEE

Salt water shows orange rust within one day. Vinegar nails look bright at first, then rust after a few days. Plain water rusts slowly. Oil and dry nails show little change.

Safety: Wear splash goggles. Wear gloves when handling nails to avoid scratches. Throw the nails in the trash after the test.

Try next: Add a nail wrapped in copper wire and compare.

COPPER COAT ON AN IRON NAIL

Displacement of copper by iron

Grades 9-12 | Challenge | Prep 20 min | Class 40 (plus 2 h wait) min | HS-PS1-2 | HS-PS1-7 | Adult handling

WHY DOES IT HAPPEN?

Vinegar and salt dissolve the copper oxide from pennies and make copper acetate in the liquid. Iron is more reactive than copper, so when an iron nail goes in, iron takes the place of copper in the solution. The copper then forms a thin pink-orange layer on the nail.

MATERIALS

- 10 pre-1982 pennies or copper wire pieces (newer pennies work more slowly)
- 1/2 cup (120 mL) white vinegar
- 1 tsp (6 g) table salt
- 1 clean iron nail (not galvanized)
- 1 glass jar
- 1 spoon
- Paper towels and a cup of rinse water

6 STEPS

1. Stir the salt into the vinegar in the jar until it dissolves.
2. Add the pennies and leave them for 15 minutes so the liquid can turn light blue-green.
3. Wipe the nail clean with a paper towel.
4. Drop the nail into the jar so it rests against the pennies.
5. Wait 1 to 2 hours, checking for a pink-orange coating.
6. Take the nail out with a spoon, rinse it, and look at it with a magnifier.

THE EQUATION, SIMPLIFIED

$$Cu(CH_3COO)_2 + Fe \rightarrow Fe(CH_3COO)_2 + Cu$$

Iron swaps places with copper in the solution, so copper metal appears on the nail.

WHAT STUDENTS WILL SEE

After about an hour the nail shows a thin pink-orange copper coat. The liquid turns pale green over time, and the pennies look brighter.

Safety: Wear splash goggles. Adult handling: the copper acetate solution is toxic if swallowed, so wear gloves, wash hands, never taste, and keep the jar away from food. The adult discards the used liquid in a labeled sealed container according to school policy.

Try next: Try a steel screw and compare the coating.

LEMON BATTERY

Electricity from metals and acid

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | HS-PS1-2 | MS-ETS1-2 | Adult handling

WHY DOES IT HAPPEN?

Zinc and copper are different metals, and zinc gives up electrons more easily. When both touch lemon juice, electrons flow from the zinc to the copper through a wire, creating a small electric current. The acid in the juice helps the charge move.

MATERIALS

- 1 lemon
- 1 galvanized (zinc-coated) nail
- 1 copper penny or copper wire strip
- 2 wires with alligator clips
- 1 digital multimeter (or low-current LED)
- 1 small knife for the adult to make slits
- Paper towel

6 STEPS

1. 
Roll the lemon on the table to loosen the juice.

2. 
Have an adult make two small slits about 1 inch (2.5 cm) apart.

3. 
Push the nail into one slit and the copper into the other without letting them touch.

4. 
Clip a wire to each metal.

5. 
Connect the other ends to the multimeter and read the voltage.

6. 
Record the voltage, then join more lemons in a row and measure again.

THE EQUATION, SIMPLIFIED

$$Zn + 2H^+ \rightarrow Zn^{2+} + H_2$$

Zinc loses electrons to become zinc ions while hydrogen ions in the lemon juice gain them; the electrons flow through the wire.

WHAT STUDENTS WILL SEE

One lemon gives about 0.8 to 1 volt. Two or three lemons in a row can light a small LED. The reading drops slowly as the metals wear.

Safety: Wear splash goggles. An adult makes the cuts. Keep metal pieces away from eyes and wash hands after touching lemon juice. Do not taste the lemon.

Try next: Replace the lemon with a potato or an apple and compare voltage.

PENNY PILE BATTERY

A homemade voltaic pile

Grades 9-12 | Challenge | Prep 20 min | Class 40 min | HS-PS1-2 | MS-ETS1-3

WHY DOES IT HAPPEN?

A voltaic pile is a stack of copper and zinc pieces separated by paper wet with salty vinegar. Each pair makes a small voltage, and stacking pairs adds the voltages. Students see how a battery grows by adding cells.

MATERIALS

- 6 pennies or copper washers
- 6 galvanized (zinc) washers of the same size
- 6 paper pieces cut to the size of the washers
- 1/4 cup (60 mL) white vinegar with 1 tsp (6 g) salt
- 1 multimeter
- 2 wires with alligator clips

6 STEPS

1. Dissolve the salt in the vinegar in a small cup.
2. Soak the paper pieces in the salty vinegar.
3. Stack a zinc washer, a wet paper, and a copper washer to make one cell.
4. Measure the voltage of this single cell.
5. Add more cells in the same order and measure after each one.
6. Plot the voltage against the number of cells.

THE EQUATION, SIMPLIFIED

$$Zn + 2H^+ \rightarrow Zn^{2+} + H_2$$

Zinc dissolves into the liquid and gives off electrons while hydrogen ions take them up near the copper.

WHAT STUDENTS WILL SEE

One cell reads about 0.6 to 0.9 volts. Six cells reach about 3 to 5 volts, enough to light an LED briefly. The voltage falls as the paper dries out.

Safety: Wear splash goggles. Do not short-circuit the pile with a wire across the ends. Wash hands after handling wet washers.

Try next: Try lemon juice instead of vinegar and compare voltage.

METALS IN VINEGAR: ACTIVITY SERIES

Reactivity of household metals

Grades 9-12 | Medium | Prep 15 min | Class 30-40 (plus 24 h check) min | HS-PS1-1 | HS-PS1-2

WHY DOES IT HAPPEN?

Metals differ in how readily they react with acid. Zinc bubbles in vinegar faster than iron, while copper barely reacts. Students rank the metals by how many bubbles form, a simple version of the activity series.

MATERIALS

- 4 clear cups
- 1/2 cup (120 mL) white vinegar per cup
- 1 galvanized nail (zinc)
- 1 clean iron nail
- 1 piece of copper wire
- 1 small piece of aluminum foil
- Hand lens and timer

6 STEPS

1. Label four cups: zinc, iron, copper, and aluminum.
2. Pour 1/2 cup (120 mL) of vinegar into each cup.
3. Put one metal into each cup.
4. Watch for 5 minutes and record bubbles on each metal.
5. Check again after 30 minutes and after 24 hours.
6. Rank the metals from most to least bubbles.

THE EQUATION, SIMPLIFIED

$$Zn + 2CH_3COOH \rightarrow Zn(CH_3COO)_2 + H_2$$

Zinc replaces hydrogen in acetic acid, forming zinc acetate and hydrogen gas.

WHAT STUDENTS WILL SEE

Zinc makes tiny bubbles within minutes, iron makes a few after a longer time, aluminum is slow because of its oxide layer, and copper shows no bubbles.

Safety: Wear splash goggles. Keep the cups open and in a ventilated room; only tiny amounts of hydrogen gas form. Wear gloves because metals can have rough edges.

Try next: Test the same four metals in lemon juice.

IRON IN YOUR CEREAL

Magnetic iron in fortified cereal

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-1 | MS-PS1-2

WHY DOES IT HAPPEN?

Some breakfast cereals are fortified with iron metal powder. Iron is attracted to magnets, so students can pull tiny iron particles out of crushed cereal. This shows that metals can be mixed into foods and separated by a physical method.

MATERIALS

- 1 cup (30 g) iron-fortified cereal
- 1 zip bag
- 2 cups (480 mL) warm water
- 1 clear bowl
- 1 strong magnet inside a zip bag
- 1 stir stick
- 1 white plate

6 STEPS

1. 
Crush the cereal into a fine powder inside the bag.

2. 
Pour the powder into the bowl and add the warm water.

3. 
Stir slowly for 10 minutes until the cereal becomes soupy.

4. 
Stir slowly with the bagged magnet for a minute.

5. 
Lift the magnet and look closely at it on a white plate.

6. 
Brush off the fuzzy particles and describe them.

THE EQUATION, SIMPLIFIED

Fe(s) → Fe(s)

No new substance forms; this is a physical separation of iron metal.

WHAT STUDENTS WILL SEE

After a few minutes tiny dark specks and fuzz cling to the end of the magnet. They look like black dust and move when the magnet moves.

Safety: Wear splash goggles. Keep strong magnets away from phones and bank cards, and watch for pinched fingers. Do not eat the cereal mixture.

Try next: Compare two cereals and see which one gives more iron.

ALUMINUM'S INVISIBLE ARMOR

Protective oxide layer on aluminum

Grades 9-12 | Medium | Prep 15 min | Class 30-40 (plus 24 h check) min | HS-PS1-2 | HS-PS1-3

WHY DOES IT HAPPEN?

Aluminum is a reactive metal, yet foil lasts for years. A very thin layer of aluminum oxide forms right away and protects the metal underneath. Students test foil in different liquids, scratch it, and see that the armor stops further reaction.

MATERIALS

- 3 foil pieces about 2 inches (5 cm) square
- 3 clear cups
- 1/2 cup (120 mL) baking soda solution (1 tsp in 1/2 cup water)
- 1/2 cup (120 mL) salt water (1 tsp in 1/2 cup)
- 1/2 cup (120 mL) white vinegar
- 1 coin to scratch the foil
- Hand lens

6 STEPS

1. Label the cups: baking soda, salt water, and vinegar.
2. Pour 1/2 cup (120 mL) of each liquid into its cup.
3. Scratch each foil piece lightly with the coin on one side.
4. Place one foil piece in each cup.
5. Look at the foil with a hand lens after 10 minutes and after 24 hours.
6. Record any bubbles or color changes and explain what the armor does.

THE EQUATION, SIMPLIFIED

$$4Al + 3O_2 \rightarrow 2Al_2O_3$$

Aluminum and oxygen form a thin aluminum oxide layer that shields the metal.

WHAT STUDENTS WILL SEE

Most of the foil looks unchanged in baking soda and salt water, which shows the oxide layer is working. The foil in vinegar may show a few tiny bubbles on scratches and dull spots after a day.

Safety: Wear splash goggles. Wear gloves because foil edges are sharp. Dispose of foil in the trash, not down the sink.

Try next: Rub a scratched spot with a drop of vinegar and see if it dulls faster.

CARBONATE DETECTIVE

Which materials fizz with acid?

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Carbonates such as calcium carbonate give off carbon dioxide when acid touches them. Eggshell, seashell, and some chalk are carbonates, while sugar and salt are not. Students act as detectives and use fizz as evidence.

MATERIALS

- 5 small cups
- 1/2 cup (120 mL) white vinegar
- 1 tsp (5 g) each of crushed eggshell, crushed seashell, chalk, table salt, and sugar
- 1 spoon
- Labels and marker
- Hand lens

6 STEPS

1. Label five cups with the sample names.
2. Add 1 tsp (5 g) of each material to its own cup.
3. Pour about 2 tbsp (30 mL) of vinegar over the first sample.
4. Watch and listen for fizzing for 30 seconds.
5. Repeat for every sample and record the bubble amount from none to lots.
6. Decide which samples are carbonates and explain how you know.

THE EQUATION, SIMPLIFIED

$$CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$$

Carbonate reacts with acid to release carbon dioxide, which makes the bubbles.

WHAT STUDENTS WILL SEE

Eggshell, seashell, and many chalks fizz right away. Salt and sugar show no bubbles. Some chalk is made of gypsum and may not fizz, which is a good surprise.

Safety: Wear splash goggles. Do not taste any sample. Wipe up vinegar spills and wash hands after the test.

Try next: Test a pinch of baking soda and compare the fizz.

CHALK ACID DETECTOR

Using chalk to identify acids

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2

WHY DOES IT HAPPEN?

Chalk made of calcium carbonate fizzes only when an acid touches it. This makes a stick of chalk a simple acid detector. Students test unknown liquids and use fizz as evidence of acid.

MATERIALS

- 4 sticks of calcium carbonate chalk (test one first)
- 4 small cups
- 2 tbsp (30 mL) each of vinegar, lemon juice, baking soda solution, and water
- 1 dropper
- 1 plate
- Labels

6 STEPS

1. Label the cups A, B, C, and D and fill them with the mystery liquids.
2. Write down the key, but keep it hidden from the group.
3. Hold a chalk stick over a plate and add 3 drops of liquid A.
4. Watch for fizz and tiny bubbles.
5. Test liquids B, C, and D on fresh spots or fresh chalk.
6. Decide which are acids, then reveal the key.

THE EQUATION, SIMPLIFIED

$$CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$$

Chalk gives off carbon dioxide where acid touches it.

WHAT STUDENTS WILL SEE

Vinegar and lemon juice make small bubbles on the chalk immediately. Water and baking soda solution make no bubbles.

Safety: Wear splash goggles. Do not taste anything. Wipe the table to remove chalk dust and wash hands afterward.

Try next: Test fruit juices to find which are strong acids.

ACID RAIN EATS THE STATUE

Acid weathering of chalk

Grades 9-12 | Medium | Prep 15 min | Class 30-40 (plus 24 h wait) min | MS-ESS3-3 | HS-ESS3-4

WHY DOES IT HAPPEN?

Acid rain slowly dissolves limestone and marble because they are mostly calcium carbonate. Chalk is a good model. By finding the mass of pieces before and after soaking in vinegar and water, students measure real mass loss.

MATERIALS

- 2 equal pieces of calcium carbonate chalk
- 1 scale reading to 0.1 g
- 2 clear cups
- 1/2 cup (120 mL) white vinegar
- 1/2 cup (120 mL) plain water
- Paper towels
- Notebook

6 STEPS

1. 
Find the mass of each chalk piece and write down the numbers.

2. 
Label cup 1 'acid rain' and cup 2 'pure water'.

3. 
Pour the vinegar into cup 1 and the water into cup 2.

4. 
Put one chalk piece in each cup.

5. 
After 24 hours, dry the chalk on a paper towel for 1 hour.

6. 
Find the masses again and compare the changes.

THE EQUATION, SIMPLIFIED

$$CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$$

Acid dissolves calcium carbonate into ions and carbon dioxide.

WHAT STUDENTS WILL SEE

The chalk in vinegar fizzes at first and loses a large part of its mass, while the chalk in water loses very little. The surface of the vinegar chalk looks rough.

Safety: Wear splash goggles. Wipe up spills. Wash hands after handling wet chalk.

Try next: Test chalk in lemon juice and compare the mass loss.

ACID RAIN AND A LEAF

Acid changes leaf pigment

Grades 9-12 | Medium | Prep 10 min | Class 20-40 (plus 1 h wait) min | MS-ESS3-3 | HS-ESS3-4

WHY DOES IT HAPPEN?

Chlorophyll gives leaves their green color. When acid reaches it, the magnesium in the middle is replaced by hydrogen, and the pigment turns an olive color. Students model how acid pollution can damage plants.

MATERIALS

- 2 fresh spinach leaves of similar size
- 2 clear cups
- 1/2 cup (120 mL) white vinegar
- 1/2 cup (120 mL) tap water
- Tweezers
- Paper towel
- Hand lens

6 STEPS

1. Label cups 'acid' and 'water'.
2. Pour the vinegar into the acid cup and water into the water cup.
3. Drop one leaf in each cup and press it under the liquid.
4. Take a photo of both leaves at the start.
5. Check after 30 minutes and again after 1 hour.
6. Compare the color and texture of the leaves.

THE EQUATION, SIMPLIFIED

Chlorophyll-Mg + 2H+ → Chlorophyll-H_2 + Mg^{2+}

Hydrogen ions replace magnesium in chlorophyll and the pigment changes color.

WHAT STUDENTS WILL SEE

The leaf in vinegar slowly changes from bright green to an olive-brown within an hour. The leaf in water stays green.

Safety: Wear splash goggles. Do not eat the leaves. Discard leaves in the trash and wash hands.

Try next: Test lemon juice or very dilute vinegar to see how strong the acid must be.

HARD WATER VS. SOAP

Calcium and magnesium stop soap from lathering

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-2 | HS-ESS2-5

WHY DOES IT HAPPEN?

Hard water contains dissolved calcium and magnesium. These ions react with soap to make a sticky solid called soap scum, so less lather forms. Students make hard water with Epsom salt and compare it with plain water.

MATERIALS

- 2 clear bottles with caps, 16 oz (475 mL)
- 2 cups (480 mL) tap water
- 2 tbsp (30 g) Epsom salt
- 1 tbsp (15 g) grated bar soap
- 1 spoon
- Labels
- Ruler

6 STEPS

1. 
Label the bottles 'soft' and 'hard'.

2. 
Pour 1 cup (240 mL) of water into each bottle.

3. 
Stir the Epsom salt into the hard bottle until dissolved.

4. 
Add 1/2 tbsp (7 g) of grated soap to each bottle.

5. 
Cap and shake both bottles for 20 seconds.

6. 
Measure the suds height and note the cloudy scum.

THE EQUATION, SIMPLIFIED

$$2C_{17}H_{35}COONa + Mg_{1}+ \rightarrow (C_{17}H_{35}COO)_{2}Mg + 2Na+$$

Soap reacts with magnesium ions and forms a solid scum instead of suds.

WHAT STUDENTS WILL SEE

The soft water bottle makes tall, thick suds. The hard water bottle makes few suds and looks cloudy with white curds.

Safety: Wear splash goggles. Open the caps slowly. Do not taste the water. Wash hands afterward.

Try next: Add more soap to the hard water until suds finally form.

WHY SILVER TARNISHES

Silver and sulfur from egg yolk

Grades 9-12 | Medium | Prep 20 min | Class 30-40 (plus 1-2 h wait) min | MS-PS1-2 | HS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Silver turns dark when it meets sulfur compounds. Egg yolk contains sulfur, and a hard-boiled egg releases a little sulfur gas. Students watch a silver spoon dull and learn that tarnish is a new substance, silver sulfide.

MATERIALS

- 1 hard-boiled egg, cooled
- 1 silver-plated spoon or ring
- 1 jar with a lid, 12 oz (355 mL)
- 1 knife (adult)
- Paper towels
- Hand lens
- Timer

6 STEPS

1. 
Have an adult boil an egg and let it cool.

2. 
Peel the egg and cut it in half with an adult.

3. 
Place the shiny spoon in the jar next to the egg half.

4. 
Close the lid and leave it for 1 to 2 hours.

5. 
Open the jar in a ventilated room and take out the spoon.

6. 
Compare the spoon with a clean one using a hand lens.

THE EQUATION, SIMPLIFIED

$$4Ag + 2H_2S + O_2 \rightarrow 2Ag_2S + 2H_2O$$

Silver reacts with hydrogen sulfide and oxygen to form dark silver sulfide.

WHAT STUDENTS WILL SEE

The part of the spoon near the egg develops a gold to dark gray tarnish after 1 to 2 hours. The clean spoon stays shiny.

Safety: Wear splash goggles. Adult handling: hot water and a knife. Do not eat the egg. Open the jar in a ventilated area because it smells like egg.

Try next: Rub a little baking soda paste on the tarnish to polish it.

VITAMIN C STOPS BROWNING

Antioxidants and oxidation

Grades 6-12 | Easy | Prep 10 min | Class 20-30 (plus 30 min wait) min | MS-PS1-2 | HS-LS1-7 | Adult handling

WHY DOES IT HAPPEN?

Apple slices turn brown when oxygen reacts with compounds in the fruit. Vitamin C is an antioxidant, so it reacts with oxygen first and protects the fruit. Students compare slices treated with vitamin C solution, lemon juice, and plain water.

MATERIALS

- 1 apple (not for eating)
- 1 vitamin C tablet (500 mg), crushed
- 1/2 cup (120 mL) water
- 2 tbsp (30 mL) lemon juice
- 4 small plates
- 1 plastic knife (adult cuts)
- Labels

6 STEPS

1. Dissolve the crushed vitamin C tablet in the water.
2. Have an adult cut the apple into four equal slices.
3. Label four plates: vitamin C, lemon juice, water, and nothing.
4. Dip the slices in the matching liquid for 10 seconds.
5. Place all slices on their plates and take a picture.
6. Check them every 10 minutes for 30 minutes and compare browning.

THE EQUATION, SIMPLIFIED

$$2C_6H_8O_6 + O_2 \rightarrow 2C_6H_6O_6 + 2H_2O$$

Vitamin C reacts with oxygen and becomes dehydroascorbic acid, protecting the fruit.

WHAT STUDENTS WILL SEE

The untreated slice browns in 10 to 15 minutes. The vitamin C and lemon juice slices stay pale for much longer. The water-dipped slice browns slightly later than the untreated one.

Safety: Wear splash goggles. Do not eat the apple slices used in class. An adult handles the knife. Wash hands after.

Try next: Test other fruits such as pears or bananas.

TITRATION-LITE: COUNT THE DROPS

Neutralizing acid with baking soda solution

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | HS-PS1-2 | HS-PS1-6

WHY DOES IT HAPPEN?

A titration finds how much base it takes to neutralize an acid. Here, students add baking soda solution drop by drop to vinegar with cabbage indicator and count the drops until the color flips. The count lets them compare how strong different acids are.

MATERIALS

- 1/4 cup (60 mL) vinegar
- 1/4 cup (60 mL) red cabbage indicator (see ACI-02)
- 1/2 cup (120 mL) baking soda solution (2 tsp in 1/2 cup water)
- 2 clear cups
- 1 dropper
- 1 stir stick
- Notebook

6 STEPS

1. 
   Pour 2 tbsp (30 mL) of vinegar into a cup and add 2 tbsp (30 mL) of indicator.
2. 
   Stir and note the starting pink color.
3. 
   Fill the dropper with baking soda solution.
4. 
   Add baking soda one drop at a time, swirling after each drop.
5. 
   Count the drops until the color turns purple or blue.
6. 
   Repeat with vinegar diluted with equal water and compare counts.

THE EQUATION, SIMPLIFIED

$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$$

Each unit of acid uses up one unit of baking soda, so the drop count shows the amount of acid.

WHAT STUDENTS WILL SEE

The liquid fizzes with each drop. After roughly 20 to 40 drops the pink color changes to purple and then to blue. The diluted vinegar needs about half as many drops.

Safety: Wear splash goggles. Use small cups so spills are tiny. Do not drink the liquids. Wash hands.

Try next: Test lemon juice and compare the drops needed.

CITRIC ACID + BAKING SODA DRY FIZZ

Solid acid and base wait for water

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-2 | HS-PS1-4

WHY DOES IT HAPPEN?

Citric acid and baking soda can sit side by side as dry powders without reacting. When water is added, the ions can move and the acid and base react quickly, making carbon dioxide bubbles. Students learn why water starts the reaction.

MATERIALS

- 2 tsp (8 g) citric acid powder
- 2 tsp (9 g) baking soda
- 2 clear cups
- 1/4 cup (60 mL) water
- 1 spoon
- 1 thermometer
- Hand lens

6 STEPS

1. Mix the citric acid and baking soda in one cup.
2. Stir the dry powders together and notice that nothing happens.
3. Place a thermometer in the powders and note the temperature.
4. Pour 1/4 cup (60 mL) of water over the powder.
5. Watch the fizz and read the temperature again after 30 seconds.
6. Record the temperature change and explain why water mattered.

THE EQUATION, SIMPLIFIED

$$H_3C_6H_5O_7 + 3NaHCO_3 \rightarrow Na_3C_6H_5O_7 + 3H_2O + 3CO_2$$

One citric acid unit reacts with three baking soda units to make sodium citrate, water, and carbon dioxide.

WHAT STUDENTS WILL SEE

The dry powders do nothing. When water is added, the mixture fizzes strongly for about 30 seconds. The cup feels cooler and the temperature drops by a few degrees.

Safety: Wear splash goggles. Citric acid powder irritates eyes, so avoid dust. Do not taste.

Try next: Try cold water versus warm water and compare the speed.

ZINC PROTECTS IRON

Sacrificial protection against rust

Grades 9-12 | Challenge | Prep 15 min | Class 40 (plus 4-day check) min | HS-PS1-2 | MS-ETS1-3

WHY DOES IT HAPPEN?

Zinc is more reactive than iron, so when both are in salty water the zinc corrodes first and protects the iron. This is how galvanizing works. Students compare nails wrapped with zinc, with copper, and bare.

MATERIALS

- 3 iron nails (not galvanized)
- 3 clear cups
- 1 galvanized hardware strip or zinc-coated washer
- 1 piece of copper wire
- 1 1/2 cups (360 mL) salt water (1 tsp per 1/2 cup)
- Masking tape and marker
- Hand lens

6 STEPS

1. Label the cups: zinc, copper, and plain.
2. Wrap a strip of galvanized metal tightly around the first nail.
3. Wrap copper wire around the second nail, and leave the third bare.
4. Pour 1/2 cup (120 mL) of salt water into each cup and add a nail.
5. Check the nails daily for four days and write notes.
6. Compare the rust and decide which wrap protects the iron.

THE EQUATION, SIMPLIFIED

$$2Zn + O_2 + 2H_2O \rightarrow 2Zn(OH)_2$$

Zinc reacts with oxygen and water before the iron can, so it is the one that corrodes.

WHAT STUDENTS WILL SEE

The nail wrapped with copper rusts fastest. The bare nail rusts at a medium speed. The nail with zinc stays much cleaner while white zinc corrosion appears on the zinc.

Safety: Wear splash goggles. Wear gloves because wires and strips are sharp. Do not taste the salt water.

Try next: Test a nail with a thin coat of clear nail polish.

CURDLING MILK WITH ACID

Acid changes milk protein

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-2 | HS-LS1-6 | Adult handling

WHY DOES IT HAPPEN?

Milk contains a protein called casein that floats as tiny clumps. When acid is added, the clumps lose their charge, stick together, and form solid curds. Students see how acid changes a protein and separates curds from liquid.

MATERIALS

- 1 cup (240 mL) whole milk
- 4 tbsp (60 mL) white vinegar
- 1 microwave-safe cup
- 1 strainer or coffee filter
- 1 spoon
- 1 bowl
- Paper towels

6 STEPS

1. 
   Have an adult warm the milk until steamy but not boiling, about 120 to 130 degrees F (50 to 55 degrees C).
2. 
   Pour the milk into the cup.
3. 
   Add the vinegar and stir gently for 10 seconds.
4. 
   Wait 3 minutes and watch the curds form.
5. 
   Pour the mixture through the strainer into a bowl.
6. 
   Press the curds with a paper towel and describe the solid and the liquid.

THE EQUATION, SIMPLIFIED

Casein- + H+ → Casein-H

Acid adds hydrogen ions to casein, so it clumps and becomes solid.

WHAT STUDENTS WILL SEE

Within a minute, white clumps appear and the liquid turns yellowish. After straining, a soft white solid remains and the leftover liquid is clear.

Safety: Wear splash goggles. An adult handles the warm milk. Do not eat the curds. Wash hands and the strainer well.

Try next: Try lemon juice as the acid and compare the curds.

RUST REMOVER SHOWDOWN

Which household acid removes rust best?

Grades 6-12 | Medium | Prep 15 min | Class 30-40 (plus 24 h wait) min | HS-PS1-2 | MS-ETS1-3

WHY DOES IT HAPPEN?

Rust is iron oxide, and acids can dissolve it. Students test vinegar, lemon juice, baking soda water, and plain water on rusty nails and use evidence to decide which works best.

MATERIALS

- 4 rusty nails of similar size
- 4 clear cups
- 1/2 cup (120 mL) each of vinegar, lemon juice, baking soda solution, and plain water
- Masking tape and marker
- Old toothbrush
- Paper towels
- Hand lens

6 STEPS

1. Label the cups with the four liquids.
2. Pour 1/2 cup (120 mL) of each liquid into its cup.
3. Put one rusty nail in each cup.
4. Wait 24 hours.
5. Remove the nails, scrub with the toothbrush, and rinse.
6. Compare the nails and rank the liquids by rust removed.

THE EQUATION, SIMPLIFIED

$$Fe_2O_3 + 6CH_3COOH \rightarrow 2Fe(CH_3COO)_3 + 3H_2O$$

Rust reacts with acetic acid to form dissolved iron acetate and water.

WHAT STUDENTS WILL SEE

The vinegar nail looks cleanest after 24 hours and the liquid turns light yellow-brown. Lemon juice also cleans somewhat. Water and baking soda do little.

Safety: Wear splash goggles. Wear gloves around rusty nails to avoid cuts. Dispose of the nails in the trash.

Try next: Add a spoon of salt to the vinegar and compare the speed.

ICE CREAM IN A BAG

Freezing-point depression with salt and ice

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-4 | HS-PS3-4

WHY DOES IT HAPPEN?

Plain ice melts at 32°F (0°C). When salt dissolves in the thin film of water on ice, the mixture can stay liquid well below that, so an ice-salt bath gets colder than ice alone. The cream gives up its heat to this super-cold bath and slowly freezes as you shake.

MATERIALS

- 1/2 cup (120 mL) whole milk or half-and-half
- 1 tbsp (12 g) sugar
- 1/4 tsp vanilla extract
- 1 pint-size zipper bag and 1 gallon-size zipper bag
- 3 cups (about 400 g) ice cubes
- 1/3 cup (90 g) rock salt or table salt
- Oven mitts or a towel
- Thermometer (optional)

6 STEPS

1. 
Measure the milk, sugar, and vanilla into the small bag and press out the air.

2. 
Seal the small bag tightly and check the seal by squeezing gently.

3. 
Fill the large bag halfway with ice and stir in the salt.

4. 
Place the sealed small bag inside the large bag and seal the large bag.

5. 
Wrap the bag in a towel and shake and roll it for 5 to 8 minutes.

6. 
Open the large bag, wipe the salt off the small bag, and look at the cream inside.

THE EQUATION, SIMPLIFIED

$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

Salt dissolving in the melting ice lowers the freezing point of the water; no new substance forms in the ice-salt bath.

WHAT STUDENTS WILL SEE

Within 5 to 8 minutes the liquid thickens into soft, frozen cream. A thermometer in the ice bath can read 10 to 20°F (about -12 to -7°C), colder than plain ice.

Safety: Wear splash goggles. Wipe the sealed bag clean before opening it. This is food-safe only if made with clean food-grade tools and away from the lab area; follow your school food policy before anyone eats it.

Try next: Compare the freezing time with no salt, 1 tbsp of salt, and 1/3 cup of salt.

ICE, WATER, AND STEAM

States of matter and particle motion

Grades 6-12 | Easy | Prep 15 min | Class 20-30 min | MS-PS1-4 | Adult handling

WHY DOES IT HAPPEN?

Matter changes state when its particles gain or lose thermal energy. In ice the particles only vibrate in fixed places, in liquid water they slide past each other, and in steam they fly apart. Heating moves particles faster; cooling slows them.

MATERIALS

- 4 ice cubes
- 1 clear heat-safe glass bowl
- 1 cup (240 mL) hot tap water, about 120°F (50°C), poured by an adult
- Thermometer
- 1 small cold metal plate or spoon
- Timer
- Notebook

6 STEPS

1. 
   Place 4 ice cubes in the bowl and record the starting temperature.
2. 
   Watch for 5 minutes and record the temperature every minute.
3. 
   An adult pours the warm water over the ice while students keep recording the temperature.
4. 
   An adult holds the cold plate above the bowl so the rising vapor can touch it.
5. 
   Look closely at the plate and the bowl for droplets and cloudy vapor.
6. 
   Draw particle pictures of ice, water, and vapor and label each state.

THE EQUATION, SIMPLIFIED

$H_2O(s) \leftrightarrow H_2O(l) \leftrightarrow H_2O(g)$

These are physical changes: the same water molecules change how they are arranged and how fast they move.

WHAT STUDENTS WILL SEE

Ice melts in a few minutes, and the temperature stays near 32°F (0°C) until it is nearly gone. Cool droplets form on the plate where water vapor condenses.

Safety: Only adults handle hot water. Keep hands away from rising vapor and wear splash goggles.

Try next: Predict what happens to the final temperature if you use twice as many ice cubes.

SALT, SUGAR, OR PLAIN? MELTING ICE RACE

Melting rate and freezing-point depression

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-PS1-4 | MS-ETS1-3

WHY DOES IT HAPPEN?

Anything that dissolves in the thin water layer on ice lowers its freezing point, so the ice melts faster. Salt breaks into two ions per unit, so it affects the ice more than the same amount of sugar, which stays as whole molecules. This is why roads are salted in winter.

MATERIALS

- 3 identical ice cubes of the same size
- 3 shallow plates
- 1 tsp (6 g) table salt
- 1 tsp (4 g) sugar
- Timer
- Paper towel
- Marker for labels

6 STEPS

1. Label the plates Plain, Salt, and Sugar.
2. Place one ice cube on each plate.
3. Sprinkle 1 tsp of salt on one cube and 1 tsp of sugar on another and leave the third plain.
4. Start the timer and check every 3 minutes.
5. Compare the puddles and look for small pits and channels in each cube.
6. Rank the cubes from fastest to slowest melting and record your data.

THE EQUATION, SIMPLIFIED

$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

Salt splits into two particles in water, while sugar stays whole ($C_{12}H_{22}O_{11}(s) \rightarrow C_{12}H_{22}O_{11}(aq)$); both are physical changes.

WHAT STUDENTS WILL SEE

The salted cube melts fastest and develops pits and channels in about 10 minutes. The sugar cube melts a little faster than the plain cube.

Safety: Wear splash goggles. Wipe up spills so no one slips. Do not taste any materials.

Try next: Test the same amount of baking soda or pepper to see if it changes the melting rate.

INSTANT ICE FROM SUPERCOOLED WATER

Supercooling and nucleation

Grades 9-12 | Challenge | Prep 15 min | Class 20-40 min | MS-PS1-4 | HS-PS3-4

WHY DOES IT HAPPEN?

Very pure water can drop below 32°F (0°C) and stay liquid because it has no tiny specks for ice crystals to start on. A bump or a touch of ice gives crystals a starting point, and the water freezes in seconds. Results vary, so recording what happens is part of the science.

MATERIALS

- 3 sealed bottles of purified water, 16 oz (500 mL) each
- Freezer with a thermometer
- Cup of crushed ice
- Clock or timer
- Zipper bag
- Notebook

6 STEPS

1. An adult lays the sealed bottles gently in the freezer so they are not touching the walls.
2. Set a timer for 2 hours 30 minutes and ask an adult to check at that time.
3. Look for bottles that are still liquid without shaking them.
4. Tap a liquid bottle firmly on the table or drop a small ice chip onto its surface after opening it.
5. Watch where the ice crystals begin and how fast they spread.
6. Record which bottles froze, the time, and how the ice spread.

THE EQUATION, SIMPLIFIED

$$H_2O(l) \rightarrow H_2O(s)$$

Water below 32°F (0°C) freezes as soon as crystals have a place to start; it is a physical change.

WHAT STUDENTS WILL SEE

In a successful trial the bottle stays liquid, then ice spreads from the tap point in 5 to 15 seconds. Some bottles freeze solid in the freezer instead.

Safety: Never leave bottles in the freezer more than 3 hours because they can burst. Put them in a bag and have an adult handle the freezer. Wear splash goggles.

Try next: Compare purified water with tap water in the same freezer for the same time.

COLD PACK FROM BAKING SODA AND VINEGAR

Endothermic reaction

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-6 | HS-PS1-4

WHY DOES IT HAPPEN?

Some reactions absorb thermal energy from their surroundings, so the mixture gets colder. When baking soda meets vinegar, the reaction takes heat from the liquid and the cup. You can measure the drop with a thermometer.

MATERIALS

- 1/2 cup (120 mL) white vinegar
- 2 tbsp (28 g) baking soda
- Clear plastic cup
- Thermometer
- Spoon
- Timer
- Notebook

6 STEPS

1. 
   Pour the vinegar into the cup.
2. 
   Record the starting temperature of the vinegar.
3. 
   Measure the baking soda into a second cup.
4. 
   Add the baking soda in three small portions and stir gently.
5. 
   Record the lowest temperature in the first 2 minutes.
6. 
   Calculate the temperature change and describe where the energy went.

THE EQUATION, SIMPLIFIED

$$NaHCO_3 + CH_3COOH \rightarrow CH_3COONa + H_2O + CO_2$$

Baking soda and acetic acid make sodium acetate, water, and carbon dioxide gas, and the system absorbs heat.

WHAT STUDENTS WILL SEE

The mixture foams for about 30 seconds. The temperature falls by about 5 to 10°F (3 to 5°C).

Safety: Wear splash goggles. Work in a tray because it foams. Do not seal the cup.

Try next: Change the amount of baking soda and see how the temperature drop changes.

WARM FOAM FROM YEAST AND PEROXIDE

Exothermic decomposition

Grades 9-12 | Medium | Prep 15 min | Class 20-30 min | HS-PS1-4 | HS-PS1-5

WHY DOES IT HAPPEN?

Yeast contains an enzyme called catalase that speeds up the breakdown of hydrogen peroxide. The reaction releases oxygen gas and thermal energy, so the mixture warms up. Dish soap traps the gas as foam.

MATERIALS

- 1/4 cup (60 mL) 3% hydrogen peroxide
- 1 tsp dry yeast
- 3 tbsp (45 mL) warm water
- 1 tsp dish soap
- Tall plastic bottle, 16 oz (500 mL)
- Thermometer
- Tray

6 STEPS

1. 
Pour the peroxide and dish soap into the bottle and swirl.

2. 
Record the starting temperature.

3. 
Mix the yeast with the warm water in a cup for 30 seconds.

4. 
Pour the yeast mix into the bottle and step back.

5. 
Watch the foam rise from the bottle into the tray.

6. 
Place the thermometer in the foam and record the highest temperature.

THE EQUATION, SIMPLIFIED

$$2\,H_2O_2 \rightarrow 2\,H_2O + O_2$$

Hydrogen peroxide breaks down into water and oxygen; yeast is the catalyst and is not used up.

WHAT STUDENTS WILL SEE

Thick foam rises in 10 to 20 seconds and the foam feels warm. The temperature rises by 5 to 15°F (3 to 8°C).

Safety: Use only 3% peroxide from a pharmacy. Wear splash goggles and gloves, keep the foam away from the face, and wash hands afterward.

Try next: Test colder and warmer water to see how temperature changes the foam rate.

STEEL WOOL HAND WARMER

Exothermic oxidation of iron

Grades 9-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-6 | HS-PS1-4

WHY DOES IT HAPPEN?

Iron reacts slowly with oxygen to form rust, and the reaction releases thermal energy. Vinegar removes the protective coating on steel wool so the iron reacts faster. A thermometer shows a small but real warming.

MATERIALS

- 1 pad of plain steel wool (no soap)
- 1/2 cup (120 mL) white vinegar
- Bowl
- Small jar with lid
- Thermometer
- Paper towel
- Disposable gloves

6 STEPS

1. 
Soak the steel wool in vinegar for 2 minutes.

2. 
Squeeze out the extra vinegar and blot the wool with a paper towel.

3. 
Wrap the wool around the thermometer bulb and place it in the jar.

4. 
Close the jar loosely and wait.

5. 
Record the temperature every 2 minutes for 12 minutes.

6. 
Graph the data and mark the highest reading.

THE EQUATION, SIMPLIFIED

$$4\,Fe + 3\,O_2 \rightarrow 2\,Fe_2O_3$$

Iron reacts with oxygen to make iron(III) oxide, or rust, and gives off thermal energy.

WHAT STUDENTS WILL SEE

The temperature rises about 5 to 20°F (3 to 11°C) in 10 minutes. Orange rust spots begin to appear.

Safety: Wear gloves and splash goggles because steel wool has sharp splinters. Do not touch your face. Throw the wool away after class.

Try next: Run the same test with unsoaked steel wool as a control.

FIZZ SPEED VS WATER TEMPERATURE

Reaction rate and temperature

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | HS-PS1-5 | MS-PS1-4 | Adult handling

WHY DOES IT HAPPEN?

Particles move faster in warm water, so they collide more often and with more energy. That makes the reaction go faster. Effervescent tablets make fizz as the acid and baking soda in them react in water.

MATERIALS

- 3 plain effervescent tablets
- 3 clear cups
- 1 cup (240 mL) cold water, about 40°F (4°C)
- 1 cup (240 mL) room-temperature water
- 1 cup (240 mL) warm water, about 110°F (43°C), poured by an adult
- Timer
- Thermometer

6 STEPS

1. Label the cups Cold, Room, and Warm.
2. Pour 1 cup of the matching water into each cup.
3. Record the exact temperature of each cup.
4. Drop one tablet into the cold cup and start the timer.
5. Stop the timer when the fizzing ends and repeat with the other cups.
6. Graph temperature against time and write a conclusion.

THE EQUATION, SIMPLIFIED

$$3\,NaHCO_3 + C_6H_8O_7 \rightarrow Na_3C_6H_5O_7 + 3\,H_2O + 3\,CO_2$$

Bicarbonate and citric acid in the tablet react to make sodium citrate, water, and carbon dioxide.

WHAT STUDENTS WILL SEE

The warm cup finishes first, in under a minute. The cold cup often takes 2 to 3 times as long.

Safety: Use warm, not hot, water poured by an adult. Do not swallow any tablet or water from the cups. Wear splash goggles.

Try next: Add a fourth cup at 90°F (32°C) and add it to your graph.

FIZZ SPEED VS SURFACE AREA

Reaction rate and surface area

Grades 6-12 | Easy | Prep 10 min | Class 20-30 min | HS-PS1-5 | MS-PS1-2

WHY DOES IT HAPPEN?

A reaction can only happen where particles touch. Crushing a tablet exposes many more surfaces, so more particles react at the same time. The crushed tablet finishes in much less time than the whole one.

MATERIALS

- 2 plain effervescent tablets
- 2 clear cups
- 1 cup (240 mL) room-temperature water for each cup
- Zipper bag
- Spoon for crushing
- Timer

6 STEPS

1. 
Place one tablet inside the zipper bag.

2. 
Crush the tablet into powder with the back of a spoon.

3. 
Pour the same amount of water into both cups.

4. 
Drop the whole tablet into the first cup and start the timer.

5. 
Stop the timer when the fizzing ends.

6. 
Repeat with the powder, adding it all at once, and compare the times.

THE EQUATION, SIMPLIFIED

$$3\ NaHCO_3 + C_6H_8O_7 \rightarrow Na_3C_6H_5O_7 + 3\ H_2O + 3\ CO_2$$

The same reaction happens in both cups; only the speed changes.

WHAT STUDENTS WILL SEE

The whole tablet takes about 60 to 90 seconds. The powder fizzes fiercely and finishes in about 10 to 20 seconds.

Safety: Wear splash goggles. Do not swallow the tablets or water. Wipe up spills.

Try next: Try halves, quarters, and powder and plot the times.

BALLOON RACE: VINEGAR STRENGTH

Reaction rate and concentration

Grades 9-12 | Medium | Prep 15 min | Class 30-40 min | HS-PS1-5 | HS-PS1-7

WHY DOES IT HAPPEN?

A more concentrated solution has more acid particles in each spoonful, so more collisions happen per second. The balloon on a bottle with stronger vinegar inflates faster. The final amount of gas depends on how much baking soda you used.

MATERIALS

- 3 small bottles, 8 to 12 oz (250 to 350 mL)
- 3 balloons
- 3 tsp (14 g) baking soda
- 1/2 cup (120 mL) white vinegar
- 1/2 cup (120 mL) water
- Measuring cups
- Tape measure or string

6 STEPS

1. Make three liquids: 1/4 cup full-strength vinegar, 1/8 cup vinegar plus 1/8 cup water, and 1/16 cup vinegar plus 3/16 cup water.
2. Pour each liquid into its own bottle and label it.
3. Put 1 tsp of baking soda in each balloon using a paper funnel.
4. Stretch each balloon over a bottle mouth, holding it up so no soda falls yet.
5. Lift the balloons and shake the soda into the bottles at the same time.
6. Measure around each balloon after 30 seconds and write down your results.

THE EQUATION, SIMPLIFIED

$$NaHCO_3 + CH_3COOH \rightarrow CH_3COONa + H_2O + CO_2$$

Baking soda and acetic acid make sodium acetate, water, and carbon dioxide gas.

WHAT STUDENTS WILL SEE

The strongest vinegar inflates its balloon the fastest. After 30 seconds the weaker mixtures have smaller balloons.

Safety: Wear splash goggles. Do not point bottles at faces. Stand the bottles in a tray.

Try next: Graph balloon size against vinegar concentration.

CATALYST SHOWDOWN: YEAST VS POTATO

Catalysts and activation energy

Grades 9-12 | Medium | Prep 15 min | Class 30-40 min | HS-PS1-5 | HS-PS1-4

WHY DOES IT HAPPEN?

A catalyst speeds up a reaction without being used up. Yeast and raw potato both contain catalase, which speeds up the breakdown of hydrogen peroxide. The faster the breakdown, the more oxygen bubbles you will see.

MATERIALS

- 1/2 cup (120 mL) 3% hydrogen peroxide
- 3 clear cups
- 1/2 tsp dry yeast
- 1 pea-sized piece of raw potato
- 1 pea-sized piece of cooked potato
- Spoon
- Timer

6 STEPS

1. 
Pour 3 tbsp of peroxide into each of three cups.

2. 
Label the cups Yeast, Raw potato, and Cooked potato.

3. 
Add the yeast to the first cup.

4. 
Add the raw potato to the second cup and the cooked potato to the third.

5. 
Time the bubbling for 1 minute.

6. 
Rank the catalysts and explain why heat might change the potato.

THE EQUATION, SIMPLIFIED

$$2\,H_2O_2 \rightarrow 2\,H_2O + O_2$$

Hydrogen peroxide changes into water and oxygen; the catalysts are not used up.

WHAT STUDENTS WILL SEE

Yeast produces bubbles in seconds. Raw potato bubbles steadily, and cooked potato barely reacts because heat damaged the catalase.

Safety: Use only 3% peroxide. Wear splash goggles and wash hands afterward. Do not taste anything.

Try next: Try other raw vegetables, such as carrot or celery, and compare.

SALT AND THE BOILING POINT

Boiling-point elevation

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | MS-PS1-4 | HS-PS1-3 | Adult handling

WHY DOES IT HAPPEN?

Dissolved particles make it harder for water molecules to escape into the air, so salty water must get hotter to boil. The change is small in a classroom, but a good thermometer can detect it. This is a physical change, not a reaction.

MATERIALS

- 2 small pots
- 2 cups (480 mL) tap water
- 2 tbsp (36 g) table salt
- Cooking thermometer (up to 250°F / 120°C)
- Hot plate or electric kettle (adult only)
- Spoon
- Oven mitts

6 STEPS

1. 
Put 1 cup of water in each pot.

2. 
Stir the salt into one pot until it dissolves.

3. 
Place the thermometer in the plain water without touching the bottom.

4. 
An adult heats the pot and students record the temperature when steady bubbles appear.

5. 
Repeat with the salty pot.

6. 
Compare the two boiling points and write a short conclusion.

THE EQUATION, SIMPLIFIED

$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$

The dissolved ions raise the boiling point; no new substance forms.

WHAT STUDENTS WILL SEE

Plain water boils near 212°F (100°C). Salt water boils about 2 to 5°F (1 to 3°C) higher.

Safety: Adult handling only for the heat source and hot water. Wear splash goggles, keep hands away from steam, and let pots cool fully.

Try next: Predict what happens to the boiling point if you double the salt, then test it.

WHICH EVAPORATES FASTER? SURFACE AREA TEST

Evaporation and surface area

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-4 | MS-ESS2-4

WHY DOES IT HAPPEN?

Only molecules at the surface of the water can escape into the air as vapor. A wider container gives more molecules a chance to escape, so the water level drops faster. This is why puddles dry faster than a deep bucket.

MATERIALS

- 1/2 cup (120 mL) water for each container
- 1 tall narrow glass
- 1 wide shallow dish
- Permanent marker
- Ruler
- Measuring cup
- Notebook

6 STEPS

1. 
Measure 1/2 cup of water into each container.

2. 
Mark the water level on each container with a marker.

3. 
Place both containers on a sunny windowsill.

4. 
Check the water levels after 1 hour, 1 day, and 2 days.

5. 
Measure how far each level has dropped with the ruler.

6. 
Compare the results and explain them using surface area.

THE EQUATION, SIMPLIFIED

$H_2O(l) \rightarrow H_2O(g)$

Liquid water changes to water vapor; the molecules stay the same.

WHAT STUDENTS WILL SEE

The shallow dish loses water much faster. After 2 days it may be nearly dry while the narrow glass has barely changed.

Safety: Wear splash goggles. Wipe up spills. Keep containers where they will not be knocked over.

Try next: Add a fan or a cover and see how the rate changes.

RAIN ON A CAN: CONDENSATION

Condensation and the dew point

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-ESS2-4 | MS-PS1-4

WHY DOES IT HAPPEN?

Warm air holds water vapor. When that air touches a cold surface, the vapor loses energy and turns back into liquid droplets. This is how dew forms and how clouds form in the sky.

MATERIALS

- 1 empty metal can, clean and dry, with no sharp edge
- Ice cubes
- 1/4 cup (60 g) salt
- 1 cup (240 mL) cold water
- Thermometer
- Paper towel
- Hand lens

6 STEPS

1. 
Fill the can halfway with cold water.

2. 
Add ice cubes until the can is nearly full.

3. 
Stir in the salt to make the can even colder.

4. 
Record the temperature of the mix.

5. 
Watch the outside of the can for 5 minutes.

6. 
Use a hand lens to look at the droplets and write down your findings.

THE EQUATION, SIMPLIFIED

$H_2O(g) \rightarrow H_2O(l)$

Water vapor from the air condenses into liquid droplets on the cold can.

WHAT STUDENTS WILL SEE

Fog forms on the outside of the can in about a minute and grows into droplets that run down the sides.

Safety: Wear splash goggles. Check that the can rim is smooth. Wipe up water to avoid slips.

Try next: Compare a metal can and a plastic cup to see which collects more water.

BALLOON ON A BOTTLE: CHARLES'S LAW

Gas volume and temperature

Grades 9-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-4 | HS-PS3-4 | Adult handling

WHY DOES IT HAPPEN?

When gas is heated its particles move faster and push outward, so a gas at constant pressure expands. When it is cooled, it shrinks. A balloon on a bottle lets you see the volume change.

MATERIALS

- 1 empty 16 oz (500 mL) plastic bottle
- 1 balloon
- Bowl of hot water, about 120°F (50°C), poured by an adult
- Bowl of ice water
- Tape measure
- Thermometer
- Towel

6 STEPS

1. 
   Stretch the balloon over the mouth of the empty bottle.
2. 
   Measure around the balloon while it is still mostly flat.
3. 
   An adult fills one bowl with hot water and another with ice water.
4. 
   Stand the bottle in the hot water for 3 minutes.
5. 
   Measure the balloon again.
6. 
   Move the bottle to the ice water for 3 minutes and measure once more.

THE EQUATION, SIMPLIFIED

V / T = constant (at constant pressure)

The volume of a trapped gas goes up as its temperature goes up; the gas itself stays the same.

WHAT STUDENTS WILL SEE

The balloon puffs up in the hot water and shrinks, even sagging into the bottle, in the ice water.

Safety: Only adults handle hot water. Wear splash goggles and keep hands dry around the bottle.

Try next: Graph the balloon size against the water temperature.

MARSHMALLOW IN A SYRINGE: BOYLE'S LAW

Gas pressure and volume

Grades 9-12 | Medium | Prep 10 min | Class 20-30 min | MS-PS1-4 | HS-PS3-4

WHY DOES IT HAPPEN?

A marshmallow is full of tiny air pockets. When you pull the plunger, the pressure drops and the trapped air expands, so the marshmallow grows. Pushing the plunger squeezes the air and shrinks it again.

MATERIALS

- 1 plastic syringe, 60 mL, without a needle
- 3 mini marshmallows
- Ruler
- Tape
- Marker
- Notebook

6 STEPS

1. Place one marshmallow in the syringe barrel and measure its height.
2. Push the plunger in until it touches the marshmallow.
3. Seal the tip with your finger and pull the plunger out slowly.
4. Watch the marshmallow while you hold the plunger.
5. Measure its size at three plunger positions.
6. Release the plunger and record what happens to the marshmallow.

THE EQUATION, SIMPLIFIED

$$P_1 \times V_1 = P_2 \times V_2$$

At constant temperature, lowering the pressure on a gas makes its volume increase.

WHAT STUDENTS WILL SEE

The marshmallow grows to about twice its size when the plunger is pulled out. It shrinks back when you release it.

Safety: Wear splash goggles. Use syringes without needles. Do not eat the marshmallows.

Try next: Try a small balloon with a little air inside the syringe.

THE COLLAPSING BOTTLE

Gas cooling and air pressure

Grades 6-12 | Medium | Prep 10 min | Class 20-30 min | MS-PS1-4 | HS-PS3-4 | Adult handling

WHY DOES IT HAPPEN?

Warm air in a bottle has plenty of pushing particles. When the bottle cools, the air inside loses energy and its pressure drops. The air outside pushes harder, so it crushes the bottle.

MATERIALS

- 1 empty 16 oz (500 mL) plastic bottle with cap
- 1 cup (240 mL) hot tap water, about 120°F (50°C), poured by an adult
- Bowl of ice water
- Towel
- Thermometer

6 STEPS

1. An adult pours the hot water into the bottle and swirls it for 30 seconds.
2. An adult pours the water out.
3. Cap the bottle tightly right away.
4. Place the bottle in the bowl of ice water.
5. Watch for 1 to 3 minutes.
6. Record what happens and sketch the air particles before and after.

THE EQUATION, SIMPLIFIED

P / T = constant (at constant volume)

The pressure of a trapped gas drops as it cools; the gas is not changed chemically.

WHAT STUDENTS WILL SEE

The bottle crinkles and crushes inward within 1 to 3 minutes of cooling.

Safety: Only adults handle hot water. Wear splash goggles and do not hold the bottle near your face.

Try next: Uncap the bottle and predict what will happen to its shape.

DOES AIR HAVE MASS?

Gas as matter

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-1 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Air is matter, so it has mass even though we cannot see it. An inflated balloon holds more air particles than a flat one. A simple balance shows that the inflated balloon is a little heavier.

MATERIALS

- 1 ruler or wooden dowel
- 2 identical balloons
- String
- Tape
- Pin (adult use only)
- Hook or desk edge for hanging

6 STEPS

1. Hang the ruler from a string so it balances level.
2. Inflate both balloons to the same size and tie them.
3. Tape one balloon to each end of the ruler and adjust until it balances.
4. Check that the ruler is level.
5. An adult pops one balloon with a pin.
6. Record how the ruler changes and explain your observations.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

The air stays air; popping a balloon simply lets that air spread into the room.

WHAT STUDENTS WILL SEE

The ruler tilts toward the full balloon after the other balloon pops.

Safety: Wear splash goggles. Only an adult uses the pin. Warn the class about the loud pop.

Try next: Use a kitchen scale to compare a flat and an inflated ball.

CARBON DIOXIDE POUR

Gas density

Grades 6-12 | Medium | Prep 15 min | Class 20-30 min | MS-PS1-2 | HS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Carbon dioxide is denser than air, so it sinks and can be poured like an invisible liquid. A flame needs oxygen, and the CO2 pushes the air away from it. An adult handles the candle and a tray catches any spills.

MATERIALS

- 1/4 cup (60 mL) white vinegar
- 2 tbsp (28 g) baking soda
- 1 clear jar
- 1 tea light candle
- Matches or lighter (adult only)
- Tray
- Spoon

6 STEPS

1. 
   Put the baking soda in the jar.
2. 
   Add the vinegar and let the fizz settle.
3. 
   Move the jar next to the candle tray.
4. 
   An adult lights the candle on the tray.
5. 
   Tip the jar toward the flame without pouring any liquid.
6. 
   Observe what happens to the flame.

THE EQUATION, SIMPLIFIED

$$NaHCO_3 + CH_3COOH \rightarrow CH_3COONa + H_2O + CO_2$$

The reaction makes carbon dioxide gas, which is heavier than air.

WHAT STUDENTS WILL SEE

The flame flickers and goes out within a few seconds as the CO2 flows over it.

Safety: Adult handling for the candle only. Tie back hair, clear paper from the area, and keep a wet towel ready. Wear splash goggles.

Try next: Pour the gas into a second empty jar and test the jar with a candle.

COLOR SPREAD: HOT VS COLD DIFFUSION

Diffusion and particle speed

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-4 | MS-PS3-4 | Adult handling

WHY DOES IT HAPPEN?

Particles in a liquid are always moving and spreading out from where there are many to where there are fewer. Warmer particles move faster, so color spreads faster in warm water. No stirring is needed.

MATERIALS

- 3 clear cups
- 1 cup (240 mL) cold water
- 1 cup (240 mL) room-temperature water
- 1 cup (240 mL) warm water, about 120°F (50°C), poured by an adult
- Food coloring
- Timer
- Thermometer

6 STEPS

1. 
   Label the cups Cold, Room, and Warm.
2. 
   Fill each cup with the matching water and let it settle.
3. 
   Record the temperatures.
4. 
   Add two drops of food coloring to each cup at the same time.
5. 
   Time how long until the color is evenly spread.
6. 
   Compare the results and draw particle diagrams.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Dye and water mix by diffusion; the dye molecules are not changed.

WHAT STUDENTS WILL SEE

The color spreads quickly in warm water, in about a minute. It takes several minutes in cold water.

Safety: Only adults pour warm water. Wear splash goggles and gloves, and protect clothes from dye.

Try next: Try the same test with milk or oil to compare liquids.

POTATO OSMOSIS

Osmosis and solute concentration

Grades 6-12 | Medium | Prep 15 min | Class 30-40 min | MS-PS1-1 | HS-PS1-3 | Adult handling

WHY DOES IT HAPPEN?

Water moves through cell walls toward the side with more dissolved salt. A potato strip in salty water loses water and gets limp, while one in plain water gains water and gets firm. This is a physical process.

MATERIALS

- 1 raw potato
- 2 cups (480 mL) water
- 2 tbsp (36 g) table salt
- 2 clear cups
- Ruler
- Knife (adult use only)
- Paper towel

6 STEPS

1. 
An adult cuts two matching potato strips, each 3 inches (8 cm) long.

2. 
Measure and record the length and flex of each strip.

3. 
Stir the salt into one cup of water.

4. 
Fill the second cup with plain water.

5. 
Put one strip in each cup for 30 minutes.

6. 
Remove, blot, and measure each strip, noting stiffness.

THE EQUATION, SIMPLIFIED

H_2O (less salty side) → H_2O (saltier side) across the cell membrane

Water moves by osmosis toward the saltier side, and no new substance forms.

WHAT STUDENTS WILL SEE

The strip in salt water becomes limp and slightly shorter. The strip in plain water is firm and slightly stiffer.

Safety: An adult handles the knife. Wear splash goggles. Do not eat the potato.

Try next: Try several salt amounts and find the one that keeps the strip unchanged.

HOT OR COLD: HOW MUCH SUGAR DISSOLVES?

Solubility and temperature

Grades 9-12 | Medium | Prep 15 min | Class 30-40 min | HS-PS1-3 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Warm water particles move faster, so more sugar can dissolve in warm water than in cold water. The amounts that dissolve can be shown on a solubility curve. Salt changes much less with temperature than sugar.

MATERIALS

- 2 clear cups
- 1/2 cup (120 mL) cold water
- 1/2 cup (120 mL) warm water, about 120°F (50°C), poured by an adult
- Sugar, about 1 cup (200 g)
- Table salt, about 1/2 cup (130 g)
- Teaspoon
- Thermometer

6 STEPS

1. 
   Pour cold water in one cup and warm water in the other.
2. 
   Record both temperatures.
3. 
   Add sugar one level teaspoon at a time to the cold cup, stirring until it no longer dissolves.
4. 
   Repeat with the warm cup.
5. 
   Count the teaspoons each cup held and graph them.
6. 
   Repeat the test with salt and compare the two solids.

THE EQUATION, SIMPLIFIED

$$C_{12}H_{22}O_{11}(s) \rightarrow C_{12}H_{22}O_{11}(aq)$$

Sugar molecules spread through the water without changing.

WHAT STUDENTS WILL SEE

The warm water takes noticeably more sugar before it stops dissolving. Salt shows a much smaller change.

Safety: Only adults pour warm water. Wear splash goggles. Do not taste anything.

Try next: Predict how much sugar a cup at 150°F (65°C) would hold and test it with an adult.

CRYSTAL GROWTH: FAST VS SLOW COOLING

Crystallization from solution

Grades 6-12 | Medium | Prep 20 min | Class 30-40 min | MS-PS1-2 | HS-PS1-3 | Adult handling

WHY DOES IT HAPPEN?

Hot water holds more dissolved Epsom salt than cold water. As the solution cools, extra salt comes out as crystals. Slow cooling gives particles time to arrange neatly into larger crystals; fast cooling makes many tiny ones.

MATERIALS

- 1/2 cup (120 mL) hot tap water, about 140°F (60°C), poured by an adult
- 1/2 cup (130 g) Epsom salt
- 2 shallow dishes
- Black construction paper
- Spoon
- Refrigerator or ice bath
- Hand lens

6 STEPS

1. 
An adult pours the hot water into a cup and students stir in the Epsom salt until no more dissolves.

2. 
Stir for 1 minute until the water looks clear.

3. 
Pour half of the solution into each dish placed on black paper.

4. 
Put one dish in the refrigerator and leave the other at room temperature.

5. 
Check after 30 minutes and again after 24 hours.

6. 
Use the hand lens to compare crystal sizes.

THE EQUATION, SIMPLIFIED

$$Mg^{2+}(aq) + SO_4^{2-}(aq) + 7\,H_2O(l) \rightarrow MgSO_4.7H_2O(s)$$

Dissolved Epsom salt comes out of the solution as solid crystals; it is a physical change.

WHAT STUDENTS WILL SEE

The refrigerated dish forms tiny, needle crystals within 30 minutes. The room-temperature dish forms fewer, larger crystals after several hours.

Safety: An adult handles hot water. Wear splash goggles. Do not eat the crystals or touch your eyes.

Try next: Add a drop of food coloring to see how it affects the crystals.

LIQUID RACE: VISCOSITY

Viscosity and intermolecular forces

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-1 | MS-PS1-4

WHY DOES IT HAPPEN?

Viscosity is how much a liquid resists flowing. Large, sticky molecules like those in honey slide past each other slowly, while water molecules slide easily. Warm liquids flow faster than cold ones.

MATERIALS

- 4 identical tall clear cups or jars
- 1/2 cup (120 mL) honey
- 1/2 cup (120 mL) dish soap
- 1/2 cup (120 mL) vegetable oil
- 1/2 cup (120 mL) water
- 4 identical marbles
- Stopwatch

6 STEPS

1. 
Fill each jar with a different liquid to the same height.

2. 
Label each jar and predict the finishing order.

3. 
Drop a marble into the first jar and start the stopwatch.

4. 
Stop the timer when the marble hits the bottom.

5. 
Repeat for the other liquids and record the times.

6. 
Rank the liquids from least viscous to most viscous.

THE EQUATION, SIMPLIFIED

No new substance forms: physical property

The marble falls through each liquid at a speed set by how strongly its molecules attract each other.

WHAT STUDENTS WILL SEE

The marble drops in about a second through water and takes about 10 seconds or more through honey.

Safety: Wear splash goggles. Do not taste the liquids. Wipe up spills at once.

Try next: Warm the honey in a bowl of warm water and retest.

HOW MANY DROPS ON A PENNY?

Surface tension

Grades 6-8 | Easy | Prep 10 min | Class 20-30 min | MS-PS1-1 | HS-PS1-3

WHY DOES IT HAPPEN?

Water molecules attract each other, so the surface acts like a stretchy skin. A penny can hold a dome of water with many drops before it spills. Soap weakens the attraction and the drops spill much sooner.

MATERIALS

- 3 pennies
- 1 eyedropper
- 1/2 cup (120 mL) water
- 1/2 cup (120 mL) water with 1 tsp dish soap
- Paper towel
- Magnifier
- Notebook

6 STEPS

1. Place a penny on a flat paper towel.
2. Add drops of plain water one at a time and count them.
3. Look at the dome shape from the side.
4. Record the number of drops when the water spills.
5. Repeat with soapy water on a fresh penny.
6. Compare the drop counts and write an explanation.

THE EQUATION, SIMPLIFIED

No new substance forms: physical property

Hydrogen bonds hold water molecules together; soap weakens them.

WHAT STUDENTS WILL SEE

A penny holds about 20 to 40 drops of plain water in a dome. Soapy water spills over after far fewer drops.

Safety: Wear splash goggles. Wash hands after handling coins. Wipe up spills.

Try next: Test salt water or sugar water and compare the drop counts.

HEATING RACE: WATER VS OIL

Specific heat

Grades 9-12 | Challenge | Prep 15 min | Class 30-40 min | MS-PS3-4 | HS-PS3-4 | Adult handling

WHY DOES IT HAPPEN?

Different materials need different amounts of energy to warm up by one degree. Water has a high specific heat, so it warms more slowly than oil. That is why lakes stay cooler than the land in summer.

MATERIALS

- 1/2 cup (120 mL) water
- 1/2 cup (120 mL) vegetable oil
- 2 identical microwave-safe cups
- 2 thermometers
- Microwave (adult use only)
- Oven mitts
- Timer

6 STEPS

1. 
   Measure equal amounts of water and oil into matching cups.
2. 
   Record their starting temperatures.
3. 
   An adult heats each cup separately in the microwave for 20 seconds.
4. 
   Stir gently and record the highest temperature.
5. 
   Calculate the temperature change of each liquid.
6. 
   Explain which liquid has the higher specific heat.

THE EQUATION, SIMPLIFIED

q = m x c x (change in temperature)

The same energy raised the oil's temperature more than the water's because the oil has a lower specific heat.

WHAT STUDENTS WILL SEE

The oil warms about twice as much as the water in the same time.

Safety: Adult handling only for the microwave and hot liquids. Use oven mitts, stay back from splashes, and wear splash goggles.

Try next: Try a salt-water solution to see if it changes the heating.

SPOON HEAT RACE: THERMAL CONDUCTIVITY

Conduction through materials

Grades 6-8 | Easy | Prep 15 min | Class 20-30 min | MS-PS3-4 | MS-ETS1-3 | Adult handling

WHY DOES IT HAPPEN?

Metals pass heat along quickly because their particles share energy easily, while wood and plastic pass it more slowly. Butter melting on a handle shows how far the heat has traveled.

MATERIALS

- 1 metal spoon
- 1 wooden spoon
- 1 plastic spoon
- Butter
- 3 small beads or raisins
- Hot tap water, about 140°F (60°C), in a tall mug, poured by an adult
- Timer

6 STEPS

1. Dab a small bit of butter on each spoon handle 3 inches (8 cm) from the bowl.
2. Press a bead onto each butter spot.
3. An adult fills the mug with hot water.
4. Stand all three spoons in the mug with the bowls in the water.
5. Time when each bead falls off.
6. Rank the materials from best conductor to worst.

THE EQUATION, SIMPLIFIED

No new substance forms: heat transfer by conduction

Thermal energy moves from the hot water along the spoon.

WHAT STUDENTS WILL SEE

The bead on the metal spoon falls first, usually within a few minutes. Wood and plastic take much longer.

Safety: An adult handles the hot water. Wear splash goggles and do not touch the spoons while hot.

Try next: Test a strip of aluminum foil or a copper spoon if you have one.

THE GREAT INSULATION CHALLENGE

Insulation and heat transfer

Grades 6-12 | Medium | Prep 20 min | Class 30-40 min | MS-PS3-4 | MS-ETS1-2 | Adult handling

WHY DOES IT HAPPEN?

Insulation slows heat transfer. Trapped air in cotton, foil wraps, or paper is a poor conductor, so these materials keep hot water warm longer. Engineers test and compare designs to find the best solution.

MATERIALS

- 4 identical cups
- 1 cup (240 mL) hot tap water per cup, about 140°F (60°C), poured by an adult
- Cotton balls
- Aluminum foil
- Newspaper
- Rubber bands
- Thermometer

6 STEPS

1. Label the cups Control, Cotton, Foil, and Newspaper.
2. Wrap three of the cups with their material and fix with rubber bands.
3. An adult fills each cup with the same amount of hot water.
4. Record the starting temperature of each.
5. Record the temperatures every 5 minutes for 20 minutes.
6. Graph the results and rank the designs.

THE EQUATION, SIMPLIFIED

No new substance forms: heat transfer

The best insulator slows the flow of thermal energy out of the water.

WHAT STUDENTS WILL SEE

The insulated cups lose heat more slowly. The control cup cools the most, often by 10 to 20°F (6 to 11°C).

Safety: An adult handles hot water. Wear splash goggles. Place cups on a tray.

Try next: Combine two materials and see if you can beat the best single material.

DENSITY TOWER

Density and immiscible liquids

Grades 6-12 | Easy | Prep 15 min | Class 20-30 min | MS-PS1-2 | HS-PS1-3

WHY DOES IT HAPPEN?

Liquids with more mass packed into the same space are denser and sink below lighter ones. Honey, dish soap, water, and oil can form layers. Oil and water do not mix because their molecules attract themselves more than each other.

MATERIALS

- 1/4 cup (60 mL) honey
- 1/4 cup (60 mL) dish soap
- 1/4 cup (60 mL) water with food coloring
- 1/4 cup (60 mL) vegetable oil
- Tall clear glass or jar
- Spoon
- Small items such as a bead, a raisin, and a grape

6 STEPS

1. 
Pour the honey slowly into the center of the glass.

2. 
Slowly pour the dish soap down the inside edge.

3. 
Add the colored water in the same way.

4. 
Add the oil last.

5. 
Drop a bead, a raisin, and a grape and watch where each floats.

6. 
Draw the tower, label the layers, and rank their densities.

THE EQUATION, SIMPLIFIED

No new substance forms: physical property

Each liquid keeps its identity; they sit in layers because of density.

WHAT STUDENTS WILL SEE

Four distinct layers form in about 5 minutes, and each object stops at a different layer.

Safety: Wear splash goggles. Do not taste anything. Wipe up spills at once.

Try next: Test other small objects and predict which layer each will stop in.

BOILING WITHOUT HEAT

Pressure and boiling point

Grades 9-12 | Challenge | Prep 15 min | Class 20-30 min | MS-PS1-4 | HS-ESS2-5 | Adult handling

WHY DOES IT HAPPEN?

Boiling happens when the vapor pressure of the liquid equals the pressure around it. If you lower the pressure above hot water, it can boil at a lower temperature. A syringe lets you drop the pressure by hand.

MATERIALS

- 1 plastic syringe, 60 mL, without a needle
- 1/4 cup (60 mL) hot tap water, about 140°F (60°C), poured by an adult
- Cup
- Thermometer
- Towel
- Splash goggles

6 STEPS

1. 
An adult pours the hot water into the cup.

2. 
Record the temperature.

3. 
Draw up 10 mL of the water into the syringe and keep the tip pointing up.

4. 
Push out any extra air and cover the tip tightly with a finger.

5. 
Pull the plunger back quickly and watch the water.

6. 
Record what you see and explain it using pressure.

THE EQUATION, SIMPLIFIED

$H_2O(l) \rightarrow H_2O(g)$

Lower pressure lets liquid water turn into vapor at a lower temperature.

WHAT STUDENTS WILL SEE

Small bubbles form and rise in the water as soon as the plunger is pulled, even though the water is below 212°F (100°C).

Safety: An adult handles hot water. Use syringes without needles. Wear splash goggles and keep the tip pointed away from faces.

Try next: Try the same experiment with room-temperature water and compare.

WATER FILTER CHALLENGE

Filtration and engineering design

Grades 6-12 | Medium | Prep 15 min | Class 20-40 min | MS-ETS1-2 | MS-ETS1-3 | Adult handling

WHY DOES IT HAPPEN?

Dirty water can hold dirt, leaves and tiny particles. A filter works like a sieve: each layer traps particles that are bigger than its gaps. Engineers test a design, measure the result, and improve it. Clear water is not always clean water.

MATERIALS

- 1 clear 2-liter plastic bottle, bottom cut off by an adult
- 1/2 cup (120 mL) sand
- 1/2 cup (120 mL) small gravel
- 1/4 cup (60 mL) cotton balls
- 1 paper coffee filter
- 1 cup (240 mL) muddy water (soil, leaves, and water only)
- 2 clear cups

6 STEPS

1. 
Hold the bottle upside down in a cup as a stand and place the coffee filter inside the neck.

2. 
Add a layer of cotton, then sand, then gravel, in any order your team designs.

3. 
Pour 1/2 cup (120 mL) of clean tap water through first to rinse the layers.

4. 
Stir the muddy water and pour it slowly onto the top layer.

5. 
Time how long the water takes to drip through and collect it in a clear cup.

6. 
Compare the filtered water with the muddy water, then redesign one layer and test again.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

The filter separates solid particles from water by size; the water molecules pass through unchanged.

WHAT STUDENTS WILL SEE

Within 2-5 minutes the water that drips out looks much clearer than the muddy water. Some fine cloudiness often remains, and a thicker sand layer slows the flow.

Safety: Filtered water is NOT safe to drink. It can still hold germs and chemicals you cannot see. Never taste it, and let only an adult cut the bottle.

Try next: Swap the layer order (all sand versus sand over gravel) and score the clarity of each result.

ACID RAIN VS CHALK

Acetic acid + calcium carbonate

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-PS1-2 | MS-ESS3-3

WHY DOES IT HAPPEN?

Rain is slightly acidic, and polluted air can make it more acidic. Chalk is made of calcium carbonate, the same material as many statues and buildings. Acid attacks it and releases carbon dioxide gas, which is why stone slowly wears away.

MATERIALS

- 2 pieces of white blackboard chalk
- 1/2 cup (120 mL) white vinegar (acid rain model)
- 1/2 cup (120 mL) tap water (normal rain model)
- 2 clear cups
- 1 spoon
- 1 paper towel
- Masking tape and marker

6 STEPS

1. 
   Label one cup 'Rain' and one cup 'Acid Rain' with tape.
2. 
   Pour the water into the Rain cup and the vinegar into the Acid Rain cup.
3. 
   Draw a quick sketch and measure the length of both chalk pieces.
4. 
   Drop one piece of chalk into each cup at the same time.
5. 
   Watch for 10 minutes and note the bubbles in each cup.
6. 
   Lift out the chalk with a spoon, blot it on the paper towel, and compare the size and surface.

THE EQUATION, SIMPLIFIED

$$CaCO_3 + 2\,CH_3COOH \rightarrow Ca(CH_3COO)_2 + H_2O + CO_2$$

Chalk and acid make a dissolved calcium salt, water and carbon dioxide gas.

WHAT STUDENTS WILL SEE

The chalk in vinegar bubbles right away and gets smaller and rough within 10 minutes. The chalk in water makes almost no bubbles.

Safety: Do not taste anything. Wash hands after handling, and wipe up spills.

Try next: Test chalk in half-strength vinegar to see how acid strength changes the speed.

ACID RAIN AND MARBLE CHIPS

Rate of reaction with surface area

Grades 9-12 | Medium | Prep 15 min | Class 20-40 min | HS-PS1-5 | HS-PS1-7 | Adult handling

WHY DOES IT HAPPEN?

Marble is calcium carbonate, like limestone buildings. Acid rain eats away at it and releases carbon dioxide. The reaction can be tracked by the mass lost as gas escapes. Smaller pieces have more surface for acid to attack, so they react faster.

MATERIALS

- 1/4 cup (about 50 g) marble chips or limestone chips from a garden store
- 1/4 cup (about 50 g) of the same chips, crushed inside a bag by an adult
- 1 cup (240 mL) white vinegar
- 2 clear cups
- Kitchen scale (0.1 g if possible)
- Timer
- Paper and pencil

6 STEPS

1. Pour 1/2 cup (120 mL) of vinegar into each cup.
2. Place each cup on the scale and record the starting mass of the cup with its contents.
3. Add whole chips to one cup and the crushed chips to the other at the same moment.
4. Start the timer and watch the bubbling in each cup.
5. Record the mass of each cup every 2 minutes for 10 minutes.
6. Graph mass lost against time and decide which sample reacted faster.

THE EQUATION, SIMPLIFIED

$$CaCO_3 + 2\,CH_3COOH \rightarrow Ca(CH_3COO)_2 + H_2O + CO_2$$

The mass drops because carbon dioxide gas leaves the cup; no atoms are destroyed.

WHAT STUDENTS WILL SEE

The crushed chips fizz fast and lose mass quickly in the first 4 minutes. Whole chips fizz slowly and lose less mass.

Safety: Wear splash goggles. Crush chips only inside a closed, heavy bag, and let an adult do the hammering.

Try next: Test warm (not hot) vinegar and compare with cold vinegar.

ACID RAIN AND A LEAF

Acid damage to living tissue

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-ESS3-3 | MS-PS1-2

WHY DOES IT HAPPEN?

Acid rain can harm more than stone. Strong acidity can damage the waxy coat and cells of leaves. A safe vinegar model shows how leaf tissue can change over a day. It is only a model, because real acid rain is much weaker than vinegar.

MATERIALS

- 3 fresh green leaves of the same kind
- 1/2 cup (120 mL) tap water
- 1/2 cup (120 mL) half-strength vinegar (1/4 cup vinegar + 1/4 cup water)
- 1/2 cup (120 mL) full-strength white vinegar
- 3 clear cups
- Masking tape and marker
- Magnifying lens (optional)

6 STEPS

1. Label three cups 'Water', 'Weak Acid' and 'Strong Acid' with tape.
2. Pour about 1/2 cup (120 mL) of the matching liquid into each cup.
3. Photograph each leaf and sketch its color and texture.
4. Place one leaf into each cup and press it under the liquid with a spoon.
5. Check the leaves after 20 minutes, then again after 24 hours.
6. Compare color, texture and spots using the lens, and record the pattern.

THE EQUATION, SIMPLIFIED

No single simple reaction: H+ ions damage leaf cell membranes and pigments

Acid breaks down the leaf's protective layers and pigments, so the tissue changes color and softens.

WHAT STUDENTS WILL SEE

After 24 hours the leaf in strong vinegar is dull, soft and olive-brown. The weak acid leaf shows milder change, and the water leaf stays green.

Safety: Do not taste anything. Wash hands after handling leaves and liquids.

Try next: Repeat with a leaf that has a thick waxy coat and one with a thin coat.

GREENHOUSE EFFECT IN BOTTLES

Heat trapping and temperature

Grades 6-12 | Medium | Prep 15 min | Class 20-40 min | MS-PS3-4 | MS-ESS3-3 | Adult handling

WHY DOES IT HAPPEN?

The sun's light passes through the air and warms the ground. The warm surface gives off heat that some gases slow down on its way out. A closed bottle models this trapping of heat. Real greenhouse gases like carbon dioxide work in a similar way.

MATERIALS

- 2 clear 1-liter plastic bottles with caps
- 2 thermometers
- 1/4 cup (60 mL) dark soil or a sheet of dark paper
- Tape
- Desk lamp with a regular bulb (not a heat lamp)
- Ruler
- Timer and graph paper

6 STEPS

1. Place the same amount of dark soil in the bottom of each bottle, with a thermometer lying inside.
2. Leave one bottle open and cap the other tightly, keeping the thermometer scale visible.
3. Place both the same distance from the lamp, about 12 inches (30 cm).
4. Record both temperatures at the start.
5. Turn on the lamp and record the temperatures every 2 minutes for 20 minutes.
6. Graph the two lines and describe which bottle warmed more and why.

THE EQUATION, SIMPLIFIED

No new substance forms: energy transfer (light → heat)

Light turns into heat on the dark soil, and the closed bottle slows the heat from escaping.

WHAT STUDENTS WILL SEE

The closed bottle usually warms several degrees more than the open one in 20 minutes. Both rise at the start, then the closed one pulls ahead.

Safety: The lamp bulb gets hot. Do not touch it, and keep it away from paper and plastic. Do not use a heat lamp.

Try next: Add a third bottle with damp soil and compare its temperature rise.

BREATH, WATER AND CABBAGE INDICATOR

CO2 + water → carbonic acid

Grades 9-12 | Medium | Prep 15 min | Class 20-40 min | HS-PS1-6 | HS-ESS2-5 | Adult handling

WHY DOES IT HAPPEN?

Carbon dioxide dissolves in water and forms a weak acid called carbonic acid. Red cabbage juice shows acids by turning pink-red and bases by turning blue-green. Your breath holds carbon dioxide, so blowing into the indicator makes it slowly change. The same chemistry slowly acidifies oceans.

MATERIALS

- 1/4 head of red cabbage
- 2 cups (480 mL) hot water (adult pours)
- 1/2 tsp (2.5 mL) baking soda
- 2 clear cups
- 1 new drinking straw per student
- 1 spoon
- Strainer

6 STEPS

1. 
Chop the cabbage, steep it in the hot water for 10 minutes, and strain to get purple juice.

2. 
Let the juice cool, then pour 1/2 cup (120 mL) into each of two cups.

3. 
Stir a pinch of baking soda into both cups until the juice turns blue-green.

4. 
Keep one cup as the control. Gently blow through a straw into the other cup.

5. 
Blow for 1-2 minutes, resting when needed, and watch the color.

6. 
Compare the two cups and record the color change.

THE EQUATION, SIMPLIFIED

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

Carbon dioxide reacts with water to make carbonic acid, which releases H+ and lowers the pH.

WHAT STUDENTS WILL SEE

After 1-3 minutes of blowing, the liquid shifts from blue-green toward purple. The control cup stays blue-green.

Safety: Wear splash goggles. Adult supervision. Blow out only; never inhale through the straw and never drink the liquid. Use a new straw for each student.

Try next: Test flat tap water and fizzy water as starting liquids.

WHY FIZZY WATER GOES FLAT

Dissolved gas and temperature

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-PS1-4 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Gas dissolves in cold water better than in warm water. In fizzy water, carbon dioxide is held in the liquid under pressure. When the bottle opens or the water warms, the gas escapes as bubbles. This helps explain why warm waters hold less dissolved gas.

MATERIALS

- 3 clear cups
- 1 cup (240 mL) cold fizzy water, freshly opened
- 1 cup (240 mL) room-temperature fizzy water, freshly opened
- 1 cup (240 mL) fizzy water in a warm tap-water bath (adult prepares)
- Thermometer
- Timer
- Dark paper for a backdrop

6 STEPS

1. 
Pour the cold, room-temperature and warmed fizzy waters into three matching cups.

2. 
Take each temperature and write it down.

3. 
Place the dark paper behind the cups to see the bubbles.

4. 
Start the timer and count bubbles rising in 10 seconds in each cup.

5. 
Repeat the counts at 5 minutes and 10 minutes.

6. 
Decide which cup lost its fizz fastest and give a reason.

THE EQUATION, SIMPLIFIED

$H_2CO_3 \rightarrow H_2O + CO_2$

The weak acid breaks down into water and carbon dioxide gas that escapes.

WHAT STUDENTS WILL SEE

The warmest cup bubbles most and goes flat first. The cold cup keeps its bubbles longer.

Safety: Do not drink the water used in the experiment. Wipe up spills, which can be slippery.

Try next: Gently stir each sample and compare how much foam forms.

OCEAN ACIDIFICATION AND SHELLS

Acids and calcium carbonate shells

Grades 9-12 | Challenge | Prep 10 min | Class 20-40 min | HS-ESS3-4 | HS-PS1-2

WHY DOES IT HAPPEN?

Oceans absorb carbon dioxide from the air, which makes the water slightly more acidic. Shells and coral are calcium carbonate, so they weaken in acid. A weak acid like vinegar models the process in a fast, exaggerated way. Students measure how much mass the shells lose.

MATERIALS

- 3 clean seashells or eggshell pieces of similar size
- 1 cup (240 mL) white vinegar
- 1 cup (240 mL) half-strength vinegar (1/2 cup vinegar + 1/2 cup water)
- 1 cup (240 mL) tap water
- 3 clear jars with loose lids
- Kitchen scale
- Masking tape and marker

6 STEPS

1. Label the jars 'Water', 'Half Acid' and 'Acid'.
2. Weigh each shell and write down its starting mass.
3. Pour 1 cup (240 mL) of the matching liquid into each jar.
4. Place one shell in each jar and rest the lid loosely on top.
5. Check the bubbles after 10 minutes, then leave overnight.
6. Dry each shell on a paper towel for 1 hour, reweigh, and calculate the percent mass lost.

THE EQUATION, SIMPLIFIED

$$CaCO_3 + 2\,H^+ \rightarrow Ca^{2+} + H_2O + CO_2$$

Acid reacts with calcium carbonate, which dissolves the shell and releases carbon dioxide gas.

WHAT STUDENTS WILL SEE

The shell in vinegar fizzes and becomes thin and rough after a day. The half-strength shell loses less, and the water shell barely changes.

Safety: Do not tighten the lids because gas is produced. Wash hands after handling.

Try next: Test shells with thick walls and thin walls.

OIL SPILL CLEANUP CHALLENGE

Cleanup design with oil, water and absorbents

Grades 6-12 | Challenge | Prep 15 min | Class 20-40 min | MS-ETS1-2 | MS-ETS1-3

WHY DOES IT HAPPEN?

Oil does not mix with water, so it floats and spreads in a slick. It sticks to feathers and fur and harms wildlife. Engineers compare cleanup tools by how much oil each removes. Students build a fair test and score the results.

MATERIALS

- 1 large tray or baking pan
- 2 cups (480 mL) water
- 2 tbsp (30 mL) vegetable oil mixed with 1/4 tsp cocoa powder for color
- 2 craft feathers
- 1 sponge piece, 1/4 cup (60 mL) cotton balls, paper towels
- 1 spoon
- 1 tbsp (15 mL) dish soap

6 STEPS

1. Pour the water into the tray to model the ocean.
2. Drop the oil on the water and watch it spread into a slick.
3. Dip a feather into the slick and see how the oil clings to it.
4. Try each cleanup tool (spoon, sponge, cotton, paper towel) on the slick for 2 minutes and score the result.
5. Add a few drops of dish soap to a second oily feather in a cup of water and note the change.
6. Write which tool removed the most oil and what trade-offs it had.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change (oil and water stay separate)

Oil floats because it is less dense than water. Soap lets oil break into tiny drops that wash away.

WHAT STUDENTS WILL SEE

The oil spreads into a thin film. Cotton and sponge pick up most of it, and the spoon leaves a thin sheen. Soap loosens oil from the feather but does not solve the spill.

Safety: Wear splash goggles. Oil makes floors slippery. Throw oily materials in the trash, never down the drain.

Try next: Add a floating barrier made of string or straws to see how it contains the slick.

PLASTIC BAGS IN SOIL

Biodegradable versus conventional plastic

Grades 6-12 | Medium | Prep 15 min | Class 20-40 min | MS-PS1-3 | MS-ESS3-3

WHY DOES IT HAPPEN?

Some bags are labeled biodegradable, but many break down only under special conditions. Microbes in soil can slowly break down certain plastics and use them for energy. Students bury samples and observe them over weeks. This is a long-term observation, and results may be small.

MATERIALS

- 1 piece of bag labeled compostable (3 × 3 inches / 8 × 8 cm)
- 1 piece of conventional plastic bag (same size)
- 1 piece of paper (same size)
- 3 clear jars
- 2 cups (480 mL) moist garden soil
- Spray bottle with water
- Masking tape and marker

6 STEPS

1. Label three jars 'Compostable', 'Regular' and 'Paper'.
2. Fill each jar with about 3/4 cup (180 mL) of moist soil.
3. Photograph each sample and measure its starting size.
4. Bury one sample in each jar and mist with water.
5. Keep the jars in a warm spot and mist when the soil dries; check weekly for 4-8 weeks.
6. Dig out the samples, rinse gently, and compare their size, strength and holes.

THE EQUATION, SIMPLIFIED

Starch-based polymer + O_2 + microbes → CO_2 + H_2O + biomass (simplified, slow)

Microbes break big plastic chains into small pieces and use them for energy, making carbon dioxide and water.

WHAT STUDENTS WILL SEE

The paper breaks down first. The compostable piece may show holes or weakness after several weeks, while the regular plastic usually looks unchanged.

Safety: Wear splash goggles. Do not sniff the soil. Wash hands after handling soil and samples.

Try next: Test a bag cut into thin strips to see if surface area matters.

DECOMPOSITION IN COMPOST JARS

Microbes breaking down food scraps

Grades 6-8 | Easy | Prep 15 min | Class 20-40 min | MS-LS1-7 | MS-ESS3-3 | Adult handling

WHY DOES IT HAPPEN?

Tiny organisms break down leftover plants and use the energy inside them. They release carbon dioxide and water as waste. Moisture, air and warmth change how fast scraps rot. A jar lets students watch decay change over days.

MATERIALS

- 3 clear jars with lids (an adult pokes air holes in the lids)
- 1 cup (240 mL) garden soil
- Small pieces of apple core, lettuce and bread (1/4 cup / 60 mL total) per jar
- Spray bottle with water
- Masking tape and marker
- Ruler
- Notebook

6 STEPS

1. 
Label the jars 'Dry', 'Moist' and 'Moist + Cool'.

2. 
Add soil to each jar to about 1 inch (2.5 cm) deep.

3. 
Place the same mix of scraps on top of the soil in each jar.

4. 
Mist two jars with water and leave the 'Dry' jar unwatered.

5. 
Keep the cool jar in a cool, shaded spot and the others at room temperature.

6. 
Observe and sketch every 2 days for 2 weeks without opening the jars.

THE EQUATION, SIMPLIFIED

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O$$

Microbes break down sugars in the scraps and release energy, carbon dioxide and water.

WHAT STUDENTS WILL SEE

The moist, warm jar shows fuzzy mold and darker, softer scraps within 5-7 days. The dry and cool jars change more slowly.

Safety: Wear splash goggles. Never open or sniff the jars, because mold can bother people with allergies. When finished, an adult seals and throws the jars away unopened.

Try next: Add a handful of dry leaves to one jar and compare.

SOIL PH WITH CABBAGE INDICATOR

Acid-base indicator in soil samples

Grades 6-12 | Easy | Prep 15 min | Class 20-40 min | MS-PS1-2 | HS-PS1-2

WHY DOES IT HAPPEN?

Soil can be acidic, neutral or basic, and plants prefer different ranges. Red cabbage juice changes color with acidity. By mixing soil with water and adding juice, students can compare soils. This is a good first look but not a lab-grade measurement.

MATERIALS

- 1 cup (240 mL) red cabbage juice (steeped and cooled cabbage)
- 3 soil samples from different places, 2 tbsp (30 mL) each
- 3 clear cups
- 3/4 cup (180 mL) distilled or tap water
- 3 coffee filters
- 3 spoons
- Masking tape and marker

6 STEPS

1. 
Label three cups with the soil locations.

2. 
Stir each soil sample with 1/4 cup (60 mL) of water for 1 minute.

3. 
Let the mixture settle, then filter it into a clean cup.

4. 
Add 1/4 cup (60 mL) of cabbage juice to each filtered liquid.

5. 
Compare the colors on white paper and rank them from acid to base.

6. 
Record your ranking and guess which plants would grow best in each soil.

THE EQUATION, SIMPLIFIED

Anthocyanin (blue-purple) + H+ <→ anthocyanin-H+ (pink-red)

The pigment in cabbage changes shape and color when acid is present.

WHAT STUDENTS WILL SEE

Acidic soil gives a pink or red shade, neutral soil gives purple, and basic soil turns blue-green. Many garden soils show purple to blue.

Safety: Wear splash goggles. Do not taste the juice or soil. Wash hands after handling soil.

Try next: Add a pinch of baking soda to the pink sample and watch it shift back toward blue.

SOIL FILTERING RACE

Water movement through soil types

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-ESS2-4 | MS-ETS1-3

WHY DOES IT HAPPEN?

Water moves through soil by traveling through the spaces between particles. Sand has large gaps and drains quickly. Clay has tiny gaps and drains slowly. Soil type decides how well water is filtered and how much is stored.

MATERIALS

- 3 clear plastic cups with small holes in the bottom (poked by an adult)
- 3 clear cups to catch water
- 1/2 cup (120 mL) sand
- 1/2 cup (120 mL) potting soil
- 1/2 cup (120 mL) clay soil or crushed clay
- 1 cup (240 mL) water with a drop of food coloring
- Timer

6 STEPS

1. 
Set each holey cup inside a catching cup.

2. 
Add each soil type to its own holey cup, packing lightly.

3. 
Pour 1/4 cup (60 mL) of colored water onto each soil at the same time.

4. 
Start the timer and watch for the first drip.

5. 
Measure the volume of water that collected after 3 minutes.

6. 
Compare the clarity of the water and the amount held by each soil.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Water moves through gaps between grains; smaller gaps slow it and trap more particles.

WHAT STUDENTS WILL SEE

Water flows through sand in under a minute and comes out nearly as colored as it went in. Clay is slow and holds much of the water.

Safety: Do not drink the collected water. Wash hands after handling soil.

Try next: Mix sand and clay in different ratios and test again.

EROSION SLOPE CHALLENGE

Rain, slope and plant cover

Grades 6-8 | Medium | Prep 15 min | Class 20-40 min | MS-ESS3-3 | MS-ETS1-2

WHY DOES IT HAPPEN?

Moving water carries soil away, which can pollute streams. Slope and plant cover change how much soil is lost. Plant roots and mulch hold soil in place. Students test designs that reduce erosion.

MATERIALS

- 2 aluminum foil pans or trays
- 3 cups (720 mL) soil
- 1/2 cup (120 mL) grass clippings or mulch
- 1 cup (240 mL) water in a cup with small holes poked in the bottom
- Books to make a slope
- 2 clear cups
- Ruler

6 STEPS

1. 
Fill both pans with soil and pack it down.

2. 
Cover one pan with mulch or grass clippings.

3. 
Prop both pans on books to make the same slope and put a clear cup at the low end.

4. 
Pour 1 cup (240 mL) of water slowly across the top of each pan, like rain.

5. 
Let the runoff collect and compare how muddy it is.

6. 
Raise the slope and repeat to see what changes.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Falling water pushes small soil grains downhill; plant cover slows the water and anchors the soil.

WHAT STUDENTS WILL SEE

The bare soil sends muddy brown water into the cup. The mulched pan sends clearer water with less soil.

Safety: Do not pour muddy water down a sink. Put soil outdoors or in the trash.

Try next: Try a design with terraces or a row of pebbles.

WATER HARDNESS SOAP SHAKE TEST

Hard water and soap scum

Grades 9-12 | Medium | Prep 15 min | Class 20-40 min | HS-PS1-2 | HS-ESS2-5

WHY DOES IT HAPPEN?

Hard water has dissolved calcium and magnesium ions. These ions react with soap to form a solid scum that stops soap from foaming. Soft water lets soap lather easily. Measuring the foam lets students rank water samples.

MATERIALS

- 3 clear bottles with lids
- 1/2 cup (120 mL) distilled or soft water
- 1/2 cup (120 mL) tap water
- 1/2 cup (120 mL) water with 1 tbsp (15 mL) Epsom salt stirred in
- 3/4 tsp (3.5 mL) pure liquid soap (not detergent) or grated bar soap
- Ruler
- Timer

6 STEPS

1. Pour 1/2 cup (120 mL) of each water sample into its own bottle.
2. Add 1/4 tsp (1 mL) of liquid soap to each bottle.
3. Cap each bottle and shake for 10 seconds.
4. Measure the height of the foam with a ruler.
5. Look at the cloudiness and any white scum in each bottle.
6. Rank the samples from softest to hardest.

THE EQUATION, SIMPLIFIED

$$Mg^{2+} + 2\,C_{17}H_{35}COO^- \rightarrow Mg(C_{17}H_{35}COO)_2$$

Magnesium ions combine with soap ions to form a solid soap scum that does not dissolve.

WHAT STUDENTS WILL SEE

The distilled water makes lots of foam and stays clear. The Epsom salt water makes little foam and looks cloudy with white flecks.

Safety: Wear splash goggles. Do not drink any samples. Use pure soap, because most dish detergents are made to resist hard water.

Try next: Test a sample of bottled mineral water.

HOMEMADE SECCHI DISK

Turbidity and light

Grades 6-12 | Medium | Prep 15 min | Class 20-40 min | MS-ESS3-3 | MS-ETS1-3 | Adult handling

WHY DOES IT HAPPEN?

Scientists lower a black-and-white disk into lakes to measure water clarity. The deeper it can be seen, the clearer the water. Cloudy water has many small particles. Students make a mini version and compare samples.

MATERIALS

- 1 white disk cut from a paper plate, 4 inches (10 cm) wide
- Black marker
- String, 24 inches (60 cm)
- 1 tall clear container, such as a 2-liter bottle with the top cut off by an adult
- 3 cups (720 mL) water
- 1 tsp (5 mL) soil
- Ruler

6 STEPS

1. 
Color two opposite quarters of the disk black to make a checkered pattern.

2. 
Tie the string to the center of the disk and mark every inch (2.5 cm) on the string.

3. 
Fill the container with clear tap water.

4. 
Lower the disk slowly until it just disappears and read the depth.

5. 
Stir in 1 tsp (5 mL) of soil and repeat the measurement.

6. 
Add more soil and graph depth against the amount of soil.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Suspended particles scatter light, so the disk disappears at a shallower depth.

WHAT STUDENTS WILL SEE

In clear water the disk stays visible all the way down. As soil is added the disappearing depth gets shallower, often by several inches.

Safety: Let only an adult cut the container. Do not drink the water.

Try next: Compare water from a pond, collected with permission, with tap water.

SOLAR STILL: SALT WATER TO FRESH

Evaporation and condensation

Grades 6-12 | Medium | Prep 15 min | Class 20-40 min | MS-ESS2-4 | MS-PS1-4 | Adult handling

WHY DOES IT HAPPEN?

When salt water evaporates, only water leaves. Salt stays behind. Warm sunlight turns the water into vapor, which cools on a surface and turns back into liquid drops. This is how some places make fresh water from seawater.

MATERIALS

- 1 large bowl
- 1 cup (240 mL) warm water with 2 tbsp (30 mL) salt stirred in
- 1 small cup (kept empty)
- Plastic wrap
- 1 small stone or coin
- Tape
- Sunny window or desk lamp

6 STEPS

1. 
Pour the salty water into the large bowl.

2. 
Place the small empty cup in the middle so its rim stays above the water.

3. 
Cover the bowl tightly with plastic wrap and tape it.

4. 
Set the stone on the plastic above the cup so the wrap dips down.

5. 
Put the setup in sun or under a lamp for 1-2 hours.

6. 
Look for drops collecting in the cup and describe them.

THE EQUATION, SIMPLIFIED

$$H_2O(l) \rightarrow H_2O(g) \rightarrow H_2O(l)$$

Water evaporates and condenses again, leaving the salt behind in the bowl.

WHAT STUDENTS WILL SEE

Within an hour droplets form on the plastic and drip into the small cup. A small amount of liquid collects.

Safety: Do not taste the collected water; it is not tested for safe drinking. Keep the lamp away from the plastic wrap and use warm, not hot, water.

Try next: Put dark paper under the bowl and see if it speeds things up.

WATER CYCLE IN A BAG

Evaporation, condensation, precipitation

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-ESS2-4 | MS-PS1-4

WHY DOES IT HAPPEN?

Water moves in a cycle on Earth. The sun heats water and turns it into vapor. When the vapor meets a cooler surface it condenses into drops. A closed bag shows these steps in one place.

MATERIALS

- 1 zip-top plastic bag (gallon size)
- 1/4 cup (60 mL) water
- 2 drops of blue food coloring
- Permanent marker
- Tape
- Sunny window
- Paper towel

6 STEPS

1. 
   Draw clouds, sun and ocean on the bag with the marker.
2. 
   Pour the water and food coloring into the bottom of the bag.
3. 
   Seal the bag, pressing out most of the air.
4. 
   Tape the bag to a sunny window.
5. 
   Check every hour and draw where drops appear.
6. 
   Describe where evaporation, condensation and precipitation took place.

THE EQUATION, SIMPLIFIED

$H_2O(l) \rightarrow H_2O(g) \rightarrow H_2O(l)$

Water changes state without becoming a new substance.

WHAT STUDENTS WILL SEE

After 1-2 hours drops cling to the bag above the water and begin to run down. The drops are clear, not blue.

Safety: Keep the bag sealed and wipe up any leak.

Try next: Place a second bag in the shade and compare.

ALGAE BLOOM SIMULATION

Nutrients and plant growth in water

Grades 9-12 | Challenge | Prep 15 min | Class 20-40 min | HS-ESS3-4 | HS-LS1-6 | Adult handling

WHY DOES IT HAPPEN?

Extra nutrients in a lake or pond can make algae and plants grow too fast. When they die and rot, they use up the oxygen. Fertilizer runoff from yards and farms can cause this. This model tracks growth over several weeks.

MATERIALS

- 3 clear jars
- 3 cups (720 mL) pond or aquarium water, collected with permission
- Plant fertilizer (none, a tiny pinch for 'Low', and 1/4 tsp / 1 mL for 'High')
- 3 small pothos cuttings or a pinch of duckweed
- Masking tape and marker
- Sunny window
- Ruler and notebook

6 STEPS

1. 
Label the jars 'None', 'Low' and 'High' fertilizer.

2. 
Pour 1 cup (240 mL) of water into each jar.

3. 
Add the matching fertilizer amount and stir gently.

4. 
Place a cutting or duckweed in each jar and set them in a sunny window.

5. 
Photograph and record color and plant growth every 3 days for 3-4 weeks.

6. 
Compare how green the water is and describe which jar changed most.

THE EQUATION, SIMPLIFIED

$$6\,CO_2 + 6\,H_2O \rightarrow C_6H_{12}O_6 + 6\,O_2 \text{ (light)}$$

With light and nutrients, algae and plants make sugar and oxygen and grow quickly.

WHAT STUDENTS WILL SEE

The 'High' jar usually turns greener faster than 'None' as algae multiply. The control jar stays clearer.

Safety: Wear splash goggles. Wear gloves with pond water and wash hands afterward. Keep fertilizer away from skin and mouth, and never pour the jars outdoors; an adult disposes of them.

Try next: Add a jar kept in the dark.

AIR PARTICULATE SAMPLING

Dust and pollution catchers

Grades 6-12 | Easy | Prep 15 min | Class 20-40 min | MS-ESS3-3 | MS-ETS1-3

WHY DOES IT HAPPEN?

Tiny particles of dust, soot and pollen float in the air. A sticky surface catches the ones that settle. By counting particles in different places students can compare air quality. This is a simple air-sampling tool.

MATERIALS

- 6 index cards
- 1/4 cup (60 mL) petroleum jelly
- 6 cotton swabs
- Tape and string
- Magnifying lens
- Marker
- Grid paper

6 STEPS

1. Draw a 1-inch (2.5 cm) square in the center of each card.
2. Spread a thin layer of petroleum jelly inside each square with a cotton swab.
3. Label each card with a location, such as classroom, hallway, playground, kitchen, window and near a road.
4. Hang or tape the cards in those places, sticky side facing out.
5. Leave them for 24-48 hours.
6. Count the particles in each square with the lens and graph the results.

THE EQUATION, SIMPLIFIED

No new substance forms: particles stick to the surface

Particles land on the sticky layer and stay, so the card holds a sample of the air.

WHAT STUDENTS WILL SEE

Cards in the busiest or dustiest places gather more specks than clean spots. Some look dark (soot) and some light (dust or pollen).

Safety: Keep petroleum jelly off skin and clothes. Wash hands after handling cards.

Try next: Test the same spot before and after cleaning.

DETERGENT AND SEED GERMINATION

Pollutants and plant growth

Grades 6-8 | Medium | Prep 15 min | Class 20-40 min | MS-ESS3-3 | MS-LS1-7

WHY DOES IT HAPPEN?

Chemicals that wash into soil and streams can change how plants grow. Seeds need water, air and the right conditions to sprout. Soapy water can interfere with seed coats and young roots. A controlled test shows how pollution can affect germination.

MATERIALS

- 30 bean or radish seeds
- 3 zip-top bags
- 6 paper towels
- 1/2 cup (120 mL) tap water
- 1/2 cup (120 mL) water with 1/8 tsp (0.6 mL) dish soap
- 1/2 cup (120 mL) water with 1 tsp (5 mL) dish soap
- Masking tape and marker

6 STEPS

1. Label three bags 'Water', 'Little Soap' and 'More Soap'.
2. Dampen two paper towels in each bag with the matching liquid.
3. Place 10 seeds in each bag between the towels.
4. Seal the bags and tape them to a window.
5. Check for sprouts each day for 7-10 days and count them.
6. Calculate the percent that sprouted in each group and compare.

THE EQUATION, SIMPLIFIED

No simple reaction: soap changes how water moves into seed tissue

Soap lowers the surface tension of water and can damage the membranes of young cells.

WHAT STUDENTS WILL SEE

The water group sprouts first and most. The strong soap group often has fewer or slower sprouts.

Safety: Do not eat the seeds. Wash hands after handling. Pour used liquids into the trash bag, not a storm drain.

Try next: Test a different cleaner, such as diluted vinegar.

LEAF DISC PHOTOSYNTHESIS

Photosynthesis in spinach discs

Grades 9-12 | Challenge | Prep 20 min | Class 20-40 min | MS-LS1-6 | HS-LS1-6

WHY DOES IT HAPPEN?

Photosynthesis makes sugar and oxygen from carbon dioxide and water, using light. The oxygen collects inside the leaf and makes it float. In this test, baking soda in water provides dissolved carbon dioxide. Brighter light should speed up the floating.

MATERIALS

- Fresh spinach leaves
- 1 drinking straw or hole punch
- 1 cup (240 mL) water
- 1/8 tsp (0.6 mL) baking soda
- 1 tiny drop of dish soap
- 1 plastic syringe, 10 mL, with no needle
- Bright lamp and timer

6 STEPS

1. Stir the baking soda and one tiny drop of dish soap gently into the water.
2. Cut 10 leaf discs with a straw or hole punch, avoiding veins.
3. Place the discs in the syringe, add 5 mL of solution, and pull the plunger back while blocking the tip so the discs sink.
4. Pour the sunk discs and liquid into a clear cup.
5. Put the cup under the bright lamp and count the floating discs every minute.
6. Repeat in dim light and graph the floating counts for both.

THE EQUATION, SIMPLIFIED

$$6\,CO_2 + 6\,H_2O \rightarrow C_6H_{12}O_6 + 6\,O_2 \text{ (light)}$$

Leaf cells use light to turn carbon dioxide and water into sugar and oxygen.

WHAT STUDENTS WILL SEE

Under bright light the discs begin to rise in 5-15 minutes as oxygen collects. Discs in dim light float more slowly.

Safety: Wear splash goggles. Use only a needleless syringe. Keep the lamp bulb away from water and hands.

Try next: Compare discs from different kinds of leaves.

THERMOS CHALLENGE

Insulation and energy saving

Grades 6-12 | Medium | Prep 15 min | Class 20-40 min | MS-PS3-4 | MS-ETS1-3 | Adult handling

WHY DOES IT HAPPEN?

Heat moves from warm things to cold things until they match. Insulation slows this movement. A good insulator saves energy because less heat is lost. Teams design containers and measure how well each keeps water warm.

MATERIALS

- 4 identical cups or jars with lids
- Hot tap water (adult pours)
- Insulating materials: cotton, foil, newspaper, bubble wrap
- Rubber bands
- 2 thermometers
- Timer
- Graph paper

6 STEPS

1. 
Wrap three cups, each in a different insulating material, and leave one bare as the control.

2. 
Ask an adult to pour 1 cup (240 mL) of the same hot tap water into each cup.

3. 
Put on lids and place thermometers inside through a small slit.

4. 
Record each temperature at the start.

5. 
Record temperatures every 5 minutes for 30 minutes.

6. 
Graph the cooling and rank the materials.

THE EQUATION, SIMPLIFIED

No new substance forms: heat energy moves from warm water to the surroundings

Better insulators slow the transfer of heat, so the water stays warm longer.

WHAT STUDENTS WILL SEE

The bare cup cools fastest. Cotton and bubble wrap typically keep the water several degrees warmer.

Safety: Use hot tap water only (not boiling). An adult pours it. Handle cups with care.

Try next: Add a second layer and compare it with the first.

RECYCLE YOUR OWN PAPER

Making new paper from scraps

Grades 6-8 | Easy | Prep 15 min | Class 20-40 min | MS-PS1-3 | MS-ESS3-3

WHY DOES IT HAPPEN?

Paper is made of tiny plant fibers. Recycling soaks old paper so the fibers separate, then lets them tangle and dry into a new sheet. It saves trees and energy. The fibers are not changed chemically in this process.

MATERIALS

- 1 cup (240 mL) torn paper scraps (plain, not glossy)
- 2 cups (480 mL) warm water
- 1 mesh screen, such as an old window screen in a frame
- 1 large tub or dishpan
- Clean towels
- Sponge
- Rolling pin or heavy book

6 STEPS

1. 
Tear the paper into tiny pieces and soak in warm water for 20 minutes.

2. 
Mash the pulp with your hands or a whisk until it looks like oatmeal.

3. 
Slide the screen under the pulp in the tub and lift it up flat.

4. 
Press with a sponge to remove extra water.

5. 
Flip the sheet onto a towel and press with a rolling pin.

6. 
Let it dry for 1-2 days, then peel it off and compare with the old paper.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change

Cellulose fibers separate in water and bond again when they dry.

WHAT STUDENTS WILL SEE

The new sheet is thicker and rougher than store paper and may have flecks. It can be written on after it dries.

Safety: Use warm, not hot, water and dry any spills on the floor. Do not pour pulp down the drain; put it in the trash.

Try next: Mix in dried flower petals before pressing.

NATURAL DYES FROM ONION SKINS

Plant pigments and fabric

Grades 6-8 | Medium | Prep 15 min | Class 20-40 min | MS-PS1-3 | MS-PS1-2 | Adult handling

WHY DOES IT HAPPEN?

Plants make colored molecules called pigments. Onion skins contain pigments that can bind to cloth fibers when heated in water. Soaking the cloth in diluted vinegar first changes how well the color sticks. This shows how natural resources can make colorful materials.

MATERIALS

- 1 cup (240 mL) dry yellow onion skins
- 2 cups (480 mL) water
- 2 white cotton cloth squares
- 1/4 cup (60 mL) white vinegar
- Pot and stove (adult use)
- Strainer and bowl
- Rubber bands

6 STEPS

1. Add the onion skins and water to a pot and have an adult simmer for 20 minutes.
2. Let the dye cool, then strain it into a bowl.
3. Soak one cloth in 1/4 cup vinegar mixed with 1 cup (240 mL) water for 10 minutes.
4. Place both cloths in the dye and tie one with rubber bands for a pattern.
5. Wait 30 minutes, stirring now and then.
6. Rinse, dry, and compare the color of the two cloths.

THE EQUATION, SIMPLIFIED

Dye molecules (in water) + cotton fibers → dye bound to fiber

Pigment molecules stick to the surface of the fibers; no new element is made.

WHAT STUDENTS WILL SEE

Both cloths turn golden yellow or orange-brown. The vinegar-soaked cloth often keeps more color after rinsing.

Safety: An adult handles the stove and the hot dye. Dye can stain, so wear an apron. Never eat or drink any of the materials.

Try next: Try the same dye on a paper towel or a wool scrap.

ROAD SALT ON ICE

Freezing point and the environment

Grades 6-12 | Medium | Prep 10 min | Class 20-40 min | MS-PS1-4 | MS-ESS3-3

WHY DOES IT HAPPEN?

Salt makes ice melt at a lower temperature. This is why roads are salted in winter. The salty water then runs into soil and streams, where it can harm plants and fish. Students compare melting with salt, with sugar and with nothing.

MATERIALS

- 3 ice cubes of the same size
- 3 shallow dishes
- 1 tsp (5 mL) table salt
- 1 tsp (5 mL) sugar
- Timer
- Thermometer
- Measuring spoon

6 STEPS

1. 
   Place one ice cube in each dish.
2. 
   Label the first dish 'Plain' as the control.
3. 
   Sprinkle salt on the second cube and sugar on the third.
4. 
   Start the timer and look at the cubes every 5 minutes.
5. 
   Measure the melted water in each dish after 20 minutes.
6. 
   Record which melted most and explain the results.

THE EQUATION, SIMPLIFIED

NaCl(s) → Na+(aq) + Cl-(aq)

Salt dissolves into ions that get in the way of water forming ice, so the freezing point drops.

WHAT STUDENTS WILL SEE

The salted ice melts fastest and makes the most water. The sugar cube also melts a bit faster than plain ice, but less than salt.

Safety: Do not taste anything. Wipe up spills and wash hands.

Try next: Test different salt amounts and graph the melted volume.

RUST RACE: IRON AND SALT WATER

Oxidation and corrosion

Grades 6-12 | Medium | Prep 15 min | Class 20-40 min | MS-PS1-2 | HS-PS1-2

WHY DOES IT HAPPEN?

Iron reacts with oxygen and water to form rust. Salt in water helps the reaction go faster by carrying charge. That is why cars and bridges rust faster near the ocean. Rust weakens metal and costs money every year.

MATERIALS

- 4 plain iron nails or small steel wool pieces (not stainless)
- 4 clear cups
- 1/2 cup (120 mL) tap water
- 1/2 cup (120 mL) salt water (1 tsp / 5 mL salt in water)
- 1/2 cup (120 mL) white vinegar
- Masking tape and marker
- Gloves

6 STEPS

1. 
Label the cups 'Dry', 'Water', 'Salt Water' and 'Vinegar'.

2. 
Pour 1/2 cup (120 mL) of the matching liquid into the last three cups and leave 'Dry' empty.

3. 
Place a nail or steel wool piece in each cup, half under the liquid.

4. 
Photograph and describe each sample.

5. 
Check every day for 5-7 days.

6. 
Rank the samples from most to least rust.

THE EQUATION, SIMPLIFIED

$4\,Fe + 3\,O_2 \rightarrow 2\,Fe_2O_3$ (simplified)

Iron combines with oxygen to make iron oxide, which is rust.

WHAT STUDENTS WILL SEE

The nail in salt water and the one in vinegar show orange-brown rust in 1-3 days. The dry nail shows none.

Safety: Wear splash goggles. Wear gloves; rusty metal can be sharp and steel wool can splinter. Do not touch your face. Throw the metal away afterward.

Try next: Paint one nail with clear nail polish and see if it resists rust.

CORNSTARCH BIOPLASTIC FILM

Natural polymer material

Grades 9-12 | Challenge | Prep 15 min | Class 20-40 min | MS-PS1-3 | HS-ETS1-2 | Adult handling

WHY DOES IT HAPPEN?

Plastics are long chains of molecules called polymers. Starch is a natural polymer that can be cooked into a flexible film. Adding glycerin makes the film bendy. Engineers explore these materials as alternatives to plastic.

MATERIALS

- 1 tbsp (8 g) cornstarch
- 4 tbsp (60 mL) water
- 1 tsp (5 mL) glycerin from a pharmacy
- 1/2 tsp (2.5 mL) white vinegar
- Small pot and stove (adult use)
- Spoon
- Foil sheet to dry on

6 STEPS

1. 
Mix cornstarch, water, glycerin and vinegar in a small pot.

2. 
Have an adult heat gently, stirring until the mix thickens like clear gel.

3. 
Remove from heat and let it cool for 2 minutes.

4. 
Pour the gel onto foil and spread it thin.

5. 
Let it dry for 1-2 days.

6. 
Peel it off and test stretch, strength and water resistance.

THE EQUATION, SIMPLIFIED

No new substance forms: physical change (starch gel network)

Starch granules swell in hot water and then form a flexible network that dries into a film.

WHAT STUDENTS WILL SEE

The mix turns from cloudy to clear gel within a minute or two of heating. After drying it forms a thin flexible film that swells in water.

Safety: An adult handles heat. The film is not food and must never be tasted. Wash the pot with warm soapy water.

Try next: Add a drop of food coloring or try more glycerin.

MULCH AND SOIL MOISTURE

Reducing evaporation

Grades 6-8 | Easy | Prep 10 min | Class 20-40 min | MS-ESS2-4 | MS-ETS1-3

WHY DOES IT HAPPEN?

Water in soil evaporates into the air. Mulch is a layer of material on top that slows evaporation. It helps plants use less water. A fair test shows how much water each cover keeps.

MATERIALS

- 3 small pots or cups filled with equal soil (1 cup / 240 mL each)
- 3/4 cup (180 mL) water
- 1/4 cup (60 mL) dry leaves or wood chips
- 1/4 cup (60 mL) pebbles
- Kitchen scale
- Tape and marker
- Sunny window

6 STEPS

1. Weigh each filled pot and write the mass.
2. Pour 1/4 cup (60 mL) of water on each pot.
3. Cover one pot with dry leaves, one with pebbles, and leave the third bare.
4. Place the pots in a sunny window.
5. Weigh each pot every day for 5 days.
6. Graph the mass lost and decide which cover saved the most water.

THE EQUATION, SIMPLIFIED

$H_2O(l) \rightarrow H_2O(g)$

Liquid water at the surface turns to vapor and leaves the soil.

WHAT STUDENTS WILL SEE

The bare pot loses the most mass each day. The mulched pot loses less.

Safety: Keep the pots on a tray to avoid spills. Wash hands after touching soil.

Try next: Add a plastic wrap cover and compare.

WARM WATER LOSES ITS GAS

Thermal pollution and dissolved gas

Grades 9-12 | Easy | Prep 10 min | Class 20-40 min | HS-PS3-4 | HS-ESS3-4 | Adult handling

WHY DOES IT HAPPEN?

Gases dissolve better in cold water than in warm water. Fish rely on dissolved oxygen. When factories release warm water into rivers, dissolved gas escapes. A glass of cold tap water shows bubbles forming as it warms.

MATERIALS

- 2 clear glasses
- 1 cup (240 mL) cold tap water, left standing a few minutes
- Warm tap-water bath in a bowl (adult prepares, below 120 F / 50 C)
- Thermometer
- Dark paper for a backdrop
- Timer
- Magnifying lens

6 STEPS

1. Fill a clear glass with cold tap water and let it settle.
2. Place the glass in the warm water bath that an adult prepared.
3. Put the dark paper behind the glass.
4. Record the temperature as it rises.
5. Look for small bubbles forming on the glass wall using the lens.
6. Compare with a second glass kept cold, and describe the difference.

THE EQUATION, SIMPLIFIED

$O_2(aq) \rightarrow O_2(g)$

Warm water holds less dissolved gas, so gas leaves the water as tiny bubbles.

WHAT STUDENTS WILL SEE

Tiny bubbles form on the sides of the warming glass within a few minutes. The cold glass has far fewer.

Safety: An adult prepares the warm bath and keeps it below 120 F (50 C). Do not use boiling water and do not drink the water.

Try next: Compare tap water that stood overnight with fresh tap water.